ConservAte: A WEB-BASED PLATFORM FOR FOOD SALES AND REDISTRIBUTION OF SURPLUS AND NEAR-EXPIRY FOOD PRODUCTS

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In Partial Fulfillment of the Requirements for the Degree of
Bachelor of Science in Information Technology
Specialized in *Service Management Program*

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ABSTRACT

Food waste remains a global challenge for food security, environmental sustainability, and economic stability. Despite these, various initiatives have not fully translated into communities having systems that can support the management of accessible surplus or near-expired food. Therefore, this research addresses the problem of increased food waste by developing ConservAte, a web-based platform for the sale and redistribution of food. Though previous studies have shown that incorporating technology supports sustainable consumption, most have failed to adopt location-based and user-centered approaches. Hence, the present research implements this by using GIS technology to connect consumers and vendors efficiently. Employing the Agile Scrum methodology for development, the system was evaluated using ISO 25010 and the Technology Acceptance Model. The test results indicated that the system met all quality standards, proving it to be efficient, usable, and reliable. This study provides evidence that ConservAte is an innovative yet practical way to advance food sustainability and community food security.

Keywords: *GIS, TAM, Redistribution, Food Security, ISO 25010, Agile Scrum*

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DEFINITION OF TERMS

Accountability	All system activities (logins, transactions) can be traced back to specific users.
Adaptability	The system can be adapted to meet changing user needs or business rules
Agile Scrum	A project management framework that uses short, iterative work cycles (called sprints) to develop and deliver products quickly, often used in software development.
Analyzability	Errors and issues in the system are easily identified and diagnosed.
Appropriateness Recognizability	The end-users can easily recognize and understand the functions and buttons of the system to complete tasks efficiently.
Authenticity	Only registered and verified users can access system features.
Availability	It is accessible to users, minimizing downtime and service interruptions..
Capacity	It can manage high data loads and numerous concurrent users while maintaining stable performance.
Cascading Style Sheets (CSS)	A language used to style and design web pages (e.g., fonts, colors, layouts).
Co-existence	It operates efficiently in a shared-resource environment, performing its tasks without degrading the performance of other applications.
Confidentiality	Only authorized users can view or use confidential data in the system.
E-commerce	It refers to commercial transactions conducted electronically on the internet.
Fail Safe	A system feature that ensures safe operation by automatically shifting to a secure state when a failure occurs.
Fail Tolerance	The system can be operated with ease, allowing users to perform functions with minimal effort.
Faultlessness	The system operates without unexpected errors or

	failure over time.
Functional Appropriateness	The system's functionalities are appropriate and well-aligned with its core objectives of reducing and redistributing food waste.
Functional Completeness	All modules (user, market, message, data analytics, product, notification, map) are working as expected.
Functional Correctness	The system provides accurate and reliable information for fresh, surplus, and near-expiry food listings, donations, and food redistribution processes.
Geographic Information System (GIS)	A system used to capture, analyze, and display spatial or geographic data (maps, locations, routes).
Hazard Warning	A system feature that automatically shifts to a safe state during failures to prevent unsafe or incorrect operations.
HyperText Markup Language (HTML)	It is the core markup language for building and arranging content on the World Wide Web.
HyperText Preprocessor (PHP)	Widely used server-side scripting language designed for developing dynamic websites and web applications.
Inclusivity	The system is intentionally designed for accessibility by a wide range of users, accommodating varying levels of technical proficiency.
Installability	It supports straightforward, conflict-free installation and removal within its target environments.
Integrity	It protects the accuracy and consistency of data against unauthorized modification, whether at rest or in transit.
Interoperability	The system interacts effectively with other applications to exchange and use information.
JavaScript	A programming language used to make web pages interactive (animations, forms, dynamic content).
Learnability	It is intuitive and easy to learn, allowing new users to operate it with little to no formal training.
Mitigation	The process of reducing or minimizing risks,

	damage, or negative effects of a problem.
Modifiability	Its components can be changed or updated in isolation, minimizing impact on the rest of the system.
Modularity	Its modular architecture facilitates simple and efficient maintenance.
Module	A self-contained unit or component in a system, software, or course that can work independently or together with others.
Non-Repudiation	Users cannot deny the actions or transactions they perform in the system.
Operability	The system can be operated with ease, allowing users to perform functions with minimal effort.
Operational Constraint	System-defined rules and limitations that guide how a system operates to ensure proper, safe, and controlled use.
Recoverability	It can recover its full functionality and information following a disruption or crash.
Redistribution	The process of distributing something again, often used in economics to describe reallocating resources, goods, or wealth.
Replaceability	It can be swapped out for a comparable software product with little difficulty or downtime.
Resistance	The system is resistant to unauthorized access and common security threats.
Resource Utilization	The system runs efficiently even on devices with lower specifications and does not experience slowdowns during use.
Retailer	A business or individual that sells products directly to consumers (end-user).
Reusability	System functions or components can be reused after modification
Risk Identification	The process of identifying potential threats or issues that may affect the system’s operations.
Safe Integration	The system safely integrates with external services without creating new safety risks for users.

Scalability	The system can be scaled up with minimal changes to its infrastructure.
Self-Descriptiveness	Each step and function within the system is clear, and the system provides appropriate feedback for user actions.
Surplus	An excess amount of goods, resources, or income beyond what is needed or used.
Technology Acceptance Model (TAM)	A theoretical framework that explains user acceptance of technology.
Testability	It can be reliably validated on different computers and in diverse technical environments.
Time Behavior	It refers to the system's performance in terms of response time.
User Assistance	The system provides adequate help and documentation to assist users when needed.
User Engagement	The system's interface is visually appealing and encourages continued use through clear feedback and an intuitive layout.
User Error Protection	The system helps prevent errors and enables end-users to identify and correct incorrect inputs.
Web-based	Software, applications, or systems that are accessed and used through a web browser rather than installed on a computer.

CHAPTER 1

INTRODUCTION

Project Context

Food waste is a worldwide crisis that every country faces. It needs comprehensive planning and understanding of these issues to provide an approach and mechanisms for developing solutions for proper food waste management (Garcia-Garcia et al., 2017; Sakaguchi et al., 2018; Reynolds et al., 2019; Lombardi & Constantino, 2020). Globally, approximately 1.3 billion tons of food are lost or end up in landfills, equivalent to almost one-third of the total food production intended for human consumption (De Los Mozos et al., 2020). Around 88 million tons of food waste are produced annually, equal to 174 kilograms per person, and, proportionately, 143 billion euros are lost economically (Lucifero, 2016). More frighteningly, global greenhouse gas emissions rose by up to 8% to 10%, which is being contributed to by food waste, drastically intensifying the climate problem (UNEP, 2021). Even with this large surplus of food, millions of people still suffer from starvation and malnutrition, underscoring an imbalance in the global food system.

Consumer behavior often contributes to food waste, as it is linked to pressing issues such as overbuying, unsafe food storage, and misinterpretation of food expiration dates. At the same time, food supply chain inefficiencies from manufacturing to distribution and retail worsen the problem (Gustavsson et al., 2011; Bond et al., 2013). The Sustainable Development Goal (SDG) 12.3, formulated by the United Nations, emphasizes reducing food waste per capita by half at the consumer and retail levels while lessening

food losses throughout the food supply chain (UN Annual Report, 2021). Prioritizing innovative and scalable solutions in addressing the food waste problem at multiple levels contributed a lot in solving these problems, as such efforts can directly tackle the root causes, create meaningful reductions in waste, and help achieve critical environmental and social targets.

Food waste in Los Baños, Laguna, is a critical issue that affects the environment and utilizes resources. As shown in Table 1, the municipality of Los Baños conducted 4,191 trips to collect garbage, amounting to 44,088 kg of compostable or food waste, averaging 3,674 kg per month. The highest amount of food waste collected was in April, with 25,165 kg in just 271 trips, overtaking other months and indicating increased food waste generation during that period. On the other hand, March recorded the lowest collection amount of 587 kg, which emphasizes minimal waste disposal. Based on the data, later months had higher collections of food waste across all establishments and households, indicating fluctuating trends similar to those in December, peaking at 3,459 kgs of compostables (LB-MENRO, 2025). Given the amount of food waste generated in the municipality, applying technology and other measures to reduce it is needed.

Though local data is limited, global studies indicate that user behavior is one factor driving food waste. Although food waste levels are lower than the world average, research indicates that individuals continue to waste considerable amounts of fruit and vegetables, calling for improved means to minimize waste (Esguerra et al., 2017). Research regarding food waste management approaches must be assessed and re-examined to determine the “best practices” for wider industrial adoption (Wu et al., 2021).

Table 1. Los Baños 2023 Compostables / Food Waste Data

Month	No. of Trips	Compostables (kgs)
January	261	1,886
February	247	729
March	266	587
April	271	25,165
May	375	768
June	389	1,307
July	367	1,481
August	370	2,343
September	400	2,157
October	366	1,673
November	423	2,569
December	456	3,459
Total	**4,191**	**44,088**
Average	**350**	**3,674**

The University of Maine proposed a solution named the Maine Food Recovery Hierarchy, from most desirable to less desirable steps to mitigate the wastage of food, including the reduction of unwanted food through inventory and conducting a waste audit, feeding hungry people through donating to food banks, establishing partnerships with organizations that distribute perishable foods, feeding animals, composting or converting to fuel, and disposal (Iannuzzi, 2021). To address the 28% of food waste from restaurants and businesses, Lewis (2023) encouraged the adoption of policies to reduce waste and redistribute surplus food to where it can be helpful. Retailers were driving to offer magazines and membership programs that help customers understand how to prevent food waste and how to use leftover food (Kor et al., 2017). With the advent of technology, systems and AI integration can be used in food waste management through demand

forecasting, quality control, shelf-life prediction, smart inventory, donation and redistribution, supply chain optimization, and consumer engagement (PECB, 2019). As most people use smartphones and are connected to the internet, having an application that helps collect food donations and gather information by contacting and collaborating with non-governmental organizations (NGOs), volunteers, and orphanages to redistribute surplus and disposable food reduced food waste (Purnachandu, 2024).

Project Objectives

This capstone project aims to develop a web-based system for selling fresh goods, managing food waste sustainably, and redistributing surplus and near-expiry food to enhance food security. The project specifically aims to:

1. Design and develop a web-based platform, integrated with Geographic Information System (GIS), to support the food sales and the redistribution of near-expiry and surplus food products.
2. To implement a tracking and analytics system to monitor food waste trends optimize redistribution, and ensure secure data management.
3. To evaluate the web-based system's abilities using ISO 25010 and Technology Acceptance Model (TAM) for Software Quality Standards.

Project Purpose

The purpose of this project corresponds to the United Nations Sustainable Development Goals (SDGs). This includes SDG 12:Responsible Consumption and Production, specifically SDG 12.3. This is the main SDG directly addressing food waste, emphasizing the need for efficient food

production, distribution, and consumption to minimize waste. SDG 2: Zero Hunger, which targets SDG 2.4, reducing waste in production and supply chains, is key to sustainable food systems. SDG 13: Climate Change targets SDG 13.2: Food waste contributes to greenhouse gas emissions (e.g., methane from rotting food in landfills), making waste reduction critical for mitigating climate change. SDG 11: Sustainable Cities and Communities, which primarily targets SDG 11.6, emphasizes that cities that reduce food waste contribute to more sustainable urban living. SDG 1: No Poverty, targeting SDG 1.2, reducing food waste can help ensure that surplus food is redirected to support low-income families. SDG 8:Decent Work and Economic Growth, targeting SDG 8.4, supports improving resource efficiency and promoting green jobs through sustainable practices, including innovations in food recovery and redistribution. Lastly, SDG 17: Partnerships for the Goals, targeting SDG 17.17, emphasizes the importance of cross-sector collaboration among governments, the private sector, and communities to scale solutions for food waste reduction and promote sustainable development, thereby improving food security and reducing household expenses. In particular, this project were beneficial to the following:

Consumers and Households. The web-based system benefits the consumers and households from cost savings, having enhanced food security as they know what the status of their purchased food product is, and better diets since avoiding waste means they save money on unused food and have access to safer, fresher foods. Their engagement in purchasing also earns a reward point to be converted to vouchers for price purchase reductions which encourage them to continuously engage in the system.

Retailers and Food Industries. Using the web-based application allowed retailers and food manufacturers to reduce their losses by monitoring the shelf life of their products, being notified of nearing expiration and spoilage. This also served as a tool to know their customer preferences to join the trend. Their inventory management was efficient, and they easily determined which products customers always purchase through data analytics. The system reduces disposal costs by enabling them to manage their surplus food for redistribution, increase profit margins, and create a positive image of the company and businesses.

Farmers and Food Producers. This study helps farmers and food producers reduce post-harvest losses by enabling them to understand market trends and demand over time, thereby increasing profits and improving the efficient use of agricultural resources.

Governments and Policymakers. The governments and policymakers were informed and given insights into the food waste situation, and they were able to formulate policies and laws to manage and reduce waste, fund technology and innovation for food sustainability, improve food security, and advance climate action goals.

NGOs and Food Aid Organizations. This study assisted non-governmental organizations (NGOs) and food aid organizations in redistributing surplus food waste to vulnerable communities and charities, thereby further enhancing food security.

The Environment. This system and its strategies were beneficial to the environment as lower food waste minimizes the production of methane and greenhouse gas emissions, saves natural resources like water and

energy, and reduces landfill pollution as it takes time before the food becomes totally composted and is used as fertilizer.

Future Researchers. They can examine and provide more innovative solutions to reduce food waste and food loss, e.g., enhanced storage methods, improved supply chain efficiency, and support for sustainable consumption practices.

Scope and Limitation of the Study

This project aims to develop a food waste management application to help reduce concerns about food loss and waste in Los Baños, Laguna. The system includes modules for user account management, product availability, food waste redistribution, data analytics, notifications, messaging, and mapping, aiming to enhance food security, improve the efficiency of inventory management, and increase the effectiveness of implementing food waste reduction strategies through the web-based system.

User Module. The user module of the web-based system supports users to create, register, authenticate, and manage their accounts in the account settings. Within this module, the role-based access feature enables the system to identify each user's role: super admin, seller (admin), or customer. It also allows customers to manage their carts. The safe login process is also supported, in which only validated accounts can view their own accounts and the system's features.

Product Module. The user module of the web-based system supports users to create, register, authenticate, and manage their accounts in the account settings. Within this module, the role-based access feature enables

the system to identify each user's role: super admin, seller (admin), or customer. It also allows customers to manage their carts. The safe login process is also supported, in which only validated accounts can view their own accounts and the system's features.

Market Module. The market module enables customers to view all available products in the market added by sellers. They can also add their desired products to their cart and have their order processed by checking out directly or from the cart. Using this module streamlined the order process by checking out, tracking, and receiving orders.

Notification Module. The market module enables customers to view all available products in the market added by sellers. They can also add their desired products to their cart and have their order processed by checking out directly or from the cart. Using this module streamlined the order process by checking out, tracking, and receiving orders.

Message Module. In this module, the system allows the vendors and customers to message each other to talk about the products and the order; the customer can directly message the seller based on the product display, while the vendors receive it and answer the customer's concerns and questions for smooth transactions and settling issues between the vendors and customers. This also includes admin support and complaint handling, where customers and vendors can connect with the admin via messages to report system issues.

Waste Management Module. This module features redistribution of products to determine what to do with near-expiring products and reduce food

waste by reusing and distributing excess food that would otherwise be thrown away, frequently to those in need, through sales or donations.

Data Analytics Module. This module provides the user with their report analytics, including sales reports, food waste statistics, and system insights. It helped users to evaluate and track their performance, identify trends, and support improvement based on the data.

Map Module. This module uses Geographic Information Systems (GIS) to present an interactive map that improves user and vendor experience. For customers, it provides a visualization of markets around them, enabling them to quickly identify the nearest vendors. For the vendors, it gives the functionality to pin and manage market locations on the map, boosting visibility and access to potential buyers.

CHAPTER 2

THEORETICAL FRAMEWORK OF THE PROJECT

Review Of Related Literature, Studies, And Systems

This chapter presents a comprehensive review of related literature and studies, both local and foreign, found in reliable sources, including hardbound thesis books, online articles, blogs, and other trustworthy published materials. It served as the researchers' guide in conducting the study. Additionally, it connects and examines several earlier studies that helped in explaining and understanding the significance of the study.

Food Security

Food security is defined as the situation in which everyone, everywhere, has physical and financial access to enough wholesome food that meets their dietary requirements and food choices for an active and healthy life (World Bank Group, 2024). Even though food is a basic human right and necessity, the number of people severely starving continues to rise today, underscoring the centrality of food security to global issues that still require solutions (UNICEF, 2024). However, a number of problems, including growing populations, climate change, increasing food prices, and environmental issues, make this increasingly difficult to achieve (Global Network Against Food Crises, 2024). Strategic measures to address land use, trade, food affordability, water management, and food waste reduction are needed to address these problems (IFPRI, 2024).

In 2050, the global population is expected to reach 9.8 billion from 7 billion in 2010, which drove food demand to grow by over 50%, with demand

for animal-based foods skyrocketing nearly 70% (Ranganathan et al., 2018). Besides, agriculture claims about half the world's vegetated area and accounts for one-quarter of global annual greenhouse gas emissions from land-use changes (Mollins, 2019).

Achieving a sustainable agricultural production system, along with the associated policies and provisions, is quite challenging in balancing food security and environmental sustainability (Searchinger et al., 2019). Despite these predicted statistics, hundreds of millions of people currently experience starvation and skipping meals (Food and Agriculture Organization, 2009).

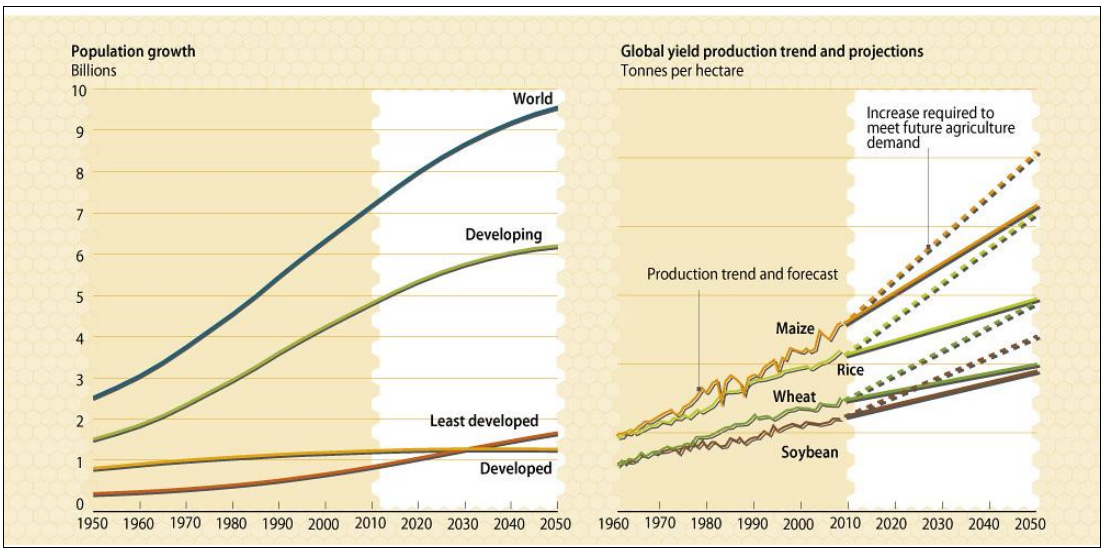


Figure 1. UN DESA 2013's Estimated Population in 2050

The urgency of food security was further exposed in 2020, marking a significant setback in the worldwide struggle against Hunger. The rate of malnutrition skyrocketed from 8.4% in 2019 to 9.9% at the start of the pandemic, an increase of 1.5% from the previous year (FAO, 2021).

In the same year, between 720 and 811 million individuals were affected, as statistics showed an increase of 118 million people compared to 2019 figures.The outcome of this event demonstrates the vulnerability of

global food security to major disturbances and underscores the importance of robust government measures to enhance crisis resilience and adaptability.

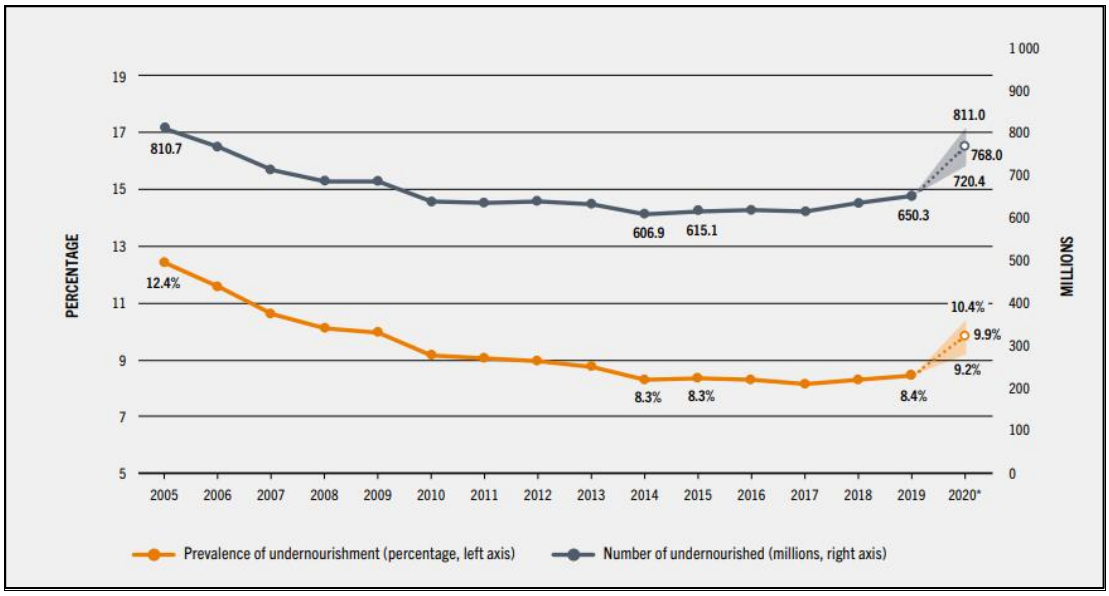


Figure 2. FAO's Malnutrition Report 2021

Food waste remains a significant factor contributing to food insecurity. In 2021, more than 10.2% of US families experienced food loss and hunger (Coleman-Jessen et. al, 2021), showing that instead of nutritious food ending up in landfills, it should have been fed to people, as there are lots of families experiencing starvation, making food waste a social challenge.

Food that was disposed of in landfills depletes a lot of essential resources, such as quality soil, clean water, renewable and non-renewable energies, and labor that was used in producing, storing, and distributing food. The disposed foods in landfills emit methane, a powerful greenhouse gas that greatly impacts environments, is one of the factors making the climate drastically change that affects all sectors of the government. Additionally, in the economy, United States food loss amounted to $161 billion annually due to loss and wastage of food at the retail and consumer household levels (US EPA, 2025).

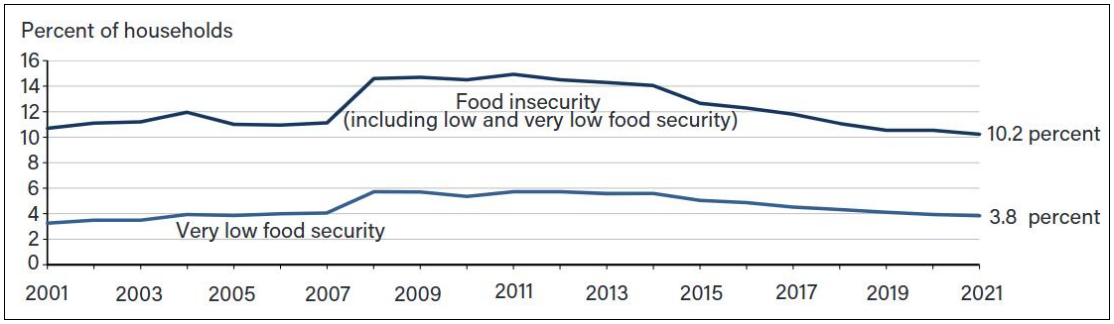


Figure 3. USDA and ERS Food Security Report 2021

According to FAO et al. (2024), the world is still far from achieving the targets of Sustainable Development Goals 2.1 (ending hunger) and 2.2 (eliminating malnutrition) based on current trends, despite some progress in select countries. Billions of people still don't have access to safe, secure, and nutritious food for their survival. Even though there is global awareness of the need to strengthen financial mechanisms to address this issue, significant gaps remain in defining, tracking, and implementing these mechanisms. With only six years left before the 2030 deadline to eradicate this, there are still insufficient resources and plans to achieve food security, especially the goal of zero hunger.

Many families in vulnerable areas, such as Malawi and Zimbabwe, have had to either skip meals or go days without eating due to droughts and poor crop and vegetation yields induced by climate change (Caritas Australia, 2023). This further emphasizes the need for sustainable solutions and stronger global policies to ensure food security for all (UNEP, 2023). Achieving food security involves addressing both environmental and economic challenges while managing natural resources efficiently (Social Science Library, 2020). Without immediate and sustained action, food insecurity continued to rise, affecting millions worldwide.

Food Loss and Food Waste (FLW)

Waste of food is a global issue, resulting in the loss of edible food from the very start to the end of the supply chain, from farm to household (Gascón, 2018; Girotto et al., 2015; Corvo, 2019). Food waste results when food is lost in quantity or quality during production, harvesting, or processing, mainly in developing countries with inadequate infrastructure and storage. Surprisingly, nearly a third of all produced food is wasted before it reaches consumers worldwide (Moračanin et al., 2024). The globe produces an enormous amount of food to sustain its people, but much of it is lost, wasted, spoiled, or damaged along the supply chain (Milijasevic, 2024). Each year, much of the food is wasted at various points along the food supply chain, from creating and producing food up to its distribution to retail and consumption (Gollnhofer, 2017).

According to the United Nations Environment Programme (UNEP) and Food and Agriculture Organization (FAO) (2022), in a global scenario where 811 million people are undernourished, and 3 billion are unable to afford a healthy diet, the percentage of food waste was sufficient to feed approximately 1 billion people out of the population.

About 17% of all food grown globally is lost and wasted, and about 14% of all food grown for human consumption fails to reach its target consumers. The level of lost food along the whole consumption and production chain is unveiled through these statistics, highlighting the necessity for better management and sustainability, and that’s the reason why accurate and detailed data on food waste are the key to implementing and monitoring

effective measures of reduction, as per the United Nations Sustainable Development Goal 12.3 (Corrado & Sala, 2018).

The economic impact of global food loss and waste is estimated at around $936 billion per year, and this amount excludes the additional societal and environmental costs (Food and Agriculture Organization, 2014). Food Waste can be considered the greatest global problem, requiring urgent attention and integrated actions from stakeholders across the food value chain to develop solutions for present and future generations (Ghosh et al., 2015). That is why Abiad et al. (2018) stated that food loss and food waste are two challenging dilemmas facing the world today that affect food security, the environment, and the regional and national economy. Beyond the lost food itself, the process squanders the vital resources invested at every stage—from production to disposal. This inefficiency harms the environment, creates economic losses, and undermines the food system's long-term viability(Garnett, 2011).

Food waste is a serious social and environmental issue, and as the population rises, its effects probably worsened. As stated by the United Nations Food and Agriculture Organization (2011), to feed a global population expected to reach 9.1 billion by mid-century, food output needed to expand by more than 70 percent. Food waste constitutes 24 percent of municipal solid waste landfilled and 22 percent of municipal solid waste combusted, the most frequently landfilled and burned material in the United States. (From Farm to Kitchen | US EPA, 2025).

Food wastFood waste is not only a problem for the food itself but also for the water, energy, and land resources used to produce it,

and it harms the environment and worsens food insecurity, especially in vulnerable countries during crises like COVID-19. According to the United Nations, the global food system is responsible for 30% of the world's energy consumption and generates 25% of its greenhouse gas emissions (Batool et al., 2023).

Scholars from a wide range of academic fields have focused their research on household food waste in industrialized countries. We can approximately distinguish two social ontologies that have guided the literature on consumer food waste, without wanting to draw too sharp disciplinary lines. According to Steg and Vlek (2009), a core focus of environmental psychology is to identify specific measurable factors, ranging from an individual's thoughts and drives to broader societal processes, which can support or hinder pro-environmental actions. Such approaches are often grounded in environmental psychology or consumer behavior. Nonetheless, intention and, to some extent, actual behavior are only partly explained by these cognitive and intrapersonal factors (Gatersleben et al., 2002).

A study by Stefan et al. (2013) found that the intention to reduce food waste does not significantly translate into less actual waste, even when accounting for planning and shopping behaviors. In other words, individuals' good intentions to avert food waste are often not carried out.

Food Waste Redistribution and Mitigation

According to the study by Galli et al. (2019), all parties in the food supply chain have a responsibility to prevent and reduce waste, including farmers, product manufacturers, sellers, and consumers. They highlighted

that food surplus redistribution is one of the solutions for mitigating food waste and alleviating poverty, though at times others view it as shifting the government's responsibility. Italian food bank research indicated the need for long-term policies to relieve poverty through food waste redistribution. In addition to this, Mir, Chang, and Hefni (2024) highlighted the need for combined efforts in reducing food waste in Saudi Awarabia and emphasized the essence of assistance from the government, financing infrastructures, engagement of related stakeholders, and providing awareness for the public. An environmentally integrated policy named "Culture of Personal Food Governance" was proposed in Morocco, which highlighted actions of controlling, promoting, and monitoring that guided the people in the academic field, decision-makers, and community-based groups to have a protocol in raising awareness against food waste in households (Zhoubi et al., 2024).

Reducing food waste and addressing food insecurity in the UK is a "win-win" solution through SFR, but it also faces criticism for not discussing the root causes of poverty and product overproduction, as these may unintentionally add to systemic issues (Papargyropoulou et al., 2022). Research focuses on surplus food distribution supply chains, and coordination with third-sector organizations and other actors is limited. They also reveal some hindrances in the food surplus supply chain, such as the insufficient regulatory framework for organizational sustainability to mitigate unforeseen issues in food quality and quantity (Karki et al., 2020). To aid this, policy-making is one factor influencing the minimization of food waste, as many intellectuals from diverse fields and organizations are interested in the issue. Food waste policies can be improved by developing systematic, organized

policies for food banks, enacting legislation, and funding scientific research projects (Shen et al., 2023).

The food system must become sustainable while adapting to climate change and helping mitigate it. They also suggested that it is necessary to address starvation through redoubling efforts, with almost one billion people still suffering from hunger and lack of nutritious food (Foresight, 2011).

The UK has limited food redistribution initiatives, perhaps because efforts to redistribute surplus food are still largely fragmented, with organizations operating independently. In solution to this, making this initiative at a national level helped make this practice doable, and governmental assistance through fiscal inducement could lead to a coherent system of surplus food redistribution and the development of a larger step in minimizing food waste and regaining food's intricate value (Facchini et al., 2018).

Ruth, Davies, and Ng (2020) conducted a study in which they reviewed Singapore's food waste management and how it supported the transition to surplus food redistribution through ecologies of participation (EOP), and found that EOP is significant in improving policy-making in the transition field. In response to the prevention and poor management that contribute to food waste, Messner et al. (2020) proposed a number of innovative approaches, including re-appraising food overproduction as a main cause of food waste.

A study by Calvo-Porral et al. (2013) found that high-income or developed countries produce substantial food waste due to poor marketing strategies, unmonitored consumer behavior, and food consumption patterns. Supporting this claim, Jungowska et al. (2021) found that purchasing too much food due to being attracted to promotions and discounts results in

buying and stocking without planning, which influences food waste in households. Aside from this, insufficient control over expiry dates and a lack of arrangements for what they did with food waste are also factors contributing to a large amount of food waste. Parallel to this, the pricing system used to promote products faced criticism because it somehow encourages consumers to overbuy, resulting in food waste. Still, the result of their study counters the allegation and shows that families who purchase discounted products waste less food, as well as people who care about the environment. In contrast, larger households and those who do not practice food waste management have more food waste (Tsalis et al., 2024).

Calvo-Pottal's study highlighted the importance of good marketing practices in reducing food waste, such as sales promotions, clear date labels, and campaigns to raise awareness about food waste. For instance, the Food Recovery Network (FRN) collaborated with Gaylord National Resort. It displayed a great impact of food waste redistribution, wherein over 67,000 pounds of excess food were recovered and given to the hungry. If not redistributed, it would be equivalent to 2 years' consumption of over 56,000 meals to nonprofits fighting hunger. In the process, 7.81 million gallons of water were saved, and 43.76 metric tons of carbon dioxide were reduced, indicating and proving the environmental advantage of food waste redistribution (FRN, 2025).

Food Waste on Sustainability

Converting food waste into a useful resource for the development of sustainability and food security. Due to the execution of composting, bioconversion, and recycling, we can reduce the issues in landfill waste,

lessen greenhouse gas emissions, and produce new economic opportunities (Sarangi et al., 2024). Furthermore, food waste is a huge global concern that can be lessened with the aid of strategies like biorefineries and sophisticated valorization techniques (Sarker et al., 2024). This study shows various approaches to minimizing food waste, such as circular economy models, sustainable farming practices, and food recovery initiatives. Each section examines the key causes of food waste and presents alternative solutions (Swetha et al., 2024).

The study highlights how sustainable technologies and zero-waste practices can promote circular bioeconomy principles. By transforming food waste into valuable resources, it highlights the potential to benefit both the environment and the economy (Pal et al., 2024; Salvin, 2024). The circular bioeconomy aims to promote sustainability by using bio-based resources, but this study highlights some potential downsides. It points out that putting more pressure on supply chains and ecosystems could end up causing more harm to the environment than the benefits it aims to deliver. (Tan & Lamers, 2021).

This study presents the effect of food waste on the environment, society, and economy, and it also features the need for sustainable food production and technological innovation. Through the analysis of precision agriculture, smart packaging, blockchain, artificial intelligence, and consumer-centric apps, it is possible to minimize food waste from the very start to the end of the entire food supply chain. (Taner, 2024). In addition, according to Browne (2023), the estimated amount of food produced worldwide that goes to waste is around one-third, making it one of the most pressing global challenges; an average of 1.3 billion tons of waste annually leads to serious

environmental, economic, and social consequences. Meanwhile, the benefits of switching from a traditional linear food system to a circular one are significant. This concept turns food waste into a new resource through improved tracking, rescue initiatives, education, infrastructure, contributions, and composting (Lewis, 2023).

In addition, a circular food system shows the combination of a secondary supply chain with the help of the Internet of Things; they can manage food waste sustainably by recycling garbage to create new products and prolong their useful lives (Iqbal & Kang, 2024). This study explains the negative side of using circular practices, such as the risk of contaminants sticking around and building up. Additionally, it is alarming to find pathogens in recycled garbage, which may be harmful to human health. Additionally, the presence of mycotoxins and heavy metals can lead to serious safety risks (Pearson et al., 2024).

This study shows several solutions in terms of minimizing food waste concerns, including the enhancement of production practices, conducting campaigns to encourage sustainable practices, and enhancing waste management activities.The study also highlights anaerobic digestion (AD) as a sustainable method that transforms food waste into valuable biogas and biohydrogen. However, this strategy is still problematic, such as supply chain issues, financial barriers, and the low value of the end products (Kohli et al., 2024). Still, Power Knot (2021) notes that while anaerobic digestion (AD) is perceived to be a sustainable way of handling food waste, it has several limitations. The process emits methane, a potent greenhouse gas, which can be harmful to the environment.

In addition, AD systems consume considerable water and energy to operate, and disposal of residual digestate is an issue since it contains hazardous pollutants. In this project, the potential utility of biorefinery technology as a viable alternative is exhibited through converting waste potato peels (RPPW) to numerous biowastes with the three-stage procedure of ultrasonic extraction, hydrolysis, dehydration, and ultimately pyrolysis. In the generation of biochar for water purification and antioxidant 5-hydroxymethylfurfural (HMF) as precursors in biobased chemicals, this method converts food waste into resources that can be utilized profitably (Ebikade, 2020).

System Application Development on Food Waste Management

According to Liorančas and Lee (2024), digital technologies are still in the starting stage of development, where experimentation, prototyping, and deployment of applications related to food waste reduction are still limited. Digital innovations solving food waste are still lacking in data regarding social, environmental, and economic advantages (Ciccullo et al., 2021).

In a study conducted by Vogels et al. (2018), they found that there must be a reward or incentives for users to achieve the success of food waste management applications, and they also noted that there is a lack of willingness on the consumer side in using the apps, as it only consumes more time, energy, and effort than the benefits they gained. Aligning with this is the previous research conducted by Elimelech et al. (2019), where this perception highlights that consumers are underestimating their food wastage. Alternatively, the risks of privacy loss are also cited to negatively affect the effectiveness of apps for consumers (Nastasi & Fraccascia, 2023). Attitudes

towards the use of these mobile applications are interrupted by these results (Doğan et al., 2023). To make the implementation of food waste management applications successful, Rizzo et al. (2023) suggested in their study that sellers and managers who have food and beverage businesses formulate effective strategies to use new technologies and stimulate sustainable behavior by considering individual differences of consumers and their desire to have a loyal customer.

Sustainability benefits have been seen in redistributing surplus food using mobile applications, demonstrating that it is necessary to develop technologies that help an in-depth understanding of factors that affect the perception of users in the field of food waste management (Apostolidis et al., 2021). Technological innovations, including intelligent packaging and IoT-enabled smart kitchen appliances, can extend food freshness. According to Liegeard and Manning (2019), this extended shelf life contributes directly to a reduction in household food waste. In line with this, K (2022) explained that in order to fill the gap of having tons of food waste, using technological advancements such as developing multiple food waste apps to fight the crisis of food being thrown away made the manufacturers, consumers, and community linked, enabling collaborations. K also indicated some of the advantages of using food waste apps, such as making inventory management easier and more efficient, and businesses may sell their near-expiry food at a discounted price, minimizing food costs.

In addition, the study of Purnachandu et al. (2024) indicated that using mobile applications in reducing food waste was positively perceived as useful and easy to use, without focusing on risks and their negative impact on the

willingness of the respondents, resulting in minimized food waste at dining, events, and other mess areas.

European Cluster Collaboration Platform (ECCP, 2018) developed Waste Master (Havikkimestari), which is another example of a web-based application designed for restaurant personnel to easily weigh and list the amount of food spoilage, resulting in a 45% decrease in food waste in just six months and minimizing groceries by around 3-9%. FoodTrek App is an application that informs consumers about the expiration dates of food items in their homes through notifications three days before and each day passing until the end date. Aside from this, the FoodTrek app monitors weight, price information, spending, and food waste costs (Phiri & Trevorrow, 2019).

Digital platforms are emerging as practical tools to combat household food waste. Misfits Market (2024) operates by sourcing "imperfect" organic produce directly from farmers to reduce systemic waste and offer consumers discounted goods.

Similarly, the app Remy (2025) helps users minimize waste by using receipt-scanning technology to automatically track pantry inventory and discourage unnecessary purchases.

According to Hong et al. (2023), food waste reduction using mobile applications is becoming a trend and growing rapidly as it is accessible in mobile application markets like Google Play and the App Store.

E-Commerce

E-commerce websites serve as vital platforms for businesses to promote their products and provide customers with easy access to product

information. However, limitations like restricted retail offerings and inefficient inventory management using manual reporting can hinder sales growth and harm online operations. To address these issues, this study developed a web-based e-commerce application using the waterfall methodology. The application provides real-time data on stock levels and sales, generates automated reports, and includes features to help shops promote their available products. After implementing this method, the result of reporting and having properly controlled inventory data is a sales increase and the shop marketing spread (Jannah et al., 2023). However, in 2024, this article shows the disadvantage of using the waterfall method, as this method can't change something they had already done previously, and being unable to change or fix some mistakes in the system can lead to an issue for system efficiency. This method accepts only what the developer visualizes for the system because the system is not capable of identifying the client's feedback or opinions, and this can also lead to an issue with user satisfaction in using the system (Auvee, 2024).

This study shows the improvement of e-commerce in terms of performance in using the Agile Scrum Method together with the Yii Framework. The use of agile scrum promotes flexibility, response to evolving market demands, and optimal stakeholder participation. Meanwhile, the Yii Framework provides a strong basis for creating scalable web applications. By combining Agile Scrum and the Yii framework, the result enhances the efficiency, scalability, and responsiveness of an e-commerce solution. It also improves the user experience, drives sales, and, lastly, gives e-commerce web development a competitive edge. (Nurhaeni et al. 2024).To develop the

solution, the study utilized the Agile Scrum framework, progressing through stages such as the product backlog, sprint planning, and daily stand-ups. This iterative process was crucial for refining the application, ultimately resulting in a system with excellent responsiveness, an intuitive interface, and a cohesive visual design that fulfilled the project's core aims (Journal JTIK, 2025). However, this study claims that there is a limitation in using the Agile Scrum method because of the reason that it is mainly used for small-scale projects, which makes it less efficient for large-scale project development (Khalid et al., 2020).

According to Al-Ebrahim et al. (2023), the use of machine learning enhances e-commerce by implementing personalized recommendations, optimizing inventory management, and helping businesses to better understand customer behavior to predict purchases and detect fraud, benefiting both businesses and customers.

In addition, the use of AI or machine learning changes e-commerce by making the shopping experience smoother, improving how businesses operate, and helping to develop businesses, making it a major key for future online retailers (Taylor, 2025). While AI makes e-commerce systems more efficient, it also causes risks, such as data breaches and biased algorithms that could serve customers unfairly, and this might also lead to job losses and lessen the human oversight that could affect trust and accountability (Shopify, 2024).

Mobile e-commerce makes shopping easier by offering more product choices and letting people buy anytime, anywhere. It also helps businesses save on advertising costs and reach customers around the world (Gupta et al.,

2023). Additionally, e-commerce apps make it easier for businesses to run smoothly, save time and money, and grow their customer base more effectively (Parikshith & Natesan, 2023). Meanwhile, this study says that e-commerce security is the most crucial part of all information security, focusing on online transactions and sensitive data. That leads to important aspects like integrity, authenticity, confidentiality, and privacy; at the same time, addressing the risks and concerns of online shopping also promotes the importance of having impenetratable security measures to ensure safety and reliable payment procedures (Dharmendra, 2013).

The State of Philippine E-Commerce

According to Clarete (2019), the Philippines has the smallest e-commerce market among selected ASEAN nations, a status largely constrained by poor internet quality. In fact, the country has the second slowest internet speed in the region, which results in the lowest performance in business facilitation. To address these systemic challenges, the Department of Trade and Industry's (2021) eCommerce Philippines 2022 Roadmap outlines strategic initiatives focused on improving Market Access, Digitalization, and Logistics Integration (MADALI) which stated their goal in eCommerce for the Philippines: “This Roadmap provides not only a general plan, but concrete and comprehensive strategies to help us shape the future of Philippine E-Commerce because we see a future where e-Commerce became easy commerce and commerce were indistinguishable from e-Commerce” (p. 2). Improving telecommunication and increasing the rate of population with the internet are some factors that are causing the Philippines to be a notable player in the Southeast Asian Market (Pua, 2023).

The Philippines faces a significant e-commerce adoption gap. Despite strong growth in the broader digital economy, data shows that as of 2015, merely 14% of businesses had adopted online sales platforms (Philippine Institute for Developing Studies, 2020). However, in 2020, e-commerce in the Philippines saw massive growth in the past five years due to the pandemic caused by the outbreak of the COVID-19 virus, which forced the country to transact digitally. GlobalData stated that e-commerce in the Philippines is foreseen to rise at a substantial Compound Annual Growth Rate (CAGR) of seventeen percent between 2021 and 2025. Furthermore, the expected growth is 2.5 times higher by the year 2025 (International Trade Organization, 2024).

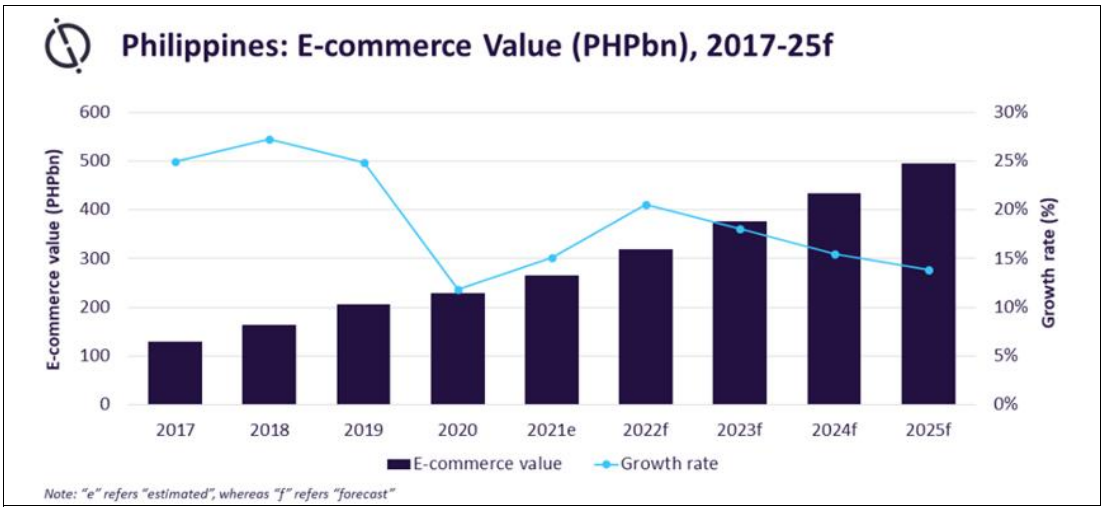


Figure 4. Compound Annual Growth Rate (CAGR) in Philippine E-Commerce

The Philippines' internet economy is expected to experience tremendous growth between 2019 and 2024, and is estimated to be 150 million U.S dollars in 2030, with an 80% population of internet users in the Philippines (Statista Research Department, 2025). In recent years, 14% of the digital economy was captured in the Philippines' eCommerce, with a growth of 18.5% annually, placing the country in the top ten countries globally for e-commerce traffic

(The Philippines Shopee and Lazada, 2025). Francisco (2024) writes that with 119 million individuals and 73–83% internet penetration, the Philippines is a huge digital economy in Southeast Asia. Its mobile-first population has seen a 15% growth in the digital economy in the region, the e-Conomy SEA report states. In terms of creating the economic digitization measures, Philippine officials are on the correct path. The government must focus more on infrastructure construction, enhance digital competence among the workforce, and strengthen the regulatory framework to ensure a glitchless process of digitization (Seah, 2024).

The Philippine eCommerce Roadmap (2022) also describes that the government is favorably disposed to broadening eCommerce and aims to either solve or redirect the issues and hindrances standing in the way of the Philippines from maximizing eCommerce's true promise. More disciplined government regulation toward proper taxation, consumer safeguards, and infrastructure establishment is part of its plans.

Integration of GIS to E-commerce and Food Waste Management

The Geographic Information System (GIS) enables people and organizations to view, analyze, and comprehend the patterns and relationships through mapping different kinds of data on just one map, such as streets, establishments, landmarks, and other locations. This data was being captured, stored, checked, and displayed with the related data about the structure of Earth's landforms through the computer system, the GIS (National Geographic, 2023). It allows users to view spatial data in its proper format, making it easier to interpret and understand (Alesheikh et al., 2002).

GIS is widely used in the disciplines of engineering, science, geography, and space technology, making industries flourish (Shastri, 2020). However, the implementation and integration of GIS in e-commerce are still lagging behind other industries (Mohammed et al., 2023).

Using GIS technology, system developers can make a system that offered an excellent user experience, maximize the use of geospatial data, make great decisions, and make the development workflows uninterrupted and easier (Andersen, 2024). In the study of Raut et al. (2015), they concluded that GIS is important in preserving data, facilitating food waste collections, and informing NGOs, resulting in reduced time and costs.

E-commerce with integrated GIS helps businesses to know their target market through customer demographics and behavior based on their positions and geographical locations. With this, they allow vendors to create marketing strategies through campaigns, personalized promotions, and discounts based on user engagement and products personalized for specific people and regions (Swain, 2025).

Using geospatial analysis of food waste can help develop food waste reduction strategies by analyzing the geographic distribution, sources, locations, and amounts of food waste, thereby enhancing the redistribution of food more conveniently and easily (Thaore et al., 2024).

Synthesis and Relevance of the Study

Studies and literature emphasize the economic, social, and environmental consequences of food waste, highlighting the need for

technological innovations, policy interventions, and behavioral changes among households and other stakeholders in the food chain. Mitigating food waste through redistribution, government policies, and digital innovation presents promising solutions. However, challenges such as contamination risks, inefficiencies in redistribution, and privacy concerns hinder effective implementation. These factors reveal a research gap in developing an accessible, efficient, and universal food waste management system.

The role of technology in reducing food waste is a key focus, with applications emerging to promote redistribution, food tracking, and smart packaging. Research suggests that integrating loyalty rewards and incentives can enhance user engagement. E-commerce plays an increasingly significant role in food waste reduction, serving as a platform for inventory tracking and surplus food sales. Studies emphasize the effectiveness of machine learning in providing personalized recommendations and optimizing supply chains. However, in the Philippines, e-commerce development lags behind other ASEAN countries due to infrastructure limitations, underscoring the need for stronger digital transformation initiatives.

A recurring challenge across studies is the gap between technological advancements and consumer adoption. The success of food waste management initiatives depends on consumer willingness, business participation, and systemic support.

Literature Matrix

Table 2. Literature Matrix

Author(s)	Year	Title	Findings & Solutions
Nurhaeni et al.	2024	Agile Scrum & Yii Framework in E-Commerce	Integrating real-time stock monitoring where food industries can lessen their waste and optimize sales by taking action to satisfy consumer demand. To ensure flexibility in e-commerce applications, Agile Scrum allows sellers to be informed and adjust immediately to inventory updates.
Mir, Chang, & Hefni	2024	Government & Industry Collaboration in Food Waste Reduction	Food waste management programs can be enhanced when there is a collaboration between businesses and government agencies. By giving incentives for donations and imposing a penalty for excessive waste, it will ensure that near-expiry or surplus food will be reallocated rather than end up in landfills.
Iqbal & Kang	2024	IoT-Enabled Circular Food System for Sustainability	IoT-based devices help track the conditions of the food when stored, avoiding food spoilage by alerting the sellers about the nearing end date of products. This will help the businesses to give importance to selling or donating the items before they get expired.
Kohli et al.	2024	Consumer Behavior and Waste Management Programs	Using an e-commerce platform can help educate consumers about food waste management by alerting them to expiring items and discounts on surplus foods. These approaches avoid unnecessary disposal and promote responsible buying.
Taner	2024	Technological Innovations in Food Supply Chains	Tracking food from manufacturers to market will improve integrity and effectivity using the Blockchain and AI, minimizing loss and ensuring better handling by optimizing food storage and logistics processes.
Papargyropoulou et al.	2022	Surplus Food Redistribution in E-Commerce	Consumers can significantly minimize their food waste when they use a digital platform that connects them with surplus food suppliers, but transporting and tracking systems are very important to make sure that food is delivered to customers before its expiration.
Shen et al.	2023	Policy-Driven Approaches to Food Waste Management	Having a government policy such as tax exemptions for donating food and imposing penalties for waste can motivate food businesses to become responsible in food waste management and compliance tools that can be integrated into e-commerce to help sellers to follow sustainability policies.
Jannah et al.	2023	Web-Based E-Commerce for Food Businesses	Providing discounts and making the near-expiry product on sale further prevents food waste. Online food platforms make the order management effective and prevent overstocking to ensure that perishable goods like food are sold before their end date.
Al-Ebrahim et al.	2023	Machine Learning in E-Commerce for Food Sustainability	Forecasting with the use of AI helps businesses to avoid producing too many products and sudden changes in stock. Using smart inventory tracking will make sure that surplus food is prioritized for selling or donating.
Galli et al.	2019	The Role of Businesses in Food Waste Responsibility	E-commerce applications and systems can help lead a connection and partnership between sellers and food waste advocate organizations to make sure that near-expiry foods are given to charities and utilized effectively. The researchers also noted that businesses must handle food responsibly, and strategies for redistribution will be effective in minimizing food waste.

Conceptual Framework

As shown in Figure 5, the conceptual framework uses an Input, Process, and Output (IPO) framework in breaking down the research processes to determine the factors used in creating the system and the study. All the data and information collected during the data gathering of the researchers were used as a guide on what kind of system and problems are needed to address through developing an application or system. Aside from this, the user data and actions also served as input that were processed and significantly improve the outcome of the project.

As for the process, several modules, including the user module, where the system made sure that the role of each user were determined, whether they are a seller, consumer, or administrator, allowing them to modify and access features based on their role. There are also modules for products to monitor the inventory, to allow the customers to view products and purchase, data analytics for insights and transactions in the system, notification for informing the users on updates, status, and announcements, messaging modules to allow communication between seller and consumers and also seller to seller, and lastly, the waste management module where the users redistributed and plan on what they are going to do on food waste and surplus foods.

To connect it and save all the processes, XAMPP was used as a database where it allowed the storing and retrieval of information contained by the application, while Visual Studio Code was used as the integrated development environment that helped in code development efficiently with interaction in programming languages of PHP, JavaScript (JS), Cascading

Style Sheets (CSS), and Hypertext Markup Language (HTML). The ConservAte application is a web-based application that were accessible through a web browser. The Agile Scrum Method were used, containing backlogs and planning, while the Technology Acceptance Model (TAM) and ISO 25010 were used in evaluating how the system works and its functionalities. With all of these inputs and processes, the ConservAte application were the output with its unique features for each kind of user.

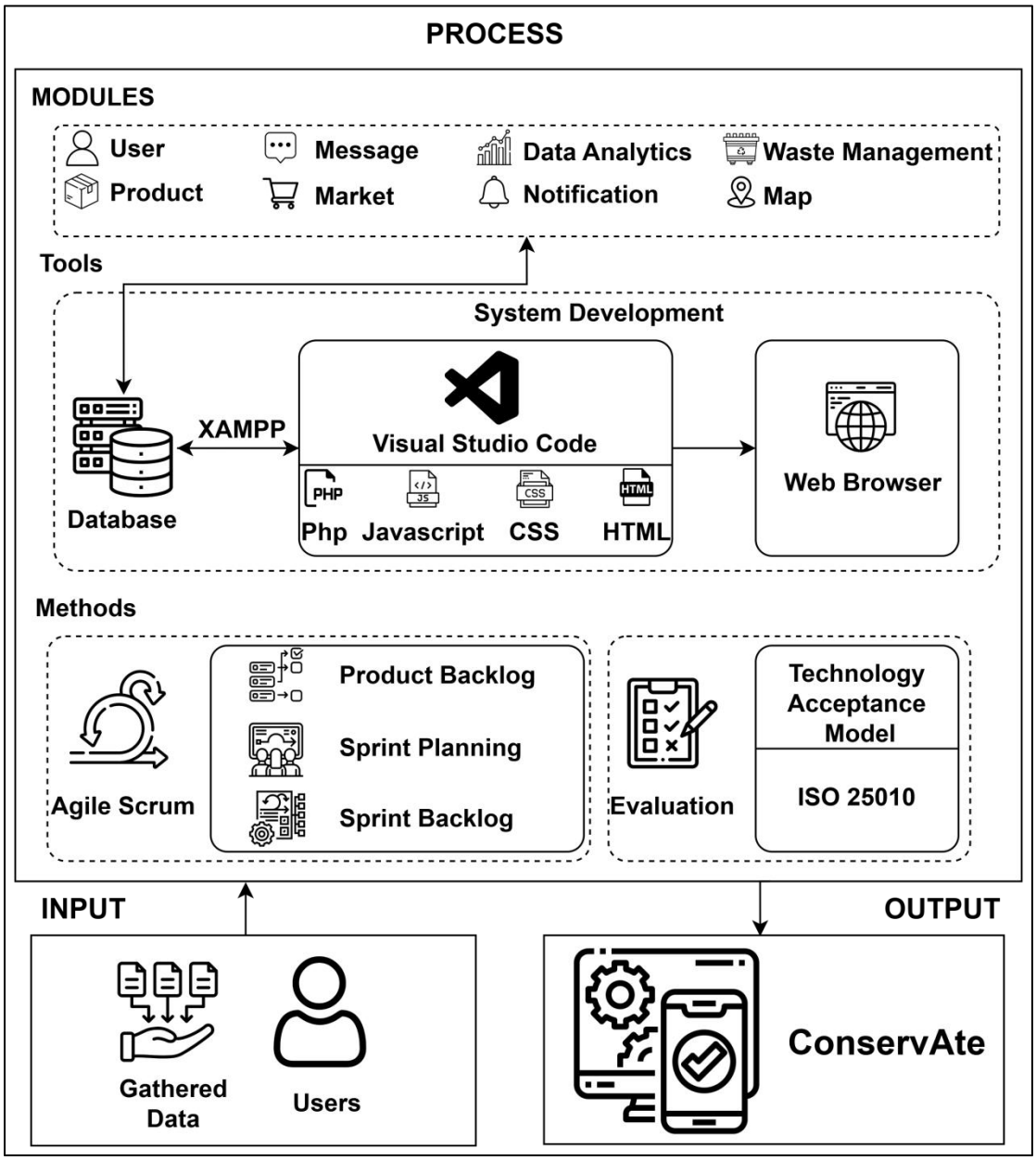


Figure 5. Conceptual Framework

CHAPTER 3

METHODOLOGY

This chapter covers the project design and development, project materials, business model, data collection instrument, statistical treatment, and project testing needed for this study and system development.

Project Design

Figure 6 presents the system's architecture of the ConservAte web-based application. It comprises modules such as the user module, product module, order module, notification module, messaging module, waste management module, and data analytics module. Integrating all these modules ensured the system's processes run smoothly and deliver a great user experience.

The User Module contains all the necessary processes for user accounts, including registration and logging. Users of the ConservAte application, including guests, customers, vendors (admin), and super admin, have their own interfaces. Each had different functions based on their roles in the system, sign out of their account for security, and modify their account information through the manage account feature. In contrast, only customers can manage their carts and the products they add to them.

The market module allows customers to view all products added by vendors in the market, check out their desired products, and add them to the cart for purchase when they decide. The Product Module was designed for vendor-type users, as they have the autonomy to manage their inventory,

monitor the status of their products, especially their shelf life, to avoid food spoilage, and supervise customer reviews of their products to improve their customer service. The system also included real-time notifications to keep users updated on announcements, reminders, expiration alerts, and order status.

The ConservAte application allows customers to message sellers, enabling them to address each other's concerns and inquiries. With this feature, they can build trust and connection during transactions, avoid misunderstandings, and make the purchasing process easier. The sellers and customers can ask the super admin for support if they have a report incident or need assistance that requires involvement with management. At the same time, the forum served as a platform for all users to share their experiences and seek support and advice from other users.

A Geographical Information System was also integrated to make it easier for customers to locate stores and vendors, and to determine which store is the nearest to their location. A unique module for waste management, specifically food waste, is designed to ensure surplus, near-expiry, and spoiled food is redistributed and logged, enabling monitoring of food waste generated by each seller. There are graphs available for visualization specifically for each user, such as sales or income graphs, surplus management, total orders for the vendors, graphs for customer spending, charts for the overview of all actions done in the system displayed at the dashboard of the admin informing them, and providing insights about processes done inside the system, helping them to make data-driven decisions..

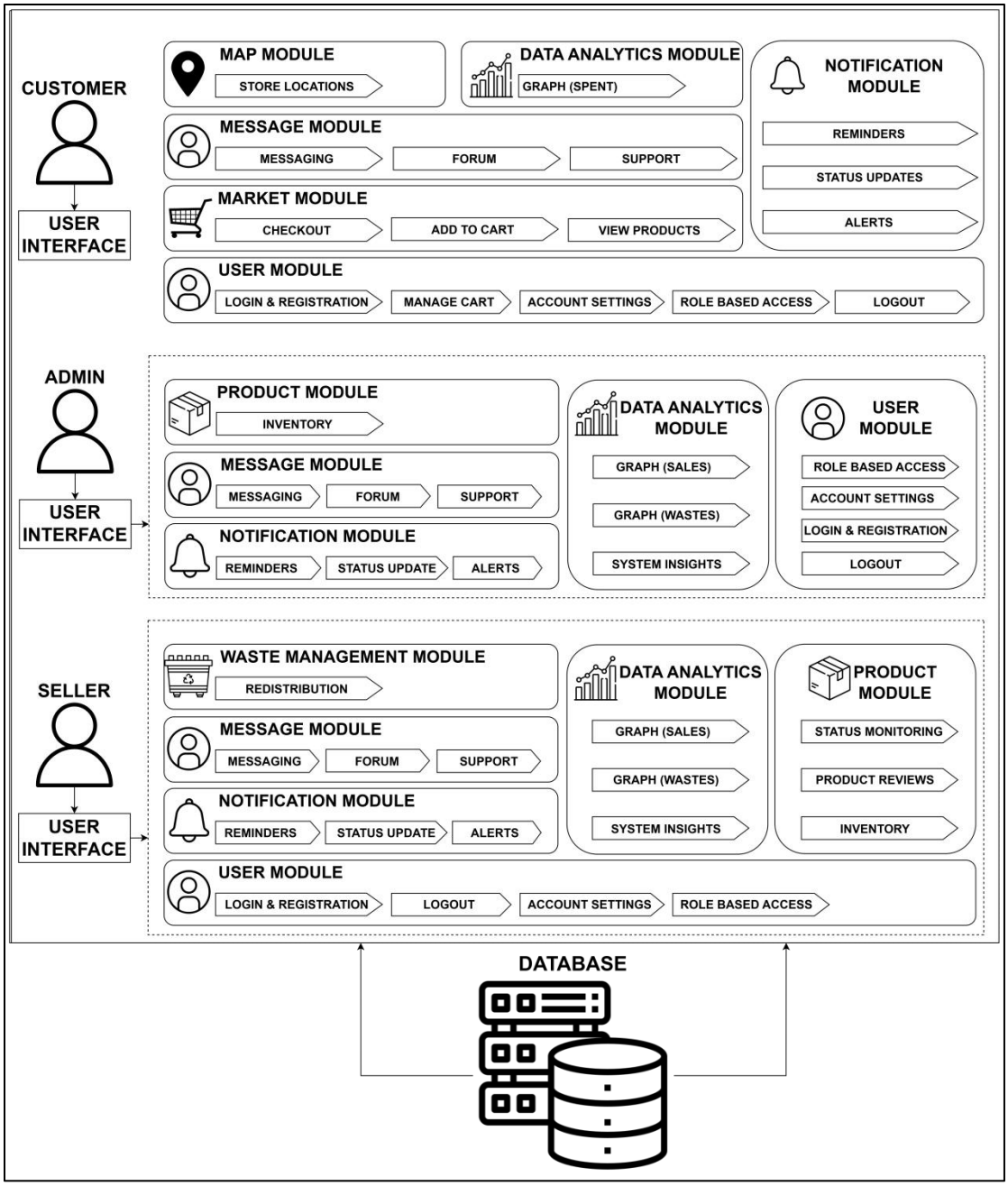


Figure 6. ConservAte's App System Architecture

Project Development

Figure 7 shows the implementation of the Agile Scrum model in the development of the ConservAte web-based system. Agile Scrum is a flexible, collaborative methodology that enables continuous feedback and improvement throughout the development process. The phase began with the creation of the Product Backlog, which included all the system's functions and

requirements. In Sprint Planning, the development team selects high-priority items from the product backlog to include in the sprint backlog, which consists of specific tasks to be completed during the sprint. Cervone (2011) notes that frameworks allow for greater project agility. Their iterative process, built on continuous review and brief development sprints, lets teams quickly adjust to the fast-evolving contexts where their systems were deployed.

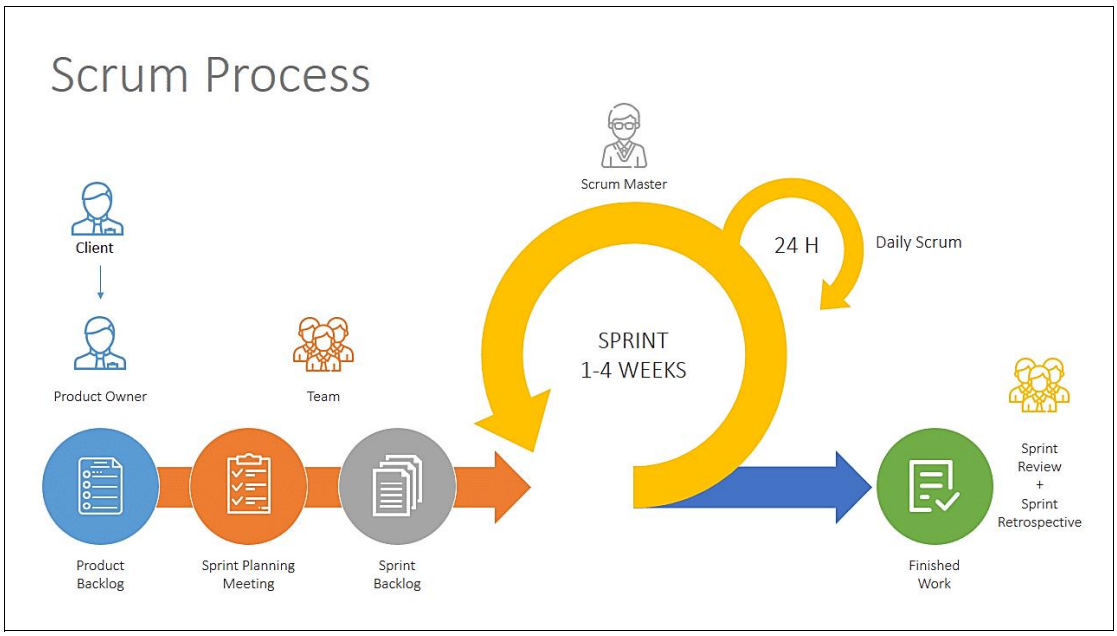


Figure 7. Agile Scrum Methodology

Product Backlogs

The product backlog plays a significant role in the development process by defining the core requirements and essential features for building the ConservAte application. It serves as a roadmap for achieving the main objectives of this study, systematically listing the tasks and functionalities to be implemented. By breaking the goals into smaller, manageable steps, the backlog guides the smooth progression of the system throughout the project lifecycle. Consequently, the backlog is a key tool for maintaining focus, ensuring each sprint contributes directly to reducing food waste.

Table 3 provides an overview of the Customer's Product Backlog, listing the most necessary functions required to operate the system efficiently. It includes features such as browsing the marketplace and placing orders. By focusing on the customer's perspective, it highlighted the features the customer needs that align with their preferences. Priority-based tasks require extra effort and time to accomplish, as they serve as the foundation for building the ConservAte system.

Table 3. Product Backlog for Customer

ID	Customer Story	Description	Priority
1	Login & Registration	As a customer, I want to register and login in the system.	High
2	Marketplace	As a customer, I want to view, purchase, add to cart, and order products	High
3	Message	As a customer, I want to message the vendor and admin	High
4	Review	As a customer, I want to provide feedbacks in products	Medium
5	Data Analytic	As a customer, I want to view the data analytic	Medium
6	Profile Management	As a customer, I want to manage my profile (update contact details, address, etc.).	Medium
7	Notifications	As a customer, I want to receive notifications about order updates	High
8	Transaction History	As a customer, I want to view my order history for tracking past purchases.	Medium

Table 4 is the overview of the vendor 's product backlog, listing the most necessary functions they require to operate the system efficiently. It contains features like managing products that catered to the system and managing customers' orders. By focusing on the vendor's perspective, it highlighted the needed features of the users.

Table 4. Product Backlog for Vendor

ID	Vendor Story	Description	Priority
1	Login & Registration	As a vendor, I want to register and log in to the system.	High
2	Product Management	As a vendor, I want to manage (view, add, edit, delete) my products.	High
3	Order Management	As a vendor, I want to view and manage customers' orders.	High
4	Communication	As a vendor, I want to message customers and admin.	High
5	Reviews & Feedbacks	As a vendor, I want to view reviews on my products.	Medium
6	Transaction History	As a vendor, I want to view my transaction history.	Medium
7	Data Analytics	As a vendor, I want to view sales and performance data.	Medium
8	Inventory & Expiry Management	As a vendor, I want to manage stock quantities and set expiry dates for products.	High
9	Notifications	As a vendor, I want to receive alerts on orders, stocks, and updates.	Medium
10	Donations & Discounts	As a vendor, I want to add discounts and donate products.	Medium

Table 5 presents an overview of the Admin's Product Backlog, listing the essential functionalities needed to operate the system efficiently. It includes key features such as user account management, messaging, product report monitoring, and data analytics. By focusing on the super admin's role, this backlog ensures administrators have the tools to oversee system operations, manage users, and take appropriate action when needed. This organized approach helps keep the entire platform running smoothly. Each user story is assigned a unique ID that details its significance in maintaining the ConservAte system's overall efficiency and usability.

Table 5. Product Backlog for Super Admin

ID	Admin Story	Description	Priority
1	Login	As an admin, I want to login in the system.	High
2	Manage Accounts	As an admin, I want to manage user accounts.	High
3	Message	As an admin, I want to message customers and vendors.	High
4	View Product Report	As an admin, I want to view the reports in any products, and take actions.	Medium
5	Data Analytics	As an admin, I want to see data analytics in my system.	Medium

Sprint Planning

Sprint planning is an integral part of developing the web-based application ConservAte, providing a guide to keep the development team organized and focused on what to build next. This helped the team prioritize assigned tasks to complete and achieve project objectives within a set timeframe, making the process more organized and efficient. Dismantling the whole project into smaller pieces makes it more manageable and easier to accomplish. The sprint planning was created by the team, outlining the specific tasks to be done. It were distributed among team members based on their expertise and availability, ensuring the work is balanced and executed efficiently.

Figure 8 illustrates the customer workflow for using the ConservAte web application, beginning with the landing page where guests can view products before registering. After email verification, customers log in and access the dashboard, which includes features such as browsing products, placing orders (with confirmation before checkout), managing their cart,

messaging vendors, reviewing transaction history, tracking orders, updating account details, locating nearby stores via map, redeeming reward points for vouchers, and participating in the refer-and-earn program. The system also provides real-time order notifications, product reviews, spending analytics through data graphs, and a forum for user discussions. Customers can securely log out, returning them to the landing page.

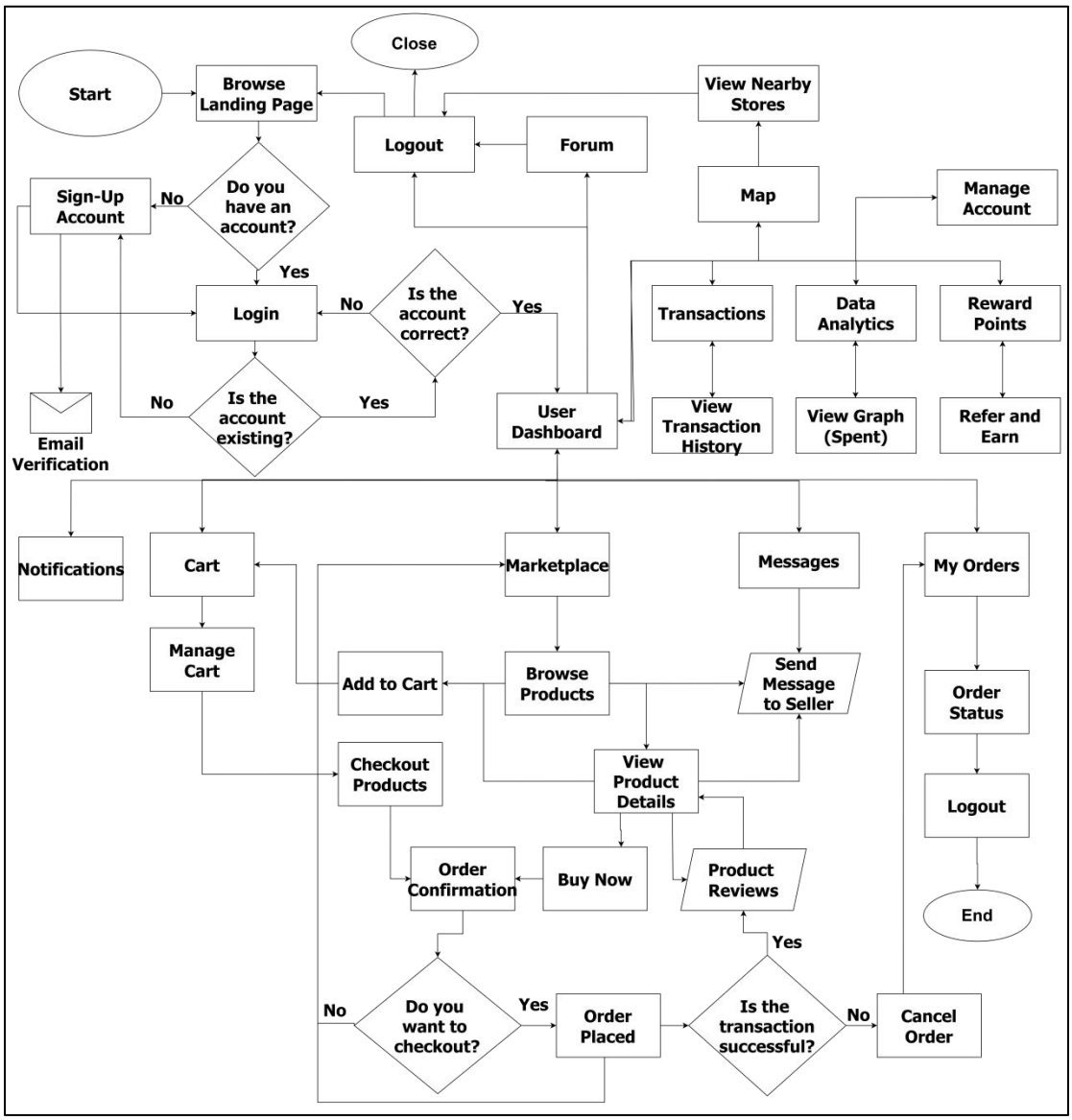


Figure 8. System Flowchart for Customer

Figure 9 outlines the vendor process for ConservAte, starting with registration requiring valid documents and email verification. Once approved,

vendors access a dashboard to manage products, orders, and customer interactions, including handling near-expiry and surplus food through discounts or donations to earn points and convert them into a reward of availing a subscription, during which their products were boosted and prioritized in the marketplace. The system provides real-time alerts and sales analytics, allowing users to visually understand the happenings in their account and transactions, and a user forum, with secure logout functionality.

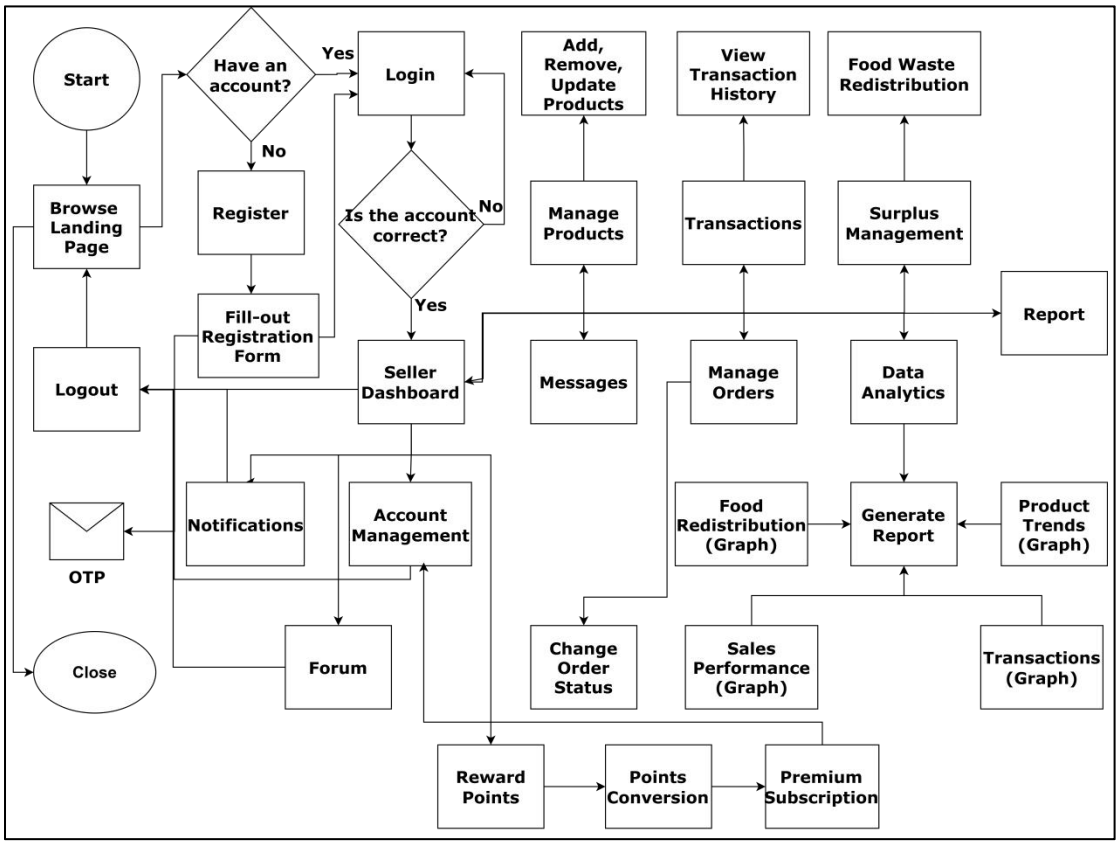


Figure 9. System Flowchart for Seller

Figure 10 describes the admin functions in ConservAte. Admins log in to access a dashboard for product monitoring, dispute resolution, transaction management, and guidance on food waste reduction. The system provides sales analytics, trend reports that can be exported for other uses, user activity data to monitor all users' actions within the system, and real-time notifications. This centralized view allows them to assist users quickly.

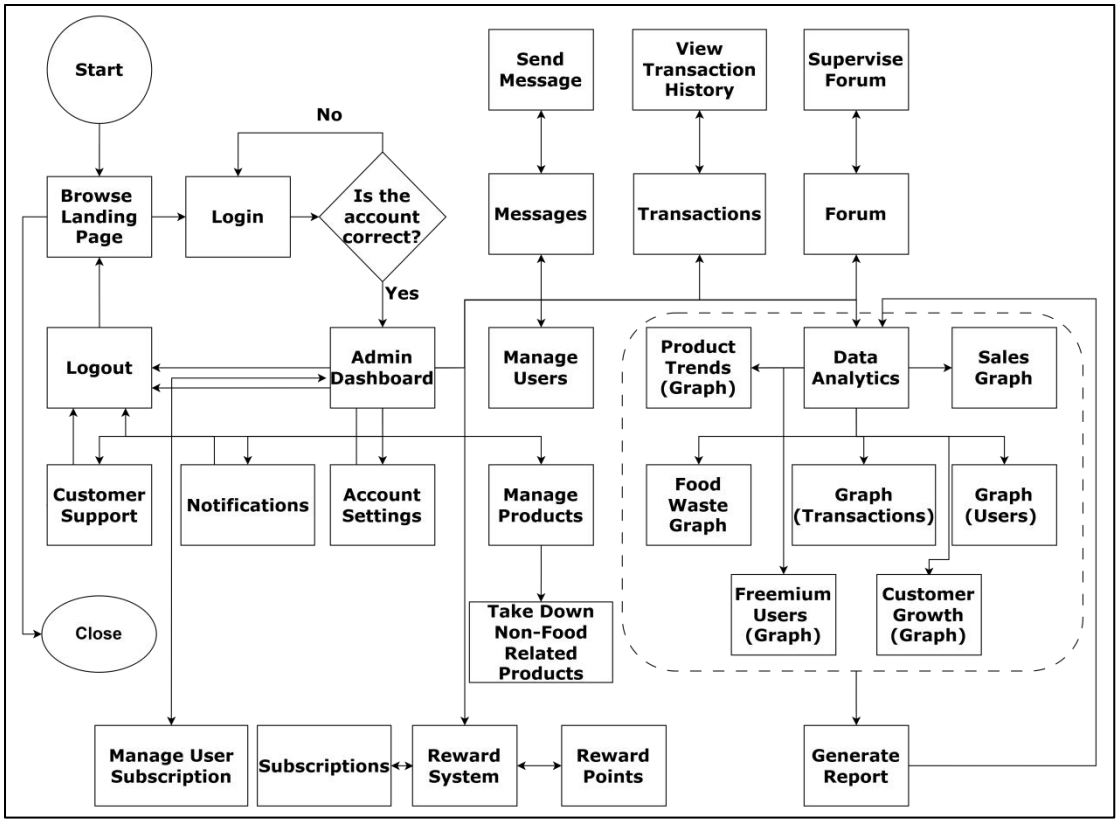


Figure 10. System Flowchart for Super Admin

Sprint Backlogs

Sprint backlogs are lists of tasks chosen from the product backlogs and represent the team's goals for the sprint. These to-do lists help developers work efficiently and stay on track in completing the planned features of the ConservAte application's web-based system.

Table 6 shows the sprint backlog for customer-related features of the ConservAte application, highlighting key user stories such as login and registration, marketplace development for product visibility, a messaging feature for communication, and order tracking and status. The development task was specified, and each key story was broken down, with estimated working hours ranging mostly from 48 to 96 hours to build the system component, guiding the development team in prioritizing the most needed features and effective time management throughout the sprint.

Table 6. Sprint Backlog for Customer

Priority	User as Customer Story	Task	Est. Work Hours
1	As a user, I want to register and login in the system.	Design and develop login and registration form	48 hrs
2	As a user, I want to view, purchase, add to cart, and order products.	Design and develop marketplace and cart section	96 hrs
3	As a user, I want to message the vendor and admin.	Design and develop message interface	48–96 hrs
4	As a user, I want to provide feedback on products.	Design and develop review section	48–96 hrs
5	As a user, I want to view the data analytic.	Design and develop data analytic interface	48–96 hrs
6	As a user, I want to manage my profile (update contact details, etc.).	Design and develop profile management section	48–96 hrs
7	As a user, I want to receive notifications about order updates.	Design and develop notifications feature	48–96 hrs
8	As a user, I want to view my order history for tracking purchases.	Design and develop order history interface	48 hrs

Table 7 shows the sprint backlog designed specifically for seller-related features in the web-based system of the ConservAte application, highlighting the key user stories as vendor or seller, such as login and registration, product and order management, messaging features for communication with other users, reviews, view transaction history, receive alerts for orders, product notifications, and inventory management with expiry monitoring to avoid food spoilage and allow them to take a quick action not to let the food end up in landfills. The performance or data analytics was also included, making the seller’s overview in their account easier to comprehend using

graphs and charts. The development task was specified and carefully listed, with each key story broken down and estimated working hours ranging mostly from 48 to 96 hours for the system component.

Table 7. Sprint Backlog for Vendor

Priority	User as Vendor Story	Task	Est. Work Hours
1	As a vendor, I want to register and log in to the system.	Design and develop vendor login and registration functionality.	48 hrs
2	As a vendor, I want to manage (view, add, edit, delete) my products.	Design and develop product management interface and database operations.	96 hrs
3	As a vendor, I want to view and manage my orders.	Develop vendor order management interface.	72 hrs
4	As a vendor, I want to message customers and admins.	Create messaging interface and backend message handling.	72–96 hrs
5	As a vendor, I want to view reviews on my products.	Integrate product review display and rating system.	48 hrs
6	As a vendor, I want to view my transaction history.	Create transaction history page with filtering and sorting.	48 hrs
7	As a vendor, I want to view sales and performance data.	Build data analytics dashboard with charts and stats.	72–96 hrs
8	As a vendor, I want to manage stock levels and set expiry dates for products.	Add inventory controls and expiry settings in product panel.	72–96 hrs
9	As a vendor, I want to receive alerts on orders, stock, and updates.	Set up notification system for vendor activities.	48–72 hrs
10	As a vendor, I want to add discounts and donate products	Develop the surplus management features to handle adding product discounts and donations	48–72 hrs

The table 8 shows the sprint backlog designed specifically for admin-related features in the web-based system of the ConservAte application, highlighting the key user stories as admin such as secure login, user account management, messaging feature for communication with vendors and customers, monitoring of product reports, and viewing system-wide data analytics. The development task was specified, and each key story was broken down with estimated working hours ranging mostly from 48 to 96 hours in making the system component, guiding the development team in prioritizing the most needed features and effective time management throughout the sprint.

Table 8. Sprint Backlog for Admin

Priority	User as Admin Story	Task	Est. Work Hours
1	As a user, I want to login in the system.	Design and develop secure admin login functionality	48 hrs.
2	As a user, I want to manage user accounts.	Create interface and backend logic for managing customer and vendor accounts	72 – 96 hrs.
3	As a user, I want to message customers and vendors.	Develop admin messaging panel for user and vendor communication	72 hrs.
4	As a user, I want to view the reports in any products, and take action.	Create report monitoring module with action capabilities	48 – 72 hrs.
5	As a user, I want to see data analytics in my system.	Build dashboard for visual analytics and system-wide performance metrics	72 – 96 hrs.

Use Case Diagram

Figure 11 presents the use case diagram with four actors: the visitor, user, vendors, and super admin. Visitors can only view products and featured

shops on the landing page, which informs them about what is inside the ConservAte system and allows them to decide whether to register as a user or vendor. Users can log in, search for and view products and their details, manage the cart, including adding and removing items, and check out. They can message vendors and the super admin, join the forum to share thoughts or experiences, receive notifications for expiring products and announcements, and view data analytics on total orders and spending. Vendors can log in, manage customer orders and products (add, edit, delete, set expiration), send and respond to messages, notify customers of updates, and use data analytics for sales and earnings, active products, and surplus management. The super admin logs in to answer concerns, manage and verify accounts, receive notifications for new messages and verifications, monitor all products, and view analytics on sales, users, vendors, and waste trends.

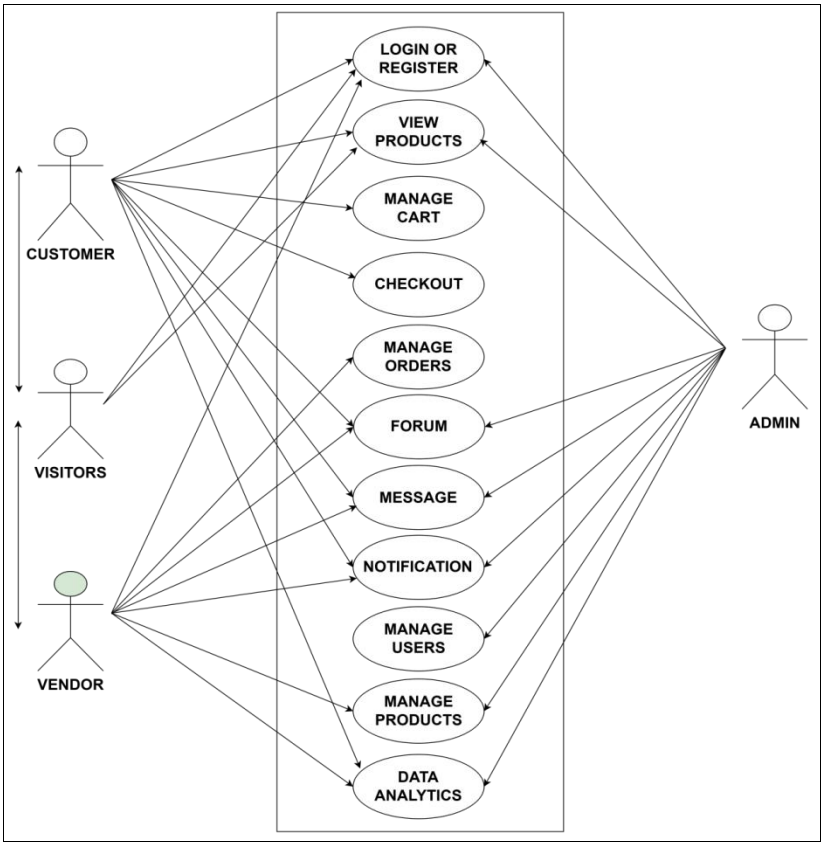


Figure 11. Use Case Diagram

Geographic Information System

Figure 12 displays how GIS operates in ConservAte to map and monitor store locations via layer mapping. The Land Usage Layer, which outlines how each territory is utilized—residential, commercial, agricultural, or industrial—and assists with placing vendors within the correct areas. The Street Layer plots all surrounding roads so that navigation and delivery planning become more convenient. Store Layer indicates vendor locations with pinpoints on the map so that it is convenient for users to locate and connect with sellers in the vicinity. With this simple integrated map, the food redistribution becomes easier and convenient for all users, promoting local economic growth while reducing food waste through improved logistical coordination and community engagement.

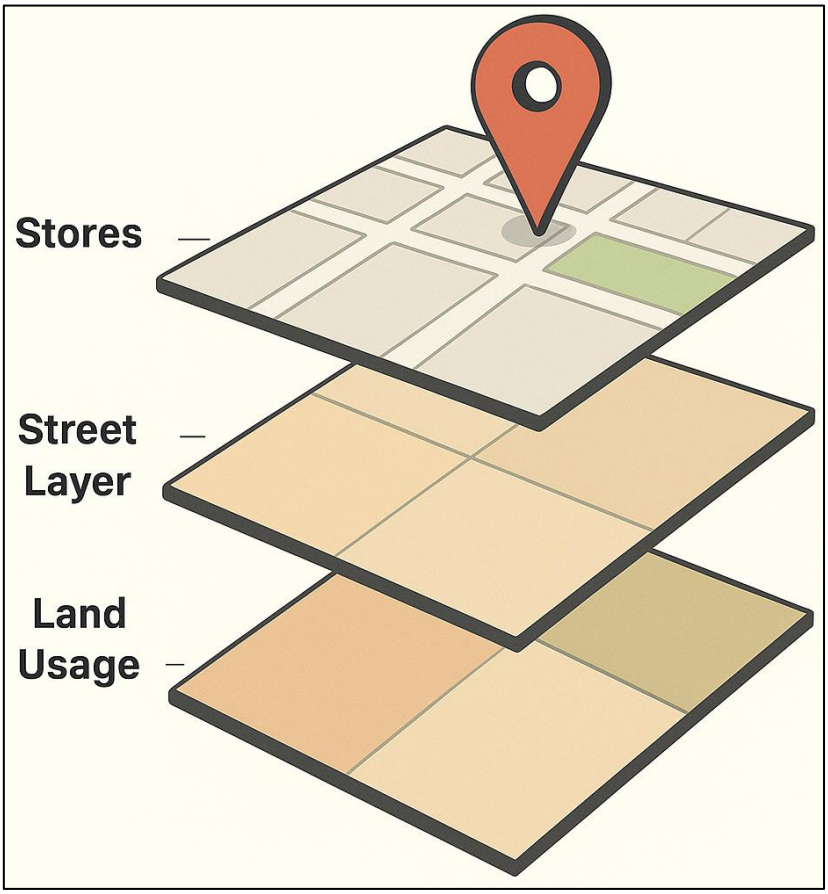


Figure 12. GIS Framework of the ConservAte Application

Project Locale

Figure 13 shows that Los Baños, Laguna, was chosen as the project locale due to its strong agricultural and research presence, high food consumption, and commitment to sustainability. The municipality fosters innovation in food security and waste management, making it the ideal location for testing and refining this project. This project helps the community minimize food waste by redistributing near-expiry products to consumers at the lowest possible price and donating them, ensuring affordability and sustainability. In addition, Los Baños has an environmentally conscious population and engaged local government programs focused on environmental preservation, providing robust stakeholder support for the system's use and success. The Los Baños municipality is the Special Science and Nature City of the Philippines, making it suitable for the research conducted.



Figure 13. Virtual Representation of the Project Locale (Los Baños, Laguna)
Maps: Google Maps

Business Model

Figure 14 highlights the mutual benefits of users from customers, vendors, and admin, illustrating the business model of the ConservAte web-based system application. Sellers can access the platform by subscribing to tiered plans, allowing them to convert reward points earned from referrals and donations into a subscription, where their products were boosted and prioritized in the marketplace, gaining an advantage over basic accounts of other sellers. Customers can browse all products, earn reward points from transactions, exchange points for discount vouchers, and benefit from the referral feature of ConservAte. The application earns revenue from transaction fees remitted by the sellers. This model ensures user engagement and promotes food accessibility and food waste reduction toward sustainability.

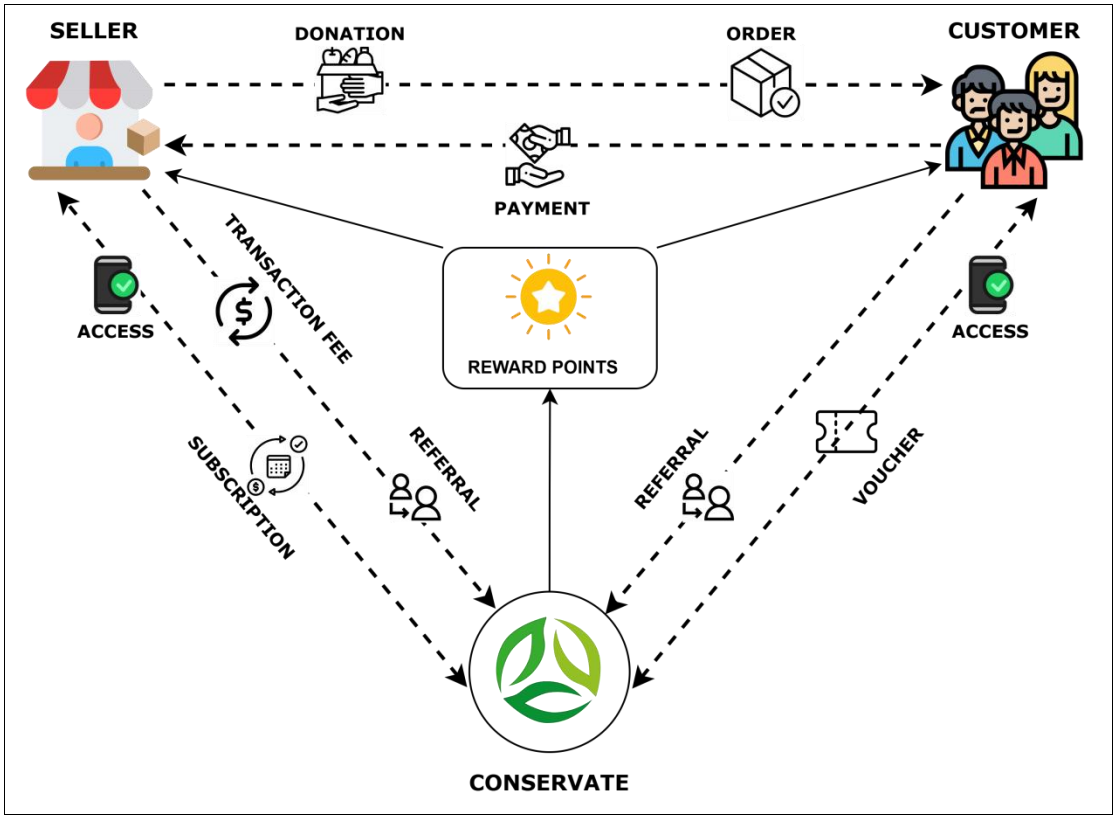


Figure 14. Business Model of ConservAte Application

Business Plan Summary

After a thorough testing and evaluation process, the ConservAte was deployed and is accessible via a hosted website. The system was successfully launched through effective marketing strategies, the identification of target customers, and competent management, ensuring continuous maintenance and providing great customer service.

Business Startup for the Project

This section discusses the strategic plan for launching and growing the system, outlining primary activities and approaches to achieve specific goals.

The ConservAte web-based system was launched in 2025 and aimed to reduce food waste by allowing users/vendors to donate their near-expiry products or food waste. It promoted zero-waste communities within its target municipality, which was Los Baños, Laguna. This system connected food waste contributors, including supermarkets, public markets, restaurants, fast food chains, etc., with interested individuals who wanted to buy products at the lowest prices or accept food waste for other purposes than putting it into landfills. The ConservAte had user-friendly interfaces that enabled all users to experience convenience and efficiency, aligning to expand the system's visibility and client base while promoting the importance of food and its accessibility through user engagement.

Core Values

ConservAte was committed to mitigating or lessening the problem of food waste, offering nearly expired products from various vendors at the lowest price, making the food more accessible to all while promoting sustainability. It also promoted the proper redistribution of the surplus, near-

expiry, and food waste so it would maximize the food usability without putting it to waste. The platform encouraged responsible consumption, supported local communities, and raised awareness to focus more on food security.

Competencies

ConservAte's key competencies included a strong technical foundation (a responsive website that was accessible on all devices, GIS mapping, and detailed analytics), a dedicated eco-friendly vendor network, and partnerships with environmental agencies and organizations. Its ability to integrate commerce, data insights, and community engagement with sustainability provided it with a competitive advantage over traditional e-commerce websites.

Products and Services Offered

The site provided sustainable and value-priced items, such as surplus food, discounted groceries, upcycled products, and eco-friendly products. Some of its features were a live marketplace, a GIS-based store locator, order tracking, reminders, and vendor control panels. It also offered buyer assurance services, vendor subscription plans, and data analysis tools that assisted businesses in monitoring waste reduction and making informed decisions.

Distribution

Products were sold mainly via vendor-store pickup points or cash on delivery (optional for sellers to choose the delivery mode). Vendors packed up products, and ConservAte offered either coordinated pickups or low-impact local delivery. Unsellable or spoiled products could be donated to users willing to claim them, making sure that the product was labelled as expired. This

could also be coordinated with authorized takers, including the Los Baños municipal government, advancing programs in sustainable waste management, composting, or agricultural reuse. All online services, such as notifications, waste monitoring, and vendor analytics, were made available through the ConservAte website.

Management

ConservAte's management was dedicated to ensuring that the platform operated efficiently while fulfilling its mission of reducing food waste and supporting communities. The team was responsible for overseeing vendor onboarding, verifying submitted credentials, and maintaining the quality of product listings. Customer support was maintained on the website, enabling users to receive prompt assistance with their concerns.

In addition, ConservAte collaborated with community partners and vendors to encourage donating expired, surplus, and near-expiry products, preventing them from ending up in landfills and instead transforming and using them in other ways. The management continuously monitored platform data and user activity to improve processes, make informed decisions, and ensure that all operations aligned with the platform's environmental and social goals.

Marketing Plan

Marketing Objectives

The project focused on gathering individuals, stores, restaurants, and public and private markets who registered into the system. These people gave feedback about their experiences in terms of the system's

functionality and user usability. Their critiques helped the developers have a foundation on what they should focus on, especially regarding the clients' preferences.

Target Customer

The primary target customers were all residents of Los Baños and beyond, especially eco-conscious consumers who were looking for sustainable products and services. Secondary customers were local farmers, vendors of sustainable products, waste management firms, environmental NGOs, and municipal governments that needed environmental data services.

Pricing Strategy

ConservAte made money from transaction fees indicated by the admin that were added to every transaction made between the sellers and consumers, remitting them weekly. Sellers could earn points by donating products, which they could convert into points and avail premium features, including boosted and prioritized listings in the marketplace, enhancing their visibility. There was an additional fee if the seller delivered the product to the customer's door. Aside from this, donated products were free of any charges and were only picked up by the recipient. This flexible pricing structure ensured inclusivity for small-scale vendors while sustaining the platform's operations.

Promotional Strategy

Since digital advertising was more effective than traditional advertising (Fuxman et al., 2014), ConservAte took the opportunity to advertise across different trending social media platforms, such as Facebook, Instagram, X, and TikTok. These ads helped target specific audiences, ensuring the app

reached potential users effectively. Limited-time promos, referral bonuses, and loyalty rewards further attracted new users and retained existing ones. Through these strategies, ConservAte had a strong digital presence and a supportive eco-conscious community.

SWOT Analysis

Figure 15 outlines the strengths, weaknesses, opportunities, and threats of the ConservAte system. It provides a comprehensive framework for identifying internal and external advantages and problems, as well as external factors that can impact the system's growth. It highlights several key points influencing the success and impact of ConservAte.

Strengths like social impact, tackling food waste and food security, a unique value proposition focusing on nearly expired products, multi-vendor support that enables sellers to cater to their near-expiry products to reduce the percentage of food waste, availability of GIS API that allows them to pinpoint the exact position of the nearest stores, restaurants, etc., cost advantage since the system offers various near-expiry products and goods that allow them to purchase low-priced items, and a data-driven system with smart inventory tracking that monitors their performance or status of their contribution in mitigating food waste.

In Weaknesses, Perceived Risk, customers might be hesitant to buy near-expiry products due to safety concerns; Regulatory Compliance, various local regulatory policies and standards might be complex; and Market Education, users, both customers and vendors, still need to be educated about how the system flow works and the benefits of buying near-expiry

products. Some users, especially sellers, might think that the more users use the system, the fewer people came to the market.

Opportunities in the eco-conscious market, raising consumer awareness around sustainability and reducing food waste in their favor, partnerships with NGOs and food banks to widen reach and credibility, corporate tie-ups with various businesses that want food surplus for their bulk buying, and government support to support food sustainability or to make an event about food sustainability.

In Threats, Traditional Retail and Supermarkets, some of them might cater to their shops with their discounted products, increasing competition; Trust and Safety Concerns: any mishap with expired or unsafe food could harm your platform's reputation; Limited Shelf Life: products need to move quickly, or they become unsellable; and Legal Issues: potential liability or disputes if buyers receive spoiled goods.

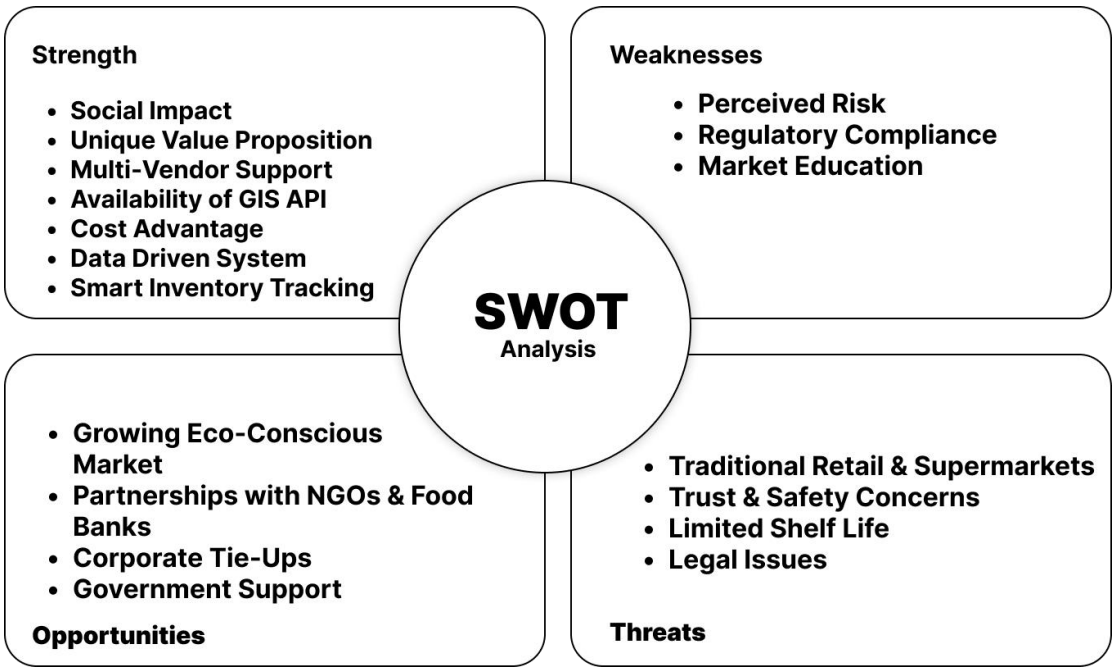


Figure 15. SWOT Analysis Diagram of ConservAte Application

Actions

To address potential challenges, ConservAte focused on building trust by ensuring sellers provided clear photos and expiry dates for all products, so buyers felt safe and confident. To stand out from big supermarkets, it offered special deals and rewards and highlighted its mission to reduce food waste — something many people cared deeply about. Since products had a short shelf life, they used features like countdowns, flash sales, and product bundles to help them sell quickly. Moreover, to stay on the safe side legally, ConservAte created easy-to-follow guidelines for sellers and gave them the support they needed to follow food safety rules.

Financial Plan

Operating and Startup Costs

The estimated startup costs for building and deploying the ConservAte web-based application ranged from around PHP 2,000 to PHP 5,000, allocated to app development, marketing and promotion, and web hosting. With this allotted budget, the efficient, user-friendly, well-behaved, and properly working web application was served for all stakeholders and users of ConservAte, providing a great user experience and promoting customer satisfaction.

Materials

In order to create the web-based system of ConservAte, the following hardware and software were selected. These tools were chosen, taking into consideration the usability, scalability, and usefulness of the system. The creation of a coherent and effective e-commerce platform was guaranteed by

the integration of design, interactivity, and database administration using the selected hardware and software. This careful selection of development tools ensures that the platform can efficiently handle real-time transactions, user interactions, and data management while maintaining optimal performance and security standards.

Software

The software selected for the development of the web-based system comprises a set of essential technologies designed to ensure usability, scalability, and functionality.

Hypertext Markup Language (HTML). It serves as the foundational building block of ConservAte's front-end interface. It is responsible for structuring the content and layout of the system, allowing developers to define essential components such as headers, footers, navigation bars, product sections, forms, and buttons. By using semantic HTML elements, the system ensures better accessibility and improved browser interpretation.

Cascading Style Sheets (CSS). It improves and personalizes the appearance of HTML elements, making the system visually appealing and user-friendly. It controls colors, fonts, spacing, alignment, and overall layout design. More importantly, CSS ensures responsiveness—meaning the ConservAte platform adjusts and functions well across devices, including smartphones, tablets, and desktops.

JavaScript. This programming language allows great interactivity and various dynamic functions to the ConservAte platform. It allows real-time updates on the interface without requiring full-page reloads, enhancing

performance and the user experience. It also plays a critical role in validating user input on the client side before sending data to the server, reducing errors and improving security.

Hypertext Preprocessor (PHP). It is the core server-side language used to develop ConservAte's backend functionality. It handles various operations, including form processing, user authentication, session management, and communication with the MySQL database. PHP retrieves and stores data related to user accounts, products, transactions, orders, and system notifications. It acts as a bridge between the user interface and the database, ensuring that all user requests and system processes are executed correctly.

Hardware

Table 9 below indicates the specifications of the computer used to develop the system. The operational computer has 8GB of RAM, an Intel(R) Core(TM) CPU that operates at speeds of 2.30GHz to 2.49GHz, and is endowed with Intel(R) HD Graphics 520 for its graphics processing. The system runs on Microsoft Windows 10 Pro, and with these specifications, the system achieved the desired output. The used device also met the base requirements of running the entire platform.

Table 9. Hardware Specification

Name	Specification
RAM (Random Access Memory)	8.00 Gigabyte
CPU (Central Processing Unit)	Intel(R) Core(TM) CPU @ 2.30GHz - 2.49GHz
GPU (Graphics Processing Unit)	Intel(R) HD Graphics 520
OS(Operating System)	Microsoft Windows 10 Pro

Table 10 below indicates the minimum requirements for a PC or laptop to run the system efficiently. These specifications ensure that the system operates smoothly without performance issues and provide end users with a high level of satisfaction when using the ConservAte web-based system. Meeting these hardware requirements guarantees that all platform features, including real-time product browsing, account and product management, and interactive map functionality, performed optimally for an enhanced user experience.

Table 10. Minimum Requirements for PC/Laptop

Name	Specification
Central Processing Unit (CPU)	At least Intel Core i3 / AMD Ryzen 3 (Recommended: Intel Core i5 / Ryzen 5)
Graphics Processing Unit (GPU)	Integrated Graphics (Intel HD Graphics / AMD Radeon Vega)
Operating System (OS)	Microsoft Windows 10 Pro
Storage	At least 128GB SSD or 256GB HDD
Random Access Memory (RAM)	At least 4GB

Table 11 below indicates the minimum requirements for a mobile device to run the system properly. These include a touchscreen display, the latest version of the operating system, at least 4 GB of RAM, and sufficient storage to run the ConservAte web-based system. These specifications help ensure smooth performance and prevent issues like slowdowns or crashes. By meeting these standards, users can expect a seamless, hassle-free experience when using the system. These requirements enable full access to mobile-specific features such as GPS for store location services and responsive design elements.

Table 11. Minimum Requirements for Mobile

Name	Specification
Central Processing Unit (CPU)	Quad Core 1.4 GHz
Display	Touchscreen Required
Operating System (OS)	Android 8.0 / IOS v.12.0
Storage	At least 500mb free space
Random Access Memory (RAM)	At least 4GB

Project Testing and Evaluation

Figure 16 shows the project testing consisting of unit, integration, system, and acceptance testing following a structural approach from the basic to the comprehensive evaluation of the project. This testing phase helped identify potential errors or bugs, ensure the system's reliability and component functionality, and provide a great user experience for the web application's users. By thoroughly analyzing and conducting software testing, the researchers developed a system that achieved the goal of providing high-quality service and user satisfaction.

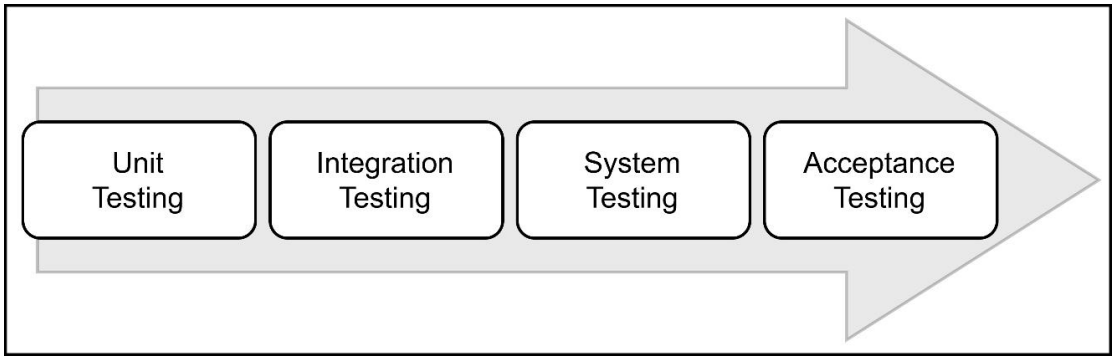


Figure 16. Level of Testing

Unit Testing

The unit testing approach, as shown in Figure 17, focuses on testing a single software module or component to verify its intended behavior. This

process, in turn, involves manipulating specific code elements, such as functions, methods, and classes, to verify that they function as intended before integrating them into the larger system. Manual testing was conducted to make sure the unit's behavior in specified situations, where potential errors can be identified and resolved as early as possible in the development cycle. This servd as a measurement to determine whether the functions and every unit of the module are working as expected.

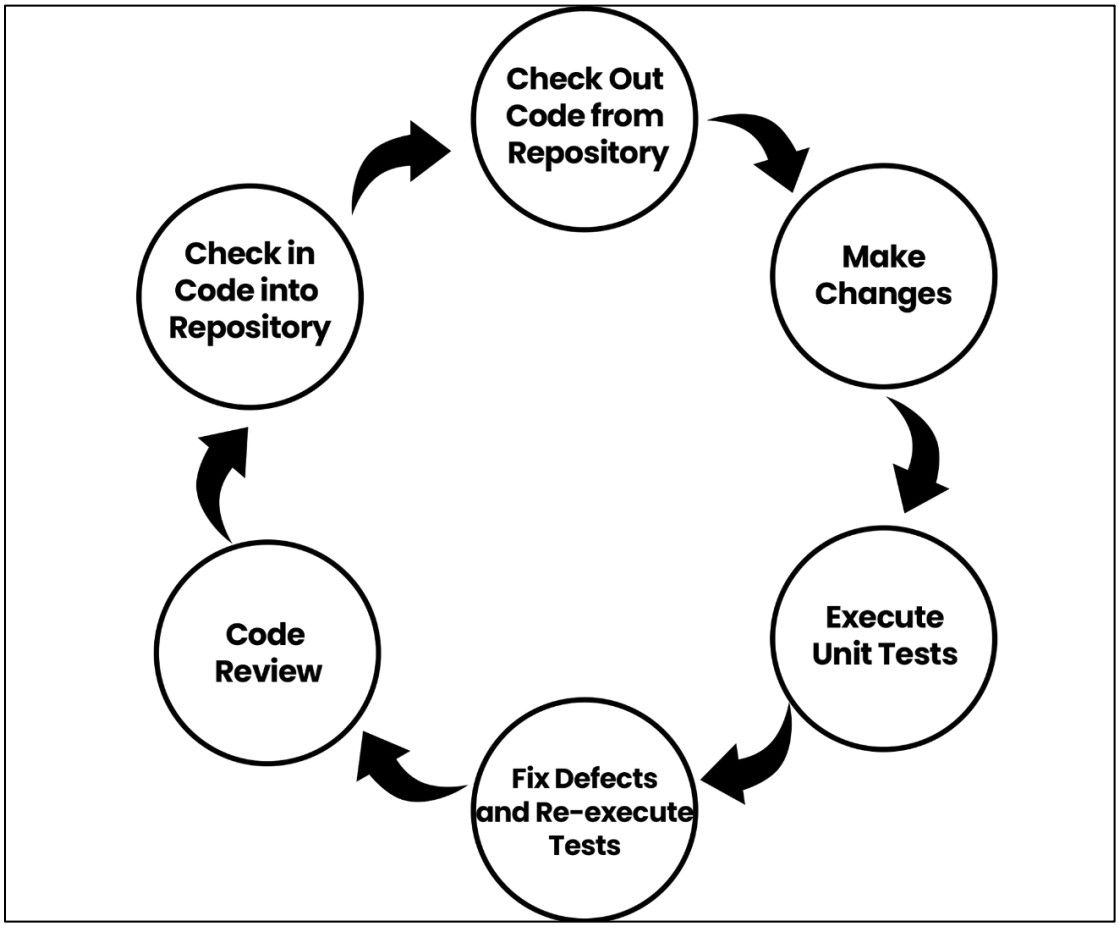


Figure 17. Unit Testing

Integration Testing

Figure 18 shows how integration testing focuses on the how modules or components interacts within a software system. For instance, in the ConservAte web application, modules responsible for user, market, message,

product, waste management, map, data analytics, and notification would be integrated and tested to ensure seamless data flow and functionality. This helped enhance code quality, make it reusable across different parts of the application, and enable early detection of failures. By combining and testing these modules, developers can identify and address issues related to data consistency, error handling, and overall system behavior. The integration of all modules helped maintain the quality, integrity, high performance, and regulatory compliance of the entire system.

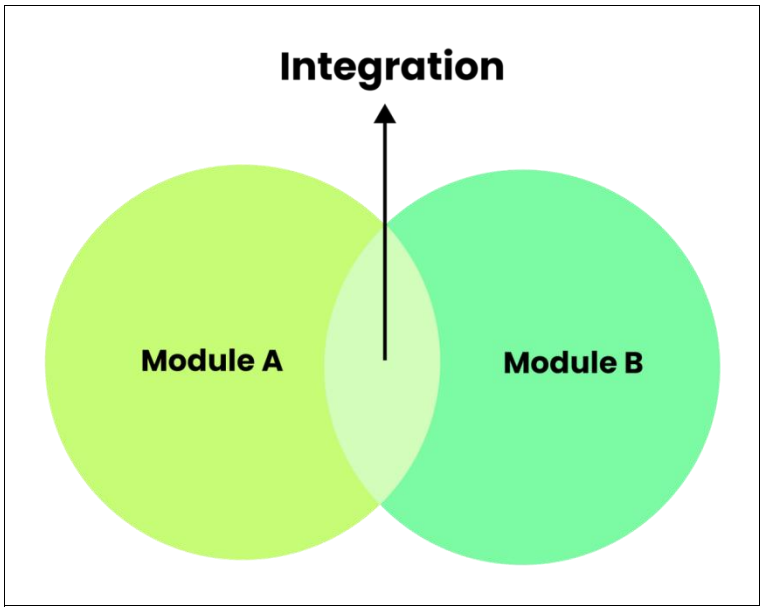


Figure 18. Integration Testing

System Testing

Figure 19 illustrates how the system worked continuously while integrating and retaining the function of the modules that compose the system. This helped evaluating the entire platform to ensure that all components meet the functional and performance requirements in a real-world environment. The researchers conducted compatibility testing, load testing, and stress testing. This comprehensive testing approach validates the complete user journey from product discovery to checkout and donation claims, confirming that all

integrated modules work harmoniously under various usage scenarios and data loads. Lambdatest was used to run the system and ensure it can still load on different platforms.

Compatibility Testing

This type of testing checks the functioning of the system such as devices, operating system, network, software, and browsers. Through conducting the compatibility testing, the developers were able to guarantee that the system were able to provide consistent performance and functionality irrespective of the hardware and software of the user. This assists in determining and solving issues that could occur as a result of variations in user environments.

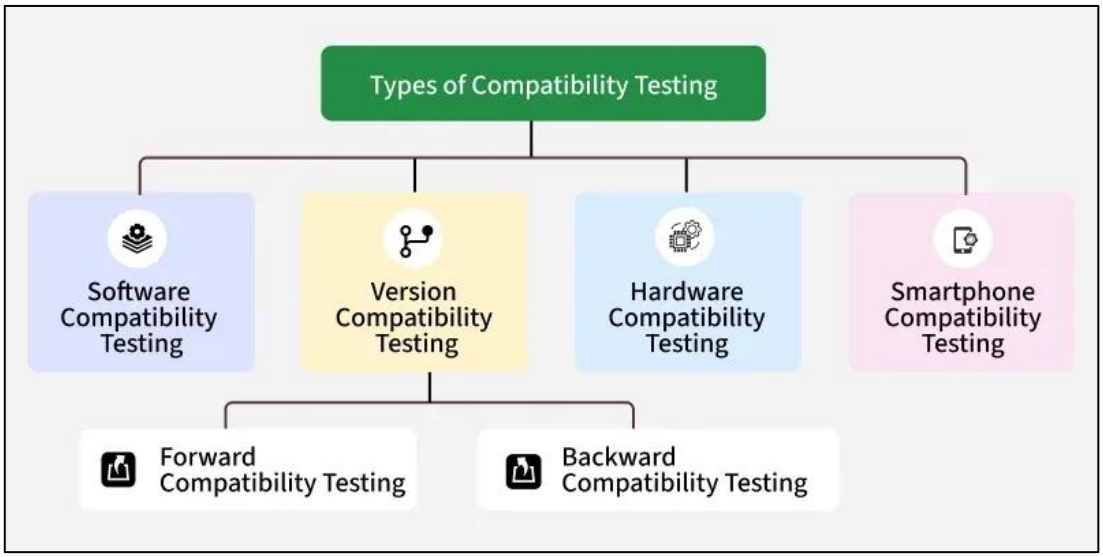


Figure 19. Compatibility Testing

Load Testing

This testing evaluates the system's performance under expected and peak heavy usage, and its acceptable performance, by simulating multiple concurrent users and assessing response time, throughput, and resource utilization. This helps identify and resolve bottlenecks, such as slow database queries or server limitations, before they impact real users.

Stress Testing

This testing evaluates the system's performance beyond its normal operational capacity by applying extreme loads, such as heavy user traffic and high request volumes. The researchers conducted the stress testing to ensure data integrity when facing failures and to improve error handling to maintain reliability. This process ultimately helps build a safe and more resilient system.

Acceptance Testing

Figure 20 highlights the final stage of development, during which acceptance testing was conducted to confirm that ConservAte meets the intended requirements. At this point, the system was reviewed based on how well it supports the needs of actual users—whether customers, vendors, or administrators. The goal was to verify that every feature works properly in practical scenarios, not just in isolated tests. The team placed orders, managed listings, tracked donations, and handled notifications, making sure each function behaved as expected and ensuring the system is complete and reliable for everyday use. This final validation phase also included stress-testing the platform under peak usage conditions to ensure stability.

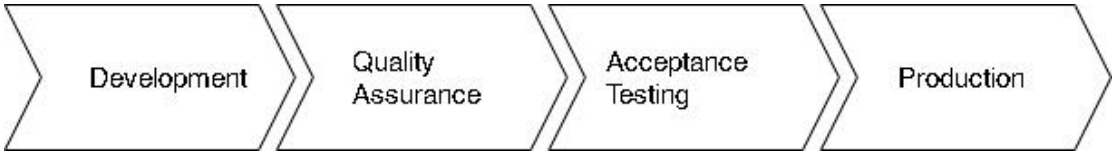


Figure 20. Acceptance Testing

Population of the Study

The study population comprises selected individuals and groups directly involved in evaluating the ConservAte system to ensure its usability, effectiveness, and relevance. These participants were considered key

stakeholders who could provide valuable insights during the assessment process.

Table 12 presents the distribution of participants based on their roles: eight (8) food establishment representatives, eight (8) public market vendors, eight (8) non-public food vendors, twenty-five (25) consumers, one (1) non-governmental organization representative, and six (6) IT Experts. These participants participated in hands-on system testing and demonstration sessions and completed questionnaires designed to evaluate the viability and functionality of the ConservAte platform.

Table 12. Respondents of the Study

Category of Respondents	Sample
Food Establishment Representatives	8
Public Market Food Vendors	8
Non - Public Food Vendors	8
Consumers	25
Non-Governmental Organization	1
IT Experts	6
TOTAL	56

Sampling Design

The research utilized purposive sampling, meaning participants were deliberately selected based on their specific, relevant knowledge or characteristics, rather than at random (Hassan, 2024). This method was chosen to ensure that feedback would come from individuals who are most likely to interact with or benefit from the ConservAte system. It greatly helped the researchers address the system's shortcomings by understanding users' preferences.

Participants were selected from different areas in Los Baños, Laguna, including food establishment representatives, public market vendors, local customers, NGO members, and IT experts. These individuals were considered suitable for the study due to their direct involvement in food services, community outreach, or system evaluation. A total of 56 respondents—eight food establishment representatives, eight public market vendors, eight non-public food vendors, twenty-five consumers, one NGO representative, and six IT experts—actively participated in testing and providing feedback for this study.

Data Collection Instruments

The researchers used several methods to gather feedback and data to inform what to do in developing the web-based application ConservAte and which problems to solve with this system. Researchers observed current trends and advancements in technology, applied them to mitigate food waste in the study area, and examined how users interacted with the system to identify usability problems and understand their overall experience.

During the planning stage and initial data gathering, questions were asked in informal conversations, allowing food establishment representatives and other related individuals and organizations to express the problems they encounter with food waste freely.

In addition, relevant materials and literature from the internet and the library were reviewed to support the research claims. To make data collection easier and reach the intended respondents more conveniently, the distribution of the survey questionnaire for system evaluation was conducted both in person and through Google Forms.

Statistical Treatment

The technique used by this study is purposive sampling, which allows the researchers to gather specific respondents to evaluate and assess the system. Collected responses from the survey questionnaires were thoroughly assessed and computed for both the TAM and the ISO surveys.

Descriptive statistics were used to compute response variability (standard deviation) and to determine central tendency (mean) for both the ISO and Technology Acceptance Model (TAM) surveys. This approach clearly describes the general perceptions of the respondents and the consistency of their answers.

a) Mean

Each survey items and components was calculated and determined its central score or the average using mean with this formula:

$$\mathbf{x} = \frac{\Sigma x}{n}$$

Mean = (ΣX = Sum of all values) **/** (**n** =Number of values)

b) Standard Deviation (SD)

The standard deviation measured the spread of respondents' answers around the calculated mean where low standard deviation highlights the same pattern of responses among the respondents while diverse opinions were suggested when the standard deviation is higher. This was also be used to determine the user level of agreement and its variations from every responses. Below is the formula of sample standard deviation:

$$\mathbf{S} = \sqrt{\frac{\Sigma x_i - x)^2}{n-1}}$$

where **S =** Sample standard deviation, $\mathbf{x_i}$ **=** Each item in the sample, **x** = mean of the sample, and **n =** sample's number of items.

Table 13 shows the 4-point Likert Scale for the Technology Acceptance Model and ISO 25010. The perceptions of the respondents were based on the mean score results from the survey and are graded from top to bottom as directed: four (4) Strongly Agree, three (3) Agree, two (2) Disagree, and one (1) Strongly Disagree. This scale helped to assess the perceptions of each target respondent. Using this kind of approach, it were ensured that the result were reliable as it is answered by well-selected respondents who align with the study objectives.

Table 13. 4-Point Likert Scale Rubrics for TAM and ISO 25010

Rating	Mean Range	Interpretation
4	3.25 - 4.00	Strongly Agree
3	2.50 - 3.24	Agree
2	1.75 - 2.49	Disagree
1	1.0 - 1.74	Strongly Disagree

Evaluation Procedure

The user acceptability of the ConservAte web-based system was evaluated using the survey questionnaire derived from the Technology Acceptance Model (TAM) (Diaz et al., 2021; Venkatesh & Davis, 2000), making a guide in formulating the questions, ensuring that feedback from the respondents were useful and align with the acceptability and usability of the system. The model includes essential elements such as the Perceived Usefulness, Perceived Ease of Use, Personal Innovativeness, Social Influence, Information Awareness, Willingness to Adopt, and Perceived Cost.

PU assesses how the system worked to address food waste and supports food redistribution efforts. PEOU focuses on how efficiently and conveniently the system is to navigate and its ability to perform tasks. PI shows the willingness of the user to try new technology. SI measures how much users are influenced by people around them to try or continue using the platform. If users see others benefiting from or recommending the system, they are more likely to adopt it themselves. IA assesses whether users are well-informed about the system's purpose and its features for achieving that purpose. WTA measures the degree of the user's readiness to use the platform, while PC evaluates the distinguished effort, time, and resources needed to use the system.

The system's quality was also evaluated using the ISO/IEC 25010 software quality product, as shown in Figure 21. This model was used to check the system's quality in areas like functional suitability, performance efficiency, reliability, flexibility, security, speed, interaction capability, compatibility, maintainability, and safety. The evaluation was conducted by collecting feedback from experts and their assessments of the system using the ISO 25010 Survey Questionnaire.

SOFTWARE PRODUCT QUALITY								
FUNCTIONAL SUITABILITY	PERFORMANCE EFFICIENCY	COMPATIBILITY	INTERACTION CAPABILITY	RELIABILITY	SECURITY	MAINTAINABILITY	FLEXIBILITY	SAFETY
FUNCTIONAL COMPLETENESS	TIME BEHAVIOUR	CO-EXISTENCE	APPROPRIATENESS RECOGNIZABILITY	FAULTLESSNESS	CONFIDENTIALITY	MODULARITY	ADAPTABILITY	OPERATIONAL CONSTRAINT
FUNCTIONAL CORRECTNESS	RESOURCE UTILIZATION	INTEROPERABILITY	LEARNABILITY	AVAILABILITY	INTEGRITY	REUSABILITY	SCALABILITY	RISK IDENTIFICATION
FUNCTIONAL APPROPRIATENESS	CAPACITY		OPERABILITY	FAULT TOLERANCE	NON-REPUDIATION	ANALYSABILITY	INSTALLABILITY	FAIL SAFE
			USER ERROR PROTECTION	RECOVERABILITY	ACCOUNTABILITY	MODIFIABILITY	REPLACEABILITY	HAZARD WARNING
			USER ENGAGEMENT		AUTHENTICITY	TESTABILITY		SAFE INTEGRATION
			INCLUSIVITY		RESISTANCE			
			USER ASSISTANCE					
			SELF-DESCRIPTIVENESS					
iso25000.com								

Figure 21. Software Quality Model using ISO 25010

CHAPTER 4

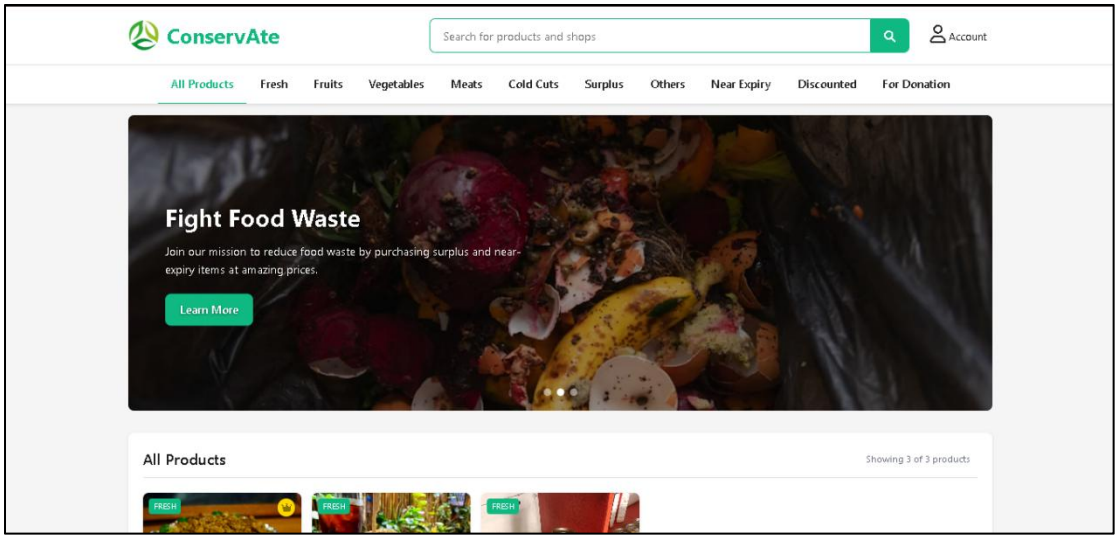
RESULTS AND DISCUSSIONS

This chapter presents and discusses the study's findings, detailing the system's functionalities and features, evaluating its effectiveness, and demonstrating how it fulfills the project's objectives through various tests.

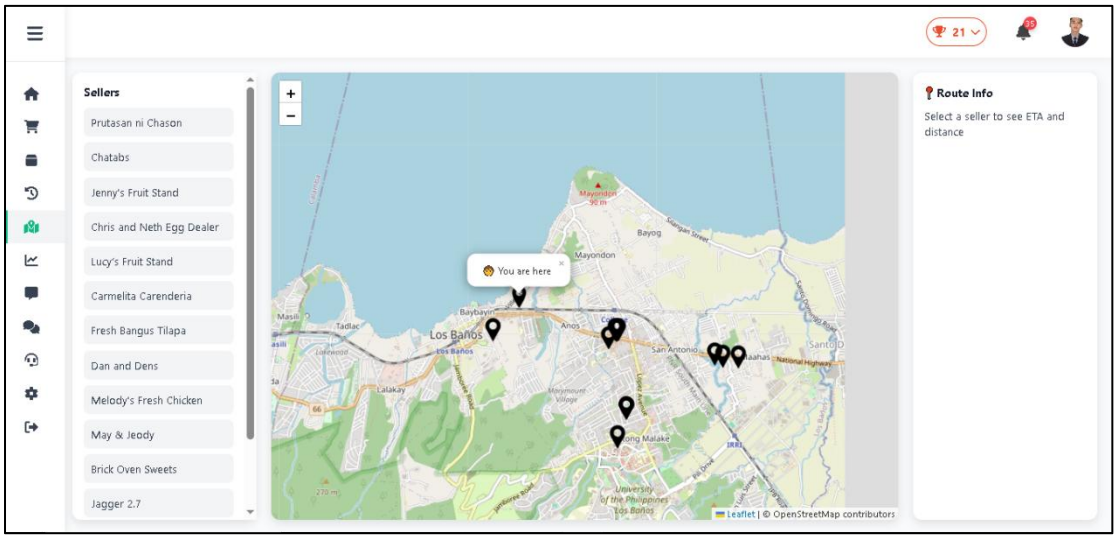
Results by the Objectives

A. Design and develop a web-based platform, integrated with Geographic Information System (GIS), to support the food sales and the redistribution of near-expiry and surplus food products.

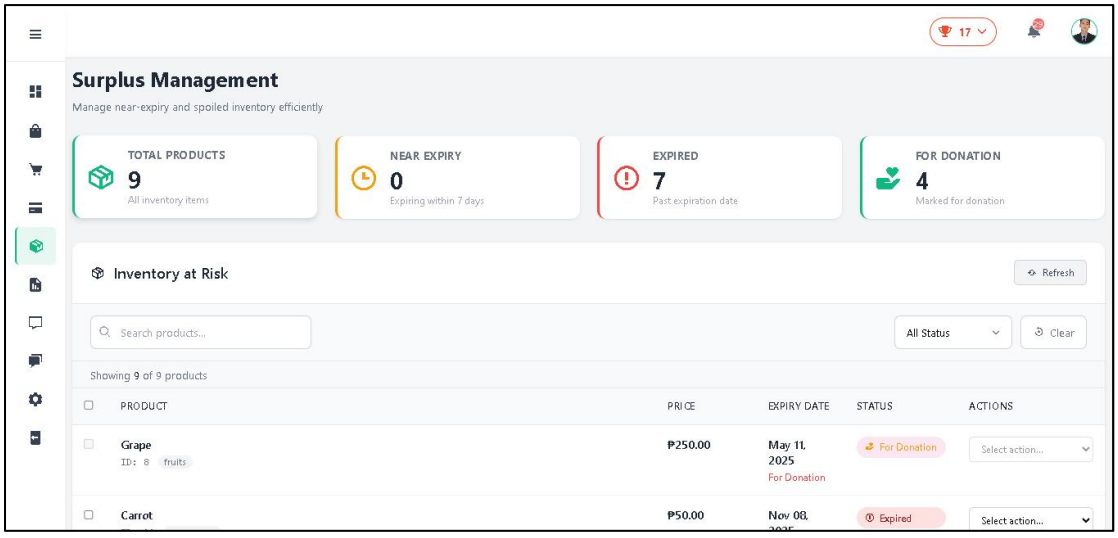
Figure 22 shows the key user interfaces of the ConservAte web platform, which fulfill the project's first objective. This includes the (a) Landing page, (b) map feature, and (c) surplus management. Every user can view the landing page even without registering. It displays information about the system's purpose, available products, and all featured sellers in the shop, along with contact information for guest users to ask questions and raise concerns. With the integration of the Geographic Information System (GIS), customers or consumers can see how many ConservAte sellers are nearby and what products they have in their stores, helping them visualize and locate shop positions. Finally, the surplus management feature allows sellers to monitor and take action on their products by checking their status and deciding whether to adjust the pricing of their products, especially near-expiry items, through discounts, or initiate donation, where users can claim products to avoid food waste.



(a)



(b)

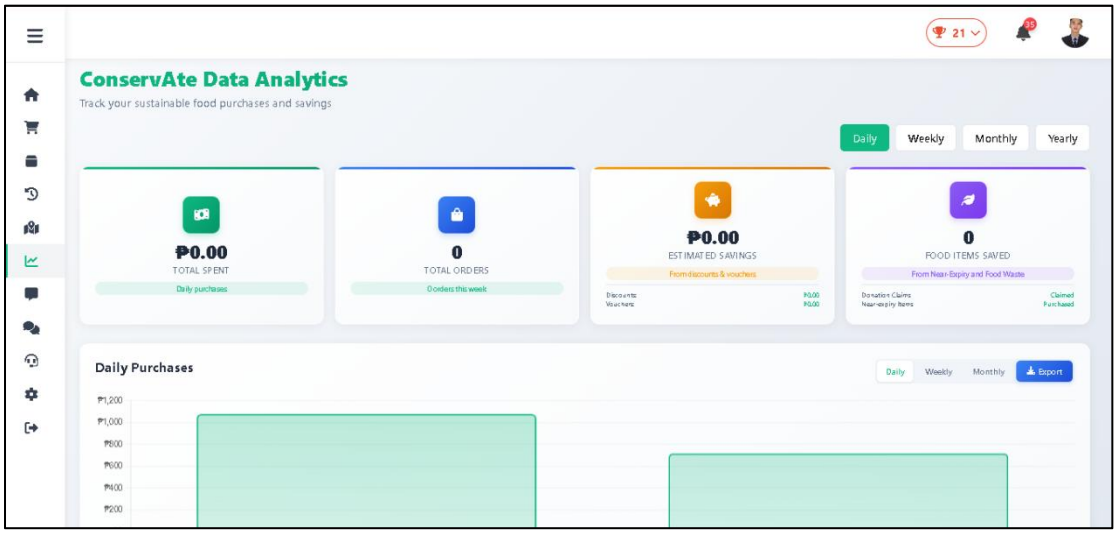


(c)

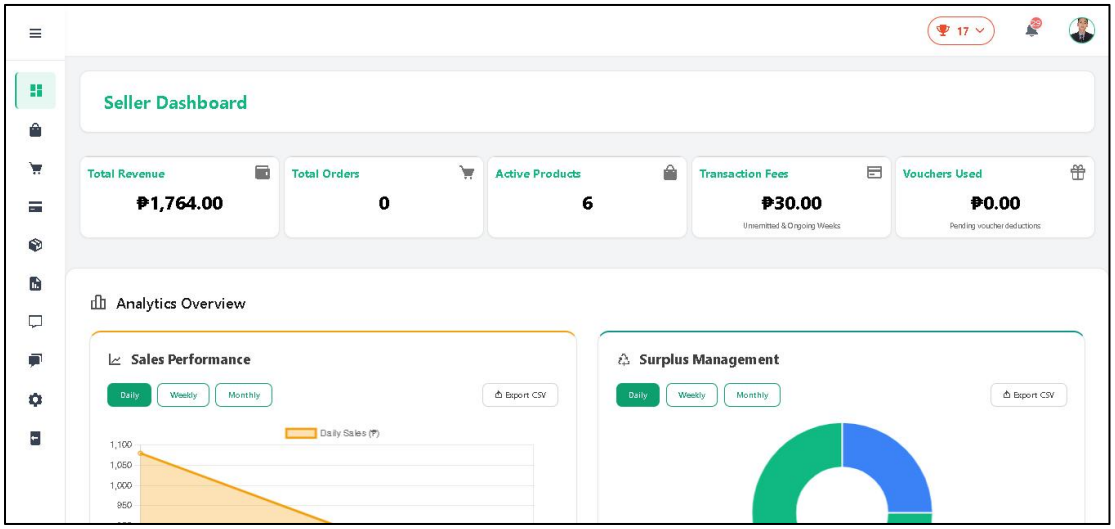
Figure 22. ConservAte System Interfaces: *(a)* Landing Page, *(b)* Customer Map, *(c)* Seller Surplus Management.

B. To implement a tracking and analytics system to monitor food waste trends, optimize redistribution, and ensure secure data management.

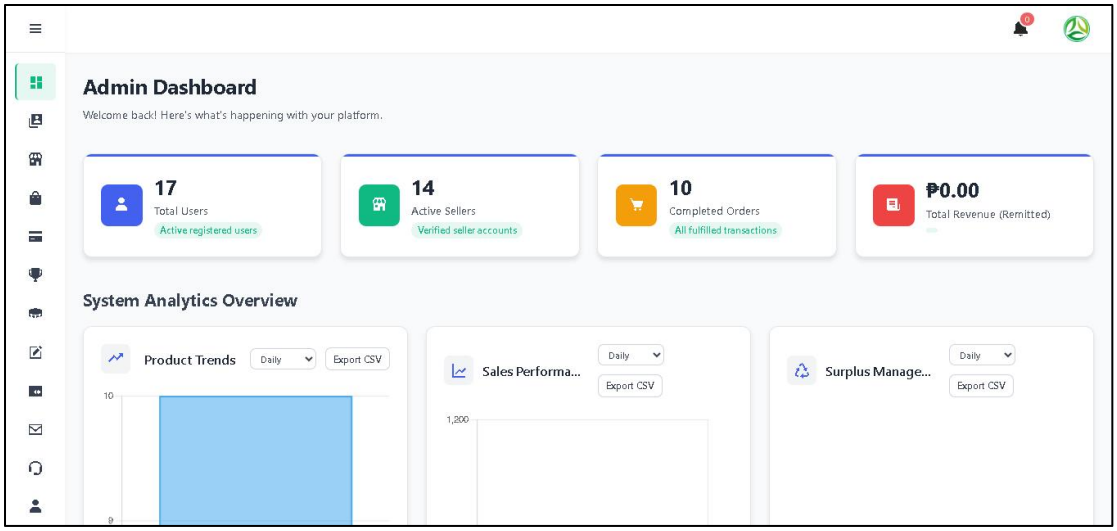
Figure 23 shows the system components where users can track and monitor the analytics stored in ConservAte. For customers (a), this includes the total amount spent, total orders, savings, and products saved. For sellers (b), it displays total revenue, active products, ongoing orders, sales performance, product trends, surplus management, and other data analysis. The super admin (c) can monitor analytics across all sellers to oversee the overall platform performance.



(a)



(b)



(c)

Figure 23. Analytics and Monitoring Dashboards: Customer Analytics *(a)*, Seller Analytics *(b)*, and Super Admin Analytics *(c)*

C. To evaluate the web-based system's abilities using ISO 25010 and Technology Acceptance Model (TAM) for Software Quality Standards.

Results of the Software Testing

This section summarizes the outcomes from checking the ConservAte system's performance. We examined individual parts, tested how they work together, and verified the complete system's operation. The findings confirm that all components function correctly as designed and the system successfully meets its goals.

Unit Testing Result

This section shows the results of testing each unit and function of the ConservAte system on its own. The tests confirm that every component works as designed, proving that the core parts of the system are reliable and can be used by users without any problems.

Customer

Table 14 presents the results of the expected outcomes and processes in the unit testing of the User Module for customers. It includes test cases for

login and registration, cart management, RBAC, account settings, and logout. Each function underwent three iterations of testing to verify accuracy and consistency. After refining the issues identified in the first two attempts, the expected and actual outputs finally aligned on the third attempt, with confirming that the module functioned as intended and received a Passed remark.

Table 14. Unit Testing for Customer's User Module

Tested Case	Expected Output	Actual Output	Process	Remarks
Login and Registra-tion	Customer will be able to log in and create account with an otp sent in their email.	Customer created their account with otp sent in email, and logged in directed to customer dashboard.	5 secs	Passed
Manage Cart	Customer will be able to manage the products added to their own cart.	Customer have managed their products in cart successfully.	3 secs	Passed
Role-Based Access	The system will automatically detect and assign the appropriate role based on the user's registration details.	The system detected and assigned the appropriate role based on the user's registration details.	3 secs	Passed
Account Setting	The customer will be able to change their informations in the settings.	The customer can edit their information in their account successfully.	3 secs	Passed
Logout	The customer's account will successfully logout in the system.	The customer's account was logged out in the system.	2 secs	Passed

Table 15 presents the results of testing for the Map Module, which was thoroughly conducted to verify whether the module's function of locating stores in the system and whether it meets the required expectations by

comparing expected and actual outputs, ensuring that all functions were performing correctly. Before achieving the remarkable feat of passing within 3 seconds, the developers made six attempts.

Table 15. Unit Testing for Customer's Map Module

Tested Case	Expected Output	Actual Output	Process	Remarks
Store Locations	Customer will be able to see the locations of the registered sellers in the system.	Registered sellers' location can see by the customer with the Estimated Time of Arrival feature.	3 secs	Passed

Table 16 presents the unit testing results for the Customer Market Module, which were verified by testing each feature within the module. The checkout function had seven attempts, the view products function had three attempts, and the Add-to-Cart function had four attempts before achieving the passed remarks. It shows the results of the test cases conducted, including order placement, add to cart, and product view and details, ensuring that all functions perform as expected based on the results shown in the actual output and the passed remarks.

Table 16. Unit Testing for Customer Market Module

Tested Case	Expected Output	Actual Output	Process	Remarks
Check-out	Customer can do checkout the product directly or from cart.	Customer can place their orders directly or from their cart.	3 secs	Passed
Add to Cart	Customer will add products to their cart.	Customer was able to add products to their cart.	3 secs	Passed
View Products	Customer will able to see the available products.	Customer was able to see products.	3 secs	Passed

Table 17 presents the results of unit testing for the Customer Message Module, showing the expected and actual outputs along with the corresponding passed remarks. Before aligning all the functions, developers noted that they passed each function after several attempts—Messaging with four attempts, support with six attempts, and forum with nine attempts. This ensures that all functions within the module, including the messaging feature, support requests, and the forum, were properly assessed and verified to perform as intended without encountering any problems.

Table 17. Unit Testing for Customer Message Module

Tested Case	Expected Output	Actual Output	Process	Remarks
Messaging	Customer will have a conversation with sellers and super admin	Customer can message sellers and super admin	3 secs	Passed
Support	Customer will be able send tickets for policy-related reports to super admin	Customer can report and ask assistance from super admin	3 secs	Passed
Forum	Customers will be allowed to post information or content they wish to share.	Customers are allowed to post information or content they wish to share.	3 secs	Passed

Table 18 presents the unit testing result for the functions of the Customer Notification Module by showing the actual output from the expected output of the tested case. Before remarking each unit as passed, functions underwent several attempts to achieve these results, such as Reminders functions with 4 attempts, Status Updates functions with 5 attempts, and Alerts functions with 4 attempts. This process ensured that all functions were operating correctly and performing as intended.

Table 18. Unit Testing for Customer Notification Module

Tested Case	Expected Output	Actual Output	Process	Remarks
Reminders	Customer will receive reminders notification within system.	Customer can receive reminders like suspended, placed orders, and account deletion.	2 secs	Passed
Status Updates	System will make the customer's account receive notifications.	Customer received notifications of their order status, account status.	2 secs	Passed
Alerts	Customer will receive alerts within system.	Customer can receive alert notifications every time they take action in the system.	2 secs	Passed

Table 19 presents the unit testing results of the Customer Data Analytics Module, which underwent testing by verifying the functionality of each feature within the module. The Graphs function required 3 attempts. The table shows the results of the test cases conducted to ensure that all functions performed according to their expected outcomes.

Table 19. Unit Testing for Customer Data Analytics Module

Tested Case	Expected Output	Actual Output	Process	Remarks
Graphs (Spent)	Customer will be able to monitor their spendings.	Customer was able to monitor their spendings.	3 secs	Passed

Seller

Table 20 presents the results of the expected outcomes and processes in the unit testing of the User Module for Seller. After refining the issues identified in the first two attempts, the expected and actual outputs finally aligned on the third attempt, with confirming that the module functioned as

intended and received a Passed remark. It presents the test cases of logging in and registration, role-based access control, account settings, and logout, that were successfully accomplished by comparing the expected and actual outputs.

Table 20. Unit Testing for Seller's User Module

Tested Case	Expected Output	Actual Output	Process	Remarks
Login and Registra-tion	Account will registered or login with correct details with verifying email address.	Account was registered or logged in with correct details with verified OTP.	5 secs	Passed
Role-Based Access	The system will automatically detect and assign role based on the user's details.	The system was able to detect and assign the correct role of the user.	3 secs	Passed
Account Setting	The seller will have a setting where they can edit their information in their account.	The seller was able to edit their information in their account.	3 secs	Passed
Logout	The seller's account will successfully logout in the system.	The account was logged out in the system.	2 secs	Passed

Table 21 presents the results of unit testing for the Seller Product Module, presenting the expected and actual outputs along with the corresponding passed remarks. Before achieving these results, functions underwent attempts such as Inventory (4 attempts), Status Monitoring (5 attempts), and Product Reviews (4 attempts). This ensures that all functions within the module were properly assessed and verified to perform as intended. Multiple testing attempts allowed developers to identify and correct errors. As a result, the module demonstrated stable, reliable performance following successful testing.

Table 21. Unit Testing for Seller Product Module

Tested Case	Expected Output	Actual Output	Process	Remarks
Inventory	Seller will have an inventory where they can their add, view, and delete products.	Seller have an inventory where they can edit their products.	3 secs	Passed
Status Monitoring	The status of the product will be able to manage and monitor of the seller.	Seller can manage and take action to their products.	3 secs	Passed
Product Reviews	Product's review can be view by the seller.	Seller can view the product reviews.	3 secs	Passed

Table 22 presents the unit testing results for the Seller Data Analytics Module, which were verified by testing each feature. The Graph (Sales) function had three attempts, System Insights had four attempts, and Graph (Wastes) had two attempts before achieving the passed remarks. It shows the results of the test cases conducted to ensure that all functions perform as expected.

Table 22. Unit Testing for Seller Data Analytics Module

Tested Case	Expected Output	Actual Output	Process	Remarks
Graphs (Sales)	Seller will be able to monitor their daily, weekly, monthly, and yearly sales.	Seller was able to monitor their daily, weekly, monthly, and yearly sales.	2 secs	Passed
System Insights	Seller will be able to monitor their transactions and placed orders	Seller was able to monitor their transactions and placed orders	2 secs	Passed
Graphs (Wastes)	Seller will be able to monitor their daily, weekly, monthly, and yearly expired and donated products	Seller was able to monitor their daily, weekly, monthly, and yearly expired and donated products	2 secs	Passed

Table 23 presents the unit testing result for the functions of the Seller Notification Module like reminders, status updates, and alerts making sure that the seller is notified by showing the actual output from the expected output of the tested case. Before aligning all the functions, developers noted that they passed each function after several attempts wherein Messaging functions with 4 attempts, Support functions with 6 attempts, and Forum functions with 9 attempts. This process ensured that all functions were operating correctly and performing as intended.

Table 23. Unit Testing for Seller Notification Module

Tested Case	Expected Output	Actual Output	Process	Remarks
Reminders	Seller will receive reminders and notifications within the system.	Seller was able to receive reminders like suspended, placed orders, and account deletion, and the status of the product	2 secs	Passed
Status Updates	Seller will receive status updates notification within system	Seller can receive notifications of their order status, account status	2 secs	Passed
Alerts	Seller will receive alerts within system	Seller can receive alert notifications every time they take action in the system	2 secs	Passed

Table 24 presents the results of unit testing for the Seller Message Module, presenting the expected and actual outputs along with the corresponding passed remarks. Before aligning all the functions, developers noted that they passed each function after several attempts—Messaging with four attempts, support with six attempts, and forum with nine attempts. This ensures that all functions within the module were properly assessed and verified to perform as intended without encountering any problems.

Table 24. Unit Testing for Seller Message Module

Tested Case	Expected Output	Actual Output	Process	Remarks
Messaging	Seller will have a conversation with customer and super admin.	Seller was able to message sellers and the super admin.	3 secs	Passed
Support	Seller will be able send tickets for policy-related reports to super admin.	Seller was able to send tickets for policy-related reports to super admin.	3 secs	Passed
Forum	Seller will be allowed to post information or content they wish to share.	Seller are allowed to post information or content they wish to share.	3 secs	Passed

Table 25 presents the unit testing results for the functions of the Seller Waste Management Module, comparing actual outputs to expected outputs for the test cases and ensuring that the redistribution features (discounts and donations) work properly and flawlessly. The functions underwent four attempts before achieving a passing remark.

Table 25. Unit Testing for Seller Waste Management Module

Tested Case	Expected Output	Actual Output	Process	Remarks
Redistribution	Seller will be able to manage their expired, near-expiry, fresh, and for discount items to redistribute.	Seller was able to manage their expired, near-expiry, fresh, and for discount items to redistribute.	2 secs	Passed

Super Admin

Table 26 shows the results of unit testing for the Super Admin Product Module, which allows the admin to manage all products added by all sellers in the system and take action to remove them if the product is non-food-related,

showing the expected and actual outputs. This function underwent four attempts before achieving a passing remark, indicating that it is working properly.

Table 26. Unit Testing for Super Admin Product Module

Tested Case	Expected Output	Actual Output	Process	Remarks
Inventory	Super Admin will have an inventory where they can view, and take down active products of the sellers.	Super Admin have an inventory where they can supervise and take down active products of the sellers.	3 secs	Passed

Table 27 shows the unit testing results for the functions of the Super Admin Notification Module, showing the actual outputs for the expected outputs in the tested cases for reminders, status updates, and alerts. Before aligning all the functions, developers noted that they passed each function after several attempts—Messaging with four attempts, support with six attempts, and forum with nine attempts.

Table 27. Unit Testing for Super Admin Notification Module

Tested Case	Expected Output	Actual Output	Process	Remarks
Reminders	Super Admin will receive reminders notification within the system	Super Admin was able to receive and send reminders to users	2 secs	Passed
Status Updates	Super Admin will receive status updates notification within the system	Super Admin was able to receive notifications of their remittance status, account and product status	2 secs	Passed
Alerts	Super Admin will receive alerts within system	Super Admin was able to receive alerts.	2 secs	Passed

Table 28 presents the results of unit testing for the Super Admin Message Module, presenting the expected and actual outputs along with all passed remarks. Admin can send messages to all users, provide support for concerns, and supervise the entire forum, where actions such as deletions and content warnings can be regulated. Before aligning all the functions, developers noted that they passed each function after several attempts—Messaging with four attempts, support with six attempts, and forum with nine attempts. This ensures that all functions within the module were properly assessed and verified to perform as intended.

Table 28. Unit Testing for Super Admin Message Module

Tested Case	Expected Output	Actual Output	Process	Remarks
Messaging	Super Admin will be able to have a conversation with all users.	Super Admin was able to message all users.	3 secs	Passed
Support	Super Admin will be able to receive tickets for reports of users.	Super Admin was able to provide support for all users.	3 secs	Passed
Forum	Super Admin will be able to supervise the forum.	Super Admin can take down contents and warn users.	3 secs	Passed

Table 29 presents the expected outcomes and processes for unit testing the User Module for Super Admin. It presents the test cases that were accomplished by comparing the expected and actual outputs. Each function underwent three iterations of testing, and after refining the issues identified in the first two attempts, all the expected and actual outputs finally aligned in the third attempt, confirming that the module functioned as intended and received a passed remark.

Table 29. Unit Testing for Super Admin's User Module

Tested Case	Expected Output	Actual Output	Process	Remarks
Login	Admin will be able to access admin account contents through logging in with correct credentials.	Admin was successfully directed to admin dashboard after logging in.	5 secs	Passed
Role-Based Access	The system will automatically detect and assign the appropriate role based on the user's registration details.	The system checked the role and redirects the admin to the admin dashboard and contents.	3 secs	Passed
Account Setting	The Super Admin will have a setting where they can edit their information in their account.	The Super Admin can edit their information in their account.	3 secs	Passed
Logout	The Super Admin's account will successfully logout in the system.	The Super Admin's account was able to logged out in the system.	2 secs	Passed

Table 30 presents the unit testing results for the Super Admin Data Analytics Module, which was tested by verifying the functionality of each feature to ensure proper operation. The Graph (Sales) function had three attempts, System Insights had four attempts, and Graph (Wastes) had two attempts before achieving the passed remarks. All tested cases for providing information for sales graphs, retrieving information for system insights, and generating a graph for waste include an indicator of 'passed' in the remarks, proving that the data analytics module is working as expected and processing in a short span of time. With this graph, sales, insights, and waste are presented more conveniently and easily understood at the daily, weekly, and yearly levels.

Table 30. Unit Testing for Super Admin Data Analytics Module

Tested Case	Expected Output	Actual Output	Process	Remarks
Graph (Sales)	Super Admin will be able to oversee their sales.	Super Admin was able to oversee their sales.	2 secs	Passed
System Insights	Super Admin will be able to oversee transactions, orders, surplus products.	Super Admin can oversee all the transactions, orders, surplus products inside the system providing insights and visualizations.	2 secs	Passed
Graph (Wastes)	Super Admin will be able to oversee the daily, weekly, monthly, and yearly expired, donated products.	Super Admin can oversee the daily, weekly, monthly, and yearly expired, donated products.	2 secs	Passed

Integration Testing Result

The results in this section show that all modules of the ConservAte system, such as the user module, data analytics module, product module, market module, message module, notification module, waste management module, and map module, connect and work together as they should. These modules must coexist and allow information and actions to synchronously perform all the tasks in the system. Each part is successfully integrated with the others and shares information correctly, ensuring the platform is working.

Table 31 confirms that all the parts of the system for customers work together smoothly. A user can easily find products on the map, buy them, and see their savings without interruptions. Each module is linked to the others, maintaining a well-established connection between them. The results indicated that all modules passed the integration testing.

Table 31. Integration Testing Results for Customer Modules

Tested Case	Expected Output	Actual Output	Process	Remarks
User Module	All user-related functions should work together seamlessly with other modules.	User authentication and profile data successfully shared across all connected modules.	3 secs	Passed
Market Module	Product browsing, cart, and checkout features should integrate smoothly with user accounts and location services.	Customers can view products, manage carts, and complete purchases using their account and location data.	3 secs	Passed
Message Module	Messaging system should connect properly with user profiles and successfully sent a message to seller and admin.	Customers can send and receive messages to the sellers and admin using their account.	3 secs	Passed
Map Module	Location services should integrate with market especially store locations.	Store locations display correctly with ETA, connecting to product availability and user location.	3 secs	Passed
Notifica-tion Module	Alert system should work across all modules (orders, messages, reminders).	Notifications delivered successfully for all customer activities and system updates	2 secs	Passed
Data Analytics Module	Spending reports and analytics should connect to user purchase history and account data.	Customer dashboards accurately display spending patterns and saved products data.	2 secs	Passed

As shown in Table 32, the different parts of the seller's system modules are well-integrated. This successful testing proves that sellers can reliably

manage their core activities. They can control their product inventory, monitor their stock, and interact with customers through messages, with confidence that all components work together as a smooth, functional unit.

Table 32. Integration Testing Results for Seller Modules

Tested Case	Expected Output	Actual Output	Process	Remarks
User Module	Seller account should work across all system modules with proper access.	Seller successfully accessed all modules with correct permissions	3 secs	Passed
Product Module	Products should sync between inventory and marketplace	All products appeared correctly across all connected systems.	3 secs	Passed
Waste Management Module	Product status changes should update across inventory and analytics	Expired and near-expiry products automatically updated in all modules	2 secs	Passed
Data Analytics Module	Sales and waste data should update in real-time from all transactions	All metrics showed current data for all products and orders	2 secs	Passed
Notification Module	Alerts should trigger for orders, status changes, and system updates	Notifications was delivered promptly for all important activities	2 secs	Passed
Message Module	Seller must be able to connect with customers and super admin	Messages was successfully sent and received.	3 secs	Passed

The results in Table 33 demonstrate that the tools for the super admin are fully integrated. This means an administrator can effectively perform their job of overseeing the platform, jumping from reviewing seller analytics to addressing a customer support ticket and then checking system health, all within a single, cohesive interface. This integration is crucial for maintaining order and providing timely support across the entire user base. All the modules in the tested case passed the integration testing.

Table 33. Integration Testing Results for Super Admin Modules

Tested Case	Expected Output	Actual Output	Process	Remarks
User Module	Super Admin should access and manage all user accounts across the platform	Super Admin was able to manage customer and seller accounts	3 secs	Passed
Product Module	Super Admin should oversee and able to manage all products listed by sellers	All seller products were visible and supervised through admin dashboard	3 secs	Passed
Data Analytics Module	Super Admin should monitor system-wide performance metrics and platform statistics	Comprehensive platform data integrated and displayed from all user activities	2 secs	Passed
Notification Module	Super Admin should send and receive system-wide alerts and updates	Platform-wide notifications delivered successfully to all user types	2 secs	Passed
Message Module	Super Admin should communicate with all users and manage support tickets	Successfully exchanged messages with both sellers and customers	3 secs	Passed

System Testing Result

System testing was conducted to evaluate the ConservAte platform's performance as a complete, unified product. The goal was to validate key end-to-end workflows a user would perform in a real-world scenario, ensuring that all the individually integrated modules function seamlessly together to accomplish the system's main objectives.

Load Testing Result

Table 34 presents the load testing results for the ConservAte system, demonstrating how the application performs under increasing concurrent user loads. The data shows how the system handled more users at the same time, measuring its speed, reliability, and overall stability. The system performed well with 5 to 100 concurrent users.

Table 34. Load Testing Results for ConservAte System

Tested Case	Number of Users	Throughput	Error	Remarks
Executed system performance under 5 concurrent users	5	2.1 / sec	0%	Stable
Executed system performance under 10 concurrent users	10	2.1 / sec	0%	Stable
Executed system performance under 20 concurrent users	20	15.5 / sec	0%	Stable
Executed system performance under 50 concurrent users	50	31.6 / sec	0%	Stable
Executed system performance under 100 concurrent users	100	54.2 / sec	0%	Stable

Stress Testing Result

The table 35 presents the outcome of the stress testing results for the ConservAte system, demonstrating how the application performs under extreme concurrent user loads. The data in the table expresses the system's breaking point, measuring its maximum capacity, failure threshold, and stability under heavy loads.

Table 35. Stress Testing Results for ConservAte System

Tested Case	Number of Users	Throughput	Error	Remarks
Extreme load	200	55.1 / sec	0%	Stable
Peak Threshold	250	52.9 / sec	53.41%	System Struggling
Breakdown Point	300	57.3 / sec	82.31%	Near Failure
System Failure	500	59 / sec	100%	System Down

Compatibility Testing Results

Table 36 presents the compatibility status of the ConservAte web platform on different Windows versions. The results indicate that Windows 11, Windows 10, Windows 8, and Windows 7 are all compatible without experiencing any error, successfully meet the expected outcomes, and are marked as passed.

Table 36. Windows Version Compatibility Testing Result

Windows Version	Expected Result	Actual Result	Remarks
Windows 11	Compatible	Compatible	Passed
Windows 10	Compatible	Compatible	Passed
Windows 8	Compatible	Compatible	Passed
Windows 7	Compatible	Compatible	Passed

Table 37 presents the runtime performance of the ConservAte web platform on different Windows versions. The results confirm that the system loads successfully across all tested operating systems. The actual load times for Windows 11 (3 seconds), Windows 10 (3 seconds), Windows 8 (4 seconds), and Windows 7 (5 seconds) all fall within the acceptable expected range of 2-5 seconds therefore, all tests were passed with a performance remark of "Fast."

Table 37. Windows Version Runtime Result

Windows Version	Expected Result	Actual Result	Remarks
Windows 11	2-5 secs	3 secs	Fast
Windows 10	2-5 secs	3 secs	Fast
Windows 8	2-5 secs	4 secs	Fast
Windows 7	2-5 secs	5 secs	Fast

Table 38 lists the browsers used to test the compatibility of the ConservAte system, including Google Chrome, Mozilla Firefox, Microsoft Edge, and Opera, ensuring it works across different browsers, especially since it is a web-based system. This test shows that the user interface, functions, responsiveness, integration, and performance standards are compatible with various popular browsers, meet expectations, and are marked as passed.

Table 38. Browsers Compatibility Testing Result

Browsers	Expected Result	Actual Result	Remarks
Google Chrome	Compatible	Compatible	Passed
Mozilla Firefox	Compatible	Compatible	Passed
Microsoft Edge	Compatible	Compatible	Passed
Opera	Compatible	Compatible	Passed

Acceptance Testing Result

The following section presents the analysis of user acceptance for the ConservAte platform. The average scores (mean) and the level of agreement among respondents (standard deviation) for each feature of the platform were discussed thoroughly. These results show the overall user perception of the system's security, safety, usefulness, ease of use, quality, and willingness to use.

Table 39 higlights that users strongly agree with the system's usefulness. Item 3 has the lowest mean of 3.46, followed by Item 5 with a mean of 3.52; Item 4 with 3.60; Item 2 with 3.62; and Item 1 with the highest mean of 3.68 among all the statements under perceived usefulness towards the ConservAte Platform. Most respondents consistently found that ConservAte helps reduce food waste and improve food access, with strong, unified positive feedback across all measures, as indicated by a composite mean of 3.58.

Table 39. Perceived Usefulness Toward ConservAte

Statements	Mean	SD	Interpretation
1. ConservAte helps users minimize food waste through its marketplace.	3.68	0.51	Strongly Agree
2. ConservAte improves access to affordable surplus and fresh food.	3.62	0.57	Strongly Agree
3. ConservAte simplifies the process of redistributing near-expiry items.	3.46	0.65	Strongly Agree
4. ConservAte contributes to local food security through its features.	3.60	0.64	Strongly Agree
5. ConservAte supports sustainable consumption and purchasing behavior.	3.52	0.65	Strongly Agree
Composite Mean	**3.58**	**0.60**	**Strongly Agree**

Legend: *3.25 - 4.00 = Strongly Agree, 2.50 - 3.24 = Agree, 1.75 - 2.49 = Disagree, 1.0 - 1.74 = Strongly Disagree*

Table 40 presents the users' level of agreement, interpreted as strongly agree, showing a composite mean score of 3.52. Based on the scores for each item, Item 1 had the lowest mean (3.44), followed by Item 3 (3.50), and both Item 4 and Item 2 had the highest mean (3.54). This implies that users perceive ConservAte as easy to use and display efficiency in performing actions. The navigation is on point, and the interface is presentable and user-friendly, making it easier for users to explore and learn how to interact with other users within the system.

Table 40. Perceived Ease Of Use Toward ConservAte

Statements	Mean	SD	Interpretation
1. It is easy to learn how to use the ConservAte application.	3.44	0.51	Strongly Agree
2. Navigating ConservAte's interface and features is user-friendly.	3.54	0.65	Strongly Agree
3. Users can quickly locate food products or vendor listings.	3.50	0.64	Strongly Agree
4. Interacting with vendors and placing orders through ConservAte is simple.	3.54	0.65	Strongly Agree
Composite Mean	**3.52**	**0.60**	**Strongly Agree**

Legend: *3.25 - 4.00 = Strongly Agree, 2.50 - 3.24 = Agree, 1.75 - 2.49 = Disagree, 1.0 - 1.74 = Strongly Disagree*

Table 41 presents the users' level of agreement, showing an average or mean score of 3.51, interpreted as "Strongly Agree,". Item 4 received the lowest score, with a mean of 3.44, followed by Item 1 with 3.48, indicating that users are looking positively toward trying new, innovative platforms like the ConservAte. Item 3 received a 3.54, and Item 2 received the highest mean score of 3.56, where users prefer using new food waste reduction technologies. This indicates that users perceived themselves as innovative in using new technologies to reduce food waste, such as the ConservAte.

Table 41. Personal Innovativeness Toward ConservAte

Statements	Mean	SD	Interpretation
1. Users are excited to try innovative platforms like ConservAte.	3.48	0.65	Strongly Agree
2. Users prefer exploring new technologies that address food waste.	3.56	0.54	Strongly Agree
3. Users are interested in discovering new e-commerce features in ConservAte.	3.54	0.58	Strongly Agree
4. Users proactively try digital solutions that promote sustainability.	3.44	0.58	Strongly Agree
Composite Mean	**3.51**	**0.59**	**Strongly Agree**

Legend: *3.25 - 4.00 = Strongly Agree, 2.50 - 3.24 = Agree, 1.75 - 2.49 = Disagree, 1.0 - 1.74 = Strongly Disagree*

Table 42 indicates a strong level of user agreement with the social influence of people on others in promoting and encouraging the use of food waste reduction applications like ConservAte, with a composite mean of 3.59. With the scores from least to highest, Item 1 is the least with a mean score of 3.48, second is Item 2 with 3.54, third is Item 3 with 3.62, while Item 4 is the highest among all statements with a mean score of 3.70, implying how social factors affect users' choice in using the system.

Table 42. Social Influence Toward ConservAte

Statements	Mean	SD	Interpretation
1. Users' family or peers encourage using ConservAte.	3.48	0.58	Strongly Agree
2. Community trends or groups influence users to adopt ConservAte.	3.54	0.71	Strongly Agree
3. Seeing others shop or donate on ConservAte inspires users to participate.	3.62	0.49	Strongly Agree
4. Users are more likely to try ConservAte if recommended by trusted people.	3.70	0.46	Strongly Agree
Composite Mean	**3.59**	**0.56**	**Strongly Agree**

Legend: *3.25 - 4.00 = Strongly Agree, 2.50 - 3.24 = Agree, 1.75 - 2.49 = Disagree, 1.0 - 1.74 = Strongly Disagree*

Table 43 presents the users' level of agreement, showing a composite mean score of 3.55, interpreted as "Strongly Agree,". Having scores by lowest to highest: Item 3 with 3.46, Item 2 with 3.54, Item 4 with 3.56, Item 1 with 3.62. This implies that the users of the ConservAte are aware and informed of the purpose of the platform, and their preference to use digital solutions promoting sustainability and exploring new technologies that address food waste.

Table 43. Information Awareness Toward ConservAte

Statements	Mean	SD	Interpretation
1. Users are aware that ConservAte promotes food waste reduction.	3.62	0.60	Strongly Agree
2. Users know how ConservAte helps communities access food affordably.	3.54	0.61	Strongly Agree
3. Users are informed about ConservAte's unique features and tools.	3.46	0.61	Strongly Agree
4. Users are aware of ConservAte's ongoing efforts and impact.	3.56	0.54	Strongly Agree
Composite Mean	**3.55**	**0.59**	**Strongly Agree**

Legend: *3.25 - 4.00 = Strongly Agree, 2.50 - 3.24 = Agree, 1.75 - 2.49 = Disagree, 1.0 - 1.74 = Strongly Disagree*

Table 44 presents the users' level of agreement, showing a composite mean score of 3.50, interpreted as "Strongly Agree,". Arranging from lowest to highest, Item 3 got the least score with a mean of 3.36, followed by Item 4 with 3.50, Item 2 with 3.54, and the highest mean is Item 1 with a mean score of 3.58, indicating that the ConservAte web-based system offers reasonable pricing for the service it gives to the users. This strongly indicates that users perceived ConservAte's benefits outweighed its associated fees and pricing, making it cost-effective and fair, as users believed that the ConservAte system offers fair pricing for surplus items and products.

Table 44. Perceived Cost Toward ConservAte

Statements	Mean	SD	Interpretation
1. ConservAte offers reasonable pricing for the value it delivers.	3.58	0.61	Strongly Agree
2. Users find ConservAte's marketplace to be cost-effective.	3.54	0.68	Strongly Agree
3. The benefits of using ConservAte outweigh any associated fees.	3.36	0.66	Strongly Agree
4. Users believe that ConservAte provides fair pricing for surplus items.	3.50	0.68	Strongly Agree
Composite Mean	**3.50**	**0.66**	**Strongly Agree**

Legend: *3.25 - 4.00 = Strongly Agree, 2.50 - 3.24 = Agree, 1.75 - 2.49 = Disagree, 1.0 - 1.74 = Strongly Disagree*

Table 45 presents the behavioral intention to use (willingness to adopt), with the results for each item listed from lowest to highest mean. Item 2 had the lowest mean score of 3.34, followed by Item 1 with 3.60, and Item 3 with 3.66. Item 4 had the highest mean with a score of 3.68, indicating that users of ConservAte were willing to recommend and encourage others to use it. Respondents also expressed a willingness to use the platform regularly, with a composite mean of 3.57.

Table 45. Willingness to Adopt Toward ConservAte

Statements	Mean	SD	Interpretation
1. Users are willing to use ConservAte for buying and redistributing food.	3.60	0.57	Strongly Agree
2. Users plan to continue using ConservAte regularly.	3.34	0.72	Strongly Agree
3. Users are willing to recommend ConservAte to others.	3.66	0.66	Strongly Agree
4. Users would encourage others to use ConservAte for sustainable food practices.	3.68	0.68	Strongly Agree
Composite Mean	**3.57**	**0.57**	**Strongly Agree**

Legend: *3.25 - 4.00 = Strongly Agree, 2.50 - 3.24 = Agree, 1.75 - 2.49 = Disagree, 1.0 - 1.74 = Strongly Disagree*

Table 46 presents all the constructs of the Technology Acceptance Model (TAM). Users perceived ConservAte as a useful platform with a mean of 3.58. The design and interface of the system made users believe that it is user-friendly and easy to use, marking a mean of 3.52. Respondents' expressed their personal preference in using new digital solutions for addressing food waste like ConservAte got a mean of 3.51. The recommendations of trusted and credible people influenced the usage of the app, where users became willing to use the platform with a mean of 3.59, marking the highest level of agreement among the constructs. Users displayed their information awareness toward actions and initiatives in fighting food waste through digital innovations like ConservAte with a mean of 3.55, while perceived cost got a mean of 3.50. The users expressed their willingness for continuous usage of the app for buying and practicing sustainable consumption with a mean of 3.57. Overall, the TAM survey got a composite mean of 3.55 with an interpretation that users strongly agree to the usefulness, effectiveness, and efficiency.

Table 46. Result of TAM Components Toward ConservAte Platform

Indicators	Mean	SD	Interpretation
Perceived Usefulness	3.58	0.60	Strongly Agree
Perceived Ease of Use	3.52	0.60	Strongly Agree
Personal Innovativeness	3.51	0.59	Strongly Agree
Social Influence	3.59	0.56	Strongly Agree
Information Awareness	3.55	0.59	Strongly Agree
Perceived Cost	3.50	0.66	Strongly Agree
Willingness to Adopt	3.57	0.57	Strongly Agree
Overall Mean	**3.55**	**0.60**	**Strongly Agree**

Legend: *3.25 - 4.00 = Strongly Agree, 2.50 - 3.24 = Agree, 1.75 - 2.49 = Disagree, 1.0 - 1.74 = Strongly Disagree*

ISO 25010 Results

This section discusses all the survey results for ISO 25010, determining that the system is properly aligned with the standard based on the assessment and evaluation by IT experts, making it more reliable and safer to use by users without compromising safety, security, or user satisfaction.

Table 47 displays the composite results for the “Functional Suitability” of ConservAte, evaluated against the ISO 25010 standard. The component achieved a composite mean score of 3.77, which is interpreted as “Strongly Agree.” Results show that Item 1 has the lowest score of 3.67, while both Item 2 and Item 3 have the highest mean of 3.83, indicating that the ConservAte provides accurate and reliable information. This result indicates a strong consensus among IT experts that the system’s core functions are working appropriately and effectively for the users, successfully meeting the essential criteria for functional suitability.

Table 47. ISO 25010 Functional Suitability Evaluation of ConservAte

Indicators	Mean	SD	Interpretation
1. Functional Completeness	3.67	0.52	Strongly Agree
2. Functional Correctness	3.83	0.41	Strongly Agree
3. Functional Appropriateness	3.83	0.41	Strongly Agree
Composite Mean	**3.77**	**0.44**	**Strongly Agree**

Legend: *3.25 - 4.00 = Strongly Agree, 2.50 - 3.24 = Agree, 1.75 - 2.49 = Disagree, 1.0 - 1.74 = Strongly Disagree*

Table 48 presents the evaluation results of the Performance Efficiency of ConservAte, based on the ISO 25010 software quality model. This component assessed whether the system performs its functions promptly and utilizes resources effectively. The composite mean of 3.83, with an

interpretation of Strongly Agree, signifies IT experts agreed that the platform runs efficiently, responds quickly, runs on various hardware, and can handle high user and data loads. The indicators of Time Behavior, Resource Utilization, and Capacity demonstrate perfect consistency, with a mean score of 3.83 and a corresponding interpretation of Strongly Agree.

Table 48. ISO 25010 Performance Efficiency Evaluation of ConservAte

Indicators	Mean	SD	Interpretation
1. Time Behavior	3.83	0.41	Strongly Agree
2. Resource Utilization	3.83	0.41	Strongly Agree
3. Capacity	3.83	0.41	Strongly Agree
Composite Mean	**3.83**	**0.41**	**Strongly Agree**

Legend: *3.25 - 4.00 = Strongly Agree, 2.50 - 3.24 = Agree, 1.75 - 2.49 = Disagree, 1.0 - 1.74 = Strongly Disagree*

Table 49 presents the evaluation results for the “Compatibility” of ConservAte, assessed against the ISO 25010 quality model. This component measures the system’s ability to coexist and interoperate with other products in its operational environment. The outcome show that Item 1 got the highest mean of 3.83, followed by Item 2 with 3.67. The composite mean score of 3.75 (Strongly Agree) indicates a strong consensus among IT experts that ConservAte performs effectively alongside other software.

Table 49. ISO 25010 Compatibility Evaluation of ConservAte

Indicators	Mean	SD	Interpretation
1. Co-existence	3.83	0.41	Strongly Agree
2. Interoperability	3.67	0.52	Strongly Agree
Composite Mean	**3.75**	**0.46**	**Strongly Agree**

Legend: *3.25 - 4.00 = Strongly Agree, 2.50 - 3.24 = Agree, 1.75 - 2.49 = Disagree, 1.0 - 1.74 = Strongly Disagree*

Table 50 presents the evaluation results for the “Usability” of ConservAte, assessed against the ISO 25010 quality model. This component

measures the system's effectiveness, efficiency, and overall satisfaction in use from a user's perspective, with the results of Item 4 and Item 8 having the lowest mean of 3.50, Item 7 with 3.67, followed by Item 2, Item 3, and Item 5 with 3.83, while Item 1 and Item 6 have the highest mean of 4.00. The composite mean score of 3.77 (Strongly Agree) indicates strong and consistent consensus among IT experts that ConservAte is a highly usable, user-centric system.

Table 50. ISO 25010 Usability Evaluation of ConservAte

Indicators	Mean	SD	Interpretation
1. Appropriateness Recognizability	4.00	0.00	Strongly Agree
2. Learnability	3.83	0.41	Strongly Agree
3. Operability	3.83	0.41	Strongly Agree
4. User Error Protection	3.50	0.55	Strongly Agree
5. User Engagement	3.83	0.41	Strongly Agree
6. Inclusivity	4.00	0.00	Strongly Agree
7. User Assistance	3.67	0.52	Strongly Agree
8. Self- Descriptiveness	3.50	0.55	Strongly Agree
Composite Mean	**3.83**	**0.41**	**Strongly Agree**

Legend: *3.25 - 4.00 = Strongly Agree, 2.50 - 3.24 = Agree, 1.75 - 2.49 = Disagree, 1.0 - 1.74 = Strongly Disagree*

Table 51 presents the evaluation results for the "Reliability" of ConservAte, assessed against the ISO 25010 quality model. This component evaluates the system's capacity to sustain a defined performance level under diverse and potentially faulty conditions. With the results from lowest to highest mean, Item 1 with 3.50, Item 3 with 3.67, Item 2 with 3.83, and Item 4 with 4.00. The composite mean score of 3.75 (Strongly Agree) indicates a strong consensus among IT experts that ConservAte is a robust and dependable system.

Table 51. ISO 25010 Reliability Evaluation of ConservAte

Indicators	Mean	SD	Interpretation
1. Faultlessness	3.50	0.84	Strongly Agree
2. Availability	3.83	0.41	Strongly Agree
3. Fault Tolerance	3.67	0.52	Strongly Agree
4. Recoverability	4.00	0.00	Strongly Agree
Composite Mean	**3.75**	**0.44**	**Strongly Agree**

Legend: *3.25 - 4.00 = Strongly Agree, 2.50 - 3.24 = Agree, 1.75 - 2.49 = Disagree, 1.0 - 1.74 = Strongly Disagree*

Table 52 presents the evaluation results for the "Security" of ConservAte, assessed against the ISO 25010 quality model. This component is responsible for safeguarding information and data, preventing unauthorized access or modification while ensuring reliable access for authorized users. Results show Item 3 has the lowest mean with a score of 3.50, followed by Item 4 and Item 6 with a corresponding mean of 3.67, Item 2 with 3.83, while both Item 1 and Item 5 have the highest mean with 4.00. The composite mean score of 3.78 (Strongly Agree) indicates a strong consensus among IT experts, demonstrating their confidence in the system's core control and security functionalities.

Table 52. ISO 25010 Security Evaluation of ConservAte

Indicators	Mean	SD	Interpretation
1. Confidentiality	4.00	0.00	Strongly Agree
2. Integrity	3.83	0.41	Strongly Agree
3. Non - Repudiation	3.50	0.55	Strongly Agree
4. Accountability	3.67	0.52	Strongly Agree
5. Authenticity	4.00	0.00	Strongly Agree
6. Resistance	3.67	0.52	Strongly Agree
Composite Mean	**3.78**	**0.33**	**Strongly Agree**

Legend: *3.25 - 4.00 = Strongly Agree, 2.50 - 3.24 = Agree, 1.75 - 2.49 = Disagree, 1.0 - 1.74 = Strongly Disagree*

Table 53 presents the evaluation results for the "Maintainability" of ConservAte, assessed against the ISO 25010 quality model. This component measures the effectiveness and efficiency with which developers can modify the system to correct errors, meet new requirements, or ease future maintenance. Item 3, Item 4, and Item 5 have the same mean score of 3.50 while Item 1 and Item 2 also have the same mean score of 3.67. The composite mean score of 3.57 (Strongly Agree) indicates a strong consensus among IT experts that the system's codebase is well-structured for long-term usage. The high scores across all items, particularly in Modularity and Reusability, confirm that the system is designed to be easily analyzed, modified, and tested.

Table 53. ISO 25010 Maintainability Evaluation of ConservAte

Indicators	Mean	SD	Interpretation
1. Modularity	3.67	0.52	Strongly Agree
2. Reusability	3.67	0.52	Strongly Agree
3. Analyzability	3.50	0.55	Strongly Agree
4. Modifiability	3.50	0.55	Strongly Agree
5. Testability	3.50	0.55	Strongly Agree
Composite Mean	**3.57**	**0.54**	**Strongly Agree**

Legend: *3.25 - 4.00 = Strongly Agree, 2.50 - 3.24 = Agree, 1.75 - 2.49 = Disagree, 1.0 - 1.74 = Strongly Disagree*

Table 54 presents the evaluation results for the "Flexibility" of ConservAte, assessed against the ISO 25010 quality model. This component assesses the system's capacity for effective and efficient transfer between different hardware, software, or operational environments. The composite mean score of 3.58 (Strongly Agree) indicates a strong consensus among IT experts that ConservAte is a highly adaptable and future-proof system. The

results are particularly strong for Installability (3.83) being the highest, while the Replaceability being the lowest with the mean score of 3.33, demonstrating expert confidence in the system.

Table 54. ISO 25010 Flexibility Evaluation of ConservAte

Indicators	Mean	SD	Interpretation
1. Adaptability	3.50	0.55	Strongly Agree
2. Scalability	3.67	0.52	Strongly Agree
3. Installability	3.83	0.41	Strongly Agree
4. Replaceability	3.33	0.52	Strongly Agree
Composite Mean	**3.58**	**0.50**	**Strongly Agree**

Legend: *3.25 - 4.00 = Strongly Agree, 2.50 - 3.24 = Agree, 1.75 - 2.49 = Disagree, 1.0 - 1.74 = Strongly Disagree*

Table 55 shows what IT experts thought about the "Safety" of the ConservAte system. For an app that deals with food, safety is about making sure users are protected from potential harms. The experts strongly agreed (with an average score of 3.60) that the system is designed responsibly with user well-being in mind. The high scores across all items, particularly for Operational Constraint, Hazard Warning, and Safe Integration (all 3.67) while both Risk Identification and Safe Fail got the lowest with the mean score of 3.50.

Table 55. ISO 25010 Safety Evaluation of ConservAte

Indicators	Mean	SD	Interpretation
1. Operational Constraint	3.67	0.52	Strongly Agree
2. Risk Identification	3.50	0.55	Strongly Agree
3. Fail Safe	3.50	0.55	Strongly Agree
4. Hazard Warning	3.67	0.52	Strongly Agree
5. Safe Integration	3.67	0.52	Strongly Agree
Composite Mean	**3.60**	**0.53**	**Strongly Agree**

Legend: *3.25 - 4.00 = Strongly Agree, 2.50 - 3.24 = Agree, 1.75 - 2.49 = Disagree, 1.0 - 1.74 = Strongly Disagree*

Table 56 summarizes all the responses of IT experts using the ISO 25010 to test and evaluate the functionalities, performance, reliability, security, safety, maintainability, flexibility, usability, and compatibility of the ConservAte web system. Performance Efficiency had the highest mean of 3.83. It emphasizes the composite mean of 3.71, indicating that they strongly agree that all areas of the platform are functional and working as they are designed to, that the purpose has been achieved, and that the software quality meets standards, making it a trustworthy, reliable, and capable system.

Table 56. ISO 25010 Components Evaluation of ConservAte

Indicators	Mean	SD	Interpretation
Functional Suitability	3.78	0.44	Strongly Agree
Performance Efficiency	3.83	0.41	Strongly Agree
Compatibility	3.75	0.46	Strongly Agree
Usability	3.77	0.35	Strongly Agree
Reliability	3.75	0.44	Strongly Agree
Security	3.78	0.33	Strongly Agree
Maintainability	3.57	0.54	Strongly Agree
Flexibility	3.58	0.50	Strongly Agree
Safety	3.60	0.53	Strongly Agree
Overall Mean	**3.71**	**0.44**	**Strongly Agree**

Legend: *3.25 - 4.00 = Strongly Agree, 2.50 - 3.24 = Agree, 1.75 - 2.49 = Disagree, 1.0 - 1.74 = Strongly Disagree*

CHAPTER 5

SUMMARY, CONCLUSION, AND RECOMMENDATION

This chapter summarizes the study's results and key findings. It provides an overview of outcomes related to the system's development, functionality, and user acceptance, generates conclusions, and recommends improvements.

Summary

ConservAte: A web-based platform for food sales and the redistribution of surplus and near-expiry food products, with a Geographic Information System (GIS), was developed to help users locate nearby sellers, markets, and shops, resulting in an efficient redistribution of surplus and near-expiry products to minimize food waste. The application allows sellers to add their products, with a real-time countdown feature that tracks each item's expiration date, enabling buyers to easily browse products, compare prices and expiration dates, and select products based on their needs and preferences. The platform also integrates shop availability, showing whether the store is open based on the day of the week and time, allowing buyers to know when the best time to pick up their orders is.

The project adopted the Agile Scrum Model, a flexible, collaborative methodology that enables continuous feedback and improvement throughout the development process. By using this model, the developers were able to quickly address the users' requirements and provide incremental updates based on feedback. The technologies employed for development included HTML as the foundation of the platform's front-end interface, JavaScript for

backend functionality, CSS for designing and building cross-platform, mobile-responsive interfaces, and PHP for database management and communication with the MySQL database. These tools enabled the development of a robust, user-friendly web-based platform that could scale and provide real-time services.

The approach was implemented to assess the system's effectiveness using the Technology Acceptance Model (TAM) and ISO 25010 Software Quality Product. The platform system was presented to 56 respondents, where 50 of them, comprising customers, sellers, and administrators, specifically answered the TAM survey, while the remaining 6, consisting of IT professionals, assessed the functionalities and other important areas of the ConservAte system using the ISO 25010. Most responses ranged only from 4.00 to 3.25, which can be interpreted as Strongly Agree. This indicates that most of the users consistently recognized that ConservAte helps reduce food waste and improve food accessibility, with the intent to recommend and use the system.

In summary, the developed web-based system successfully addressed ConservAte's purpose of reducing food waste through the redistribution of surplus, near-expiry, and food waste by offering discounts on products and accepting donations. The system provides an efficient, user-friendly, reliable, secure, and inclusive platform for all users, enabling them to purchase products while promoting sustainable consumption.

Conclusion

In conclusion, the ConservAte web-based platform successfully achieved its following objectives:

1. Design and develop a web-based platform, integrated with Geographic Information System (GIS), to support the food sales and the redistribution of near-expiry and surplus food products.

The web-based platform for food sales and redistribution of surplus and near-expiry food products successfully creates an accessible and efficient web-based platform that connects sellers, buyers, and beneficiaries to promote sustainable food distribution. With the integration of Geographic Information System (GIS) technology, the platform emphasizes the location-based services, allowing the customers to locate the nearby seller, markets, and redistribution points with ease and optimize the redistribution routes. This integration not only bridges the gap between the seller and buyers but also improves the efficiency of food redistributions by mapping out the delivery and pickup routes. Which helps to ensure that surplus and near-expiry food products are properly redirected to where they are most needed, may significantly reduce waste and will support communities that face food insecurity.

2. To implement a tracking and analytics system to monitor food waste trends optimize redistribution, and ensure secure data management.

The implementation of a tracking and analytics system was also accomplished. The system’s built-in analytics module records and visualizes important data, including sales performance, food waste statistics, and redistribution patterns. This capability empowers users to make data-driven decisions that enhance business performance, improve inventory management, and optimize resource allocation. For administrators, the

analytics tools provide valuable insights into system operations and user behavior, which can inform future strategies to reduce waste and promote sustainable food practices.

3. To evaluate the web-based system's abilities using ISO 25010 and Technology Acceptance Model (TAM) for Software Quality Standards.

The ConservAte web-based platform delivered reliable, user-friendly, secure, and scalable results based on responses to the TAM and ISO 25010 surveys. Respondents believed that the system is a cost-effective platform whose benefits outweigh the associated transaction fees. Positive user perceptions of the system can be directly interpreted as a strong behavioral intention to adopt it. Users showed a clear willingness to use ConservAte regularly and recommend it to others, confirming its business potential and widespread adoption. The Technology Acceptance Model had a mean of 3.55 and a standard deviation of 0.60, indicating that users strongly agree with the system's acceptance, while the ISO 25010 had a mean of 3.71 and a standard deviation of 0.44, suggesting that the system met all requirements for good-quality software.

In conclusion, the system not only meets high technical standards but also successfully fosters user acceptance, necessary to achieve its core mission of reducing food waste.

Recommendation

The following recommendations, derived from the project's findings, outline several impactful technological opportunities. To enhance the web-based platform for selling and redistributing surplus and near-expiry food, the

researchers propose exploring the following specific improvements based on the study's insights:

1) It is recommended to integrate a camera-based image recognition feature that automatically identifies whether a product is fresh, near expiry, or expired, enhancing accuracy and efficiency in product monitoring and management.

2) To increase accessibility and user engagement, it is recommended to develop native iOS and Android applications.

3) To optimize the redistribution of near-expiry products, integrating a dynamic pricing algorithm is recommended. This system could automatically adjust product discounts based on factors such as the time until expiration and current demand, helping sell products faster and further reducing potential waste.

4) For future researchers, it is recommended to enhance the Geographic Information System (GIS) integration to include real-time delivery tracking, allowing users to monitor and identify delivery locations accurately and efficiently.

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LIST OF APPENDICES

LETTER	Title
A	Planning and Requirement Analysis Phase
B	Source Code
C	Communication Letters and Approvals
D	Publishable Paper
E	Plagiarism Report
F	Grammarly Report
G	Curriculum Vitae

APPENDIX

A

Planning and Requirement Analysis Phase

Republic of the Philippines
Laguna State Polytechnic University
College of Computer Studies
Province of Laguna

ISO 25010 Survey Questionnaire

Dear Respondents,

We are students from Laguna State Polytechnic University – Los Baños Campus under the College of Computer Studies, currently conducting a capstone project entitled **" CONSERVATE: A WEB-BASED PLATFORM FOR FOOD SALES AND REDISTRIBUTION OF SURPLUS AND NEAR-EXPIRY FOOD PRODUCTS."**

As part of our research, we would like to gather your feedback regarding the system's functionality, usability, performance, and overall quality. Your opinions will help assess the system's effectiveness and ensure it meets the intended goals of sustainable food waste management and food security enhancement.

Please evaluate the statements below by checking the appropriate number where, **four (4) = Strongly Acceptable; three (3) = Acceptable; two (2) = Unacceptable; one (1) = Strongly Unacceptable**. Be reassured that any of your information that you provide will be treated with utmost confidentiality (RA 10173. Data Privacy Act) and will be used for academic purposes only.

Thank you for your cooperation.

Regards:
Bigtas, Linory L.
De Vera, Charles I.
Meltran, Jhon-Rick L.

Name:_____

Category of the Respondent:

☐ IT Experts

Questions		Ratings			
FUNCTIONAL SUITABILITY		**4**	**3**	**2**	**1**
1. Functional Completeness	All modules (user, market, message, data analytics, product, notification, map) are working as expected.				
2. Functional Correctness	The system provides accurate and reliable information for fresh, surplus, and near-expiry food listings, donations, and food redistribution processes.				
3. Functional Appropriate-	The system's functionalities are appropriate and aligned with its goal of food waste				

ness	reduction and redistribution.				
PERFORMANCE EFFICIENCY		**4**	**3**	**2**	**1**
4. Time Behavior	ConservAte responds quickly when browsing available foods in the marketplace, making donations, or accessing user profiles.				
5. Resource Utilization	The system runs efficiently even on devices with lower specifications and does not experience slowdowns during use.				
6. Capacity	The system can handle a high volume of data and multiple concurrent users without performance issues.				
COMPATIBILITY		**4**	**3**	**2**	**1**
7. Co-existence	The system performs its required functions efficiently while sharing resources with other applications, without negatively affecting their performance.				
8. Interoperability	The system interacts effectively with other applications (map services) to exchange and use information.				
USABILITY		**4**	**3**	**2**	**1**
9. Appropriate-ness Recognizabilty	The users can easily recognize and understand the functions and buttons of the system to complete tasks efficiently.				
10. Learnability	The system is user-friendly and requires minimal training for new users to learn and operate it.				
11. Operability	The system can be operated with ease, allowing users to perform functions with minimal effort.				
12. User Error Protection	The system helps prevent errors and enables users to identify and correct incorrect inputs.				
13. User Engagement	The system's interface is visually appealing and encourages continued use through clear feedback and an intuitive layout.				
14. Inclusivity	The system is designed to be accessible to a diverse range of users, including those with varying levels of technical skill.				
15. User Assistance	The system provides adequate help and				

	documentation (e.g., tooltips, FAQs, contact support) to assist users when needed.				
16. Self-Descriptiveness	Each step and function within the system is clear, and the system provides appropriate feedback for user actions.				
RELIABILITY		**4**	**3**	**2**	**1**
17. Faultlessness	The system operates without unexpected errors or failure over time.				
18. Availability	The system is accessible and operational whenever needed.				
19. Fault Tolerance	The system continues to function properly even when some errors occur.				
20. Recoverability	The system restores performance and information (e.g., user data, transaction history) after a failure.				
SECURITY		**4**	**3**	**2**	**1**
21. Confidentiality	Only authorized users can view or use confidential data (e.g., personal information, payment details) in the system.				
22. Integrity	The system ensures all data remains accurate and unaltered during storage and transmission.				
23. Non-Repudiation	Users cannot deny the actions or transactions (e.g., completing a purchase, confirming a donation) they perform in the system.				
24. Accountability	All system activities (logins, transactions) can be traced back to specific users.				
25. Authenticity	Only registered and verified users can access system features.				
26. Resistance	The system is resistant to unauthorized access and common security threats (e.g., brute-force attacks, SQL injection).				
MAINTAINABILITY		**4**	**3**	**2**	**1**
27. Modularity	The system is easy to maintain due to its modular design.				
28. Reusability	System functions or components can be reused after modification.				

29. Analyzability	Errors and issues in the system are easily identified and diagnosed.				
30. Modifiability	The system's components can be modified without affecting others.				
31. Testability	The system can be effectively tested across different computers and setups.				
FLEXIBILITY		**4**	**3**	**2**	**1**
32. Adaptability	The system can be adapted to meet changing user needs or business rules.				
33. Scalability	The system can be scaled up (e.g., to support more users or a wider geographic area) with minimal changes to its infrastructure.				
34. Installability	The system can be installed and uninstalled in its target environments (e.g., web servers) without conflict.				
35. Replaceability	The system can be replaced by another similar software solution with minimal effort and disruption.				
SAFETY		**4**	**3**	**2**	**1**
36. Operational Constraint	The system includes operational constraints to promote safety, such as clear labeling for near-expiry/expired products and disclaimers.				
37. Risk Identification	The system helps identify potential risks, such as transactions involving expired products or unreliable users.				
38. Fail Safe	In case of a system failure, it defaults to a safe state that prevents the distribution of unsafe or inaccurately labeled food products.				
39. Hazard Warning	The system provides clear warnings and information about the potential hazards of consuming near-expiry or expired food products.				
40. Safe Integration	The system safely integrates with external services (e.g., maps for delivery) without creating new safety risks for users.				

Republic of the Philippines
Laguna State Polytechnic University
College of Computer Studies
Province of Laguna

Technology Acceptance Model (TAM) Survey Questionnaire

Dear Respondents,

We are students from Laguna State Polytechnic University-Los Baños Campus under the College of Computer Studies conducting a capstone project entitled **CONSERVATE: A WEB-BASED PLATFORM FOR FOOD SALES AND REDISTRIBUTION OF SURPLUS AND NEAR-EXPIRY FOOD PRODUCTS.** We would like to know your opinion on the acceptability of the project.

Please evaluate the statements below by checking the appropriate number where, **four (4) = Strongly Agree; three (3) = Agree; two (2) = Disagree; one (1) = Strongly Disagree**. Be reassured that any of your information that you provide will be treated with utmost confidentiality (RA 10173. Data Privacy Act) and will be used for academic purposes only.
Thank you for your cooperation.

Regards:
Bigtas, Linory L.
De Vera, Charles I.
Meltran, Jhon-Rick L.

Name:_____

Category of the Respondent:

- ○ Food Establishment Representative
- ○ Non-Public Food Vendors
- ○ Public Market Food Vendors
- ○ Consumers
- ○ NGOs / Food Aid Organization Representatives

	Ratings			
Perceived Usefulness (PU) of Conservate	**4**	**3**	**2**	**1**
1. ConservAte helps users minimize food waste through its marketplace.				
2. ConservAte improves access to affordable surplus and fresh food.				
3. ConservAte simplifies the process of redistributing near-expiry items.				
4. ConservAte contributes to local food security through its features.				
5. ConservAte supports sustainable consumption and purchasing behavior.				
Perceived Ease of Use (PEOU) of ConservAte	**4**	**3**	**2**	**1**
1. It is easy to learn how to use the ConservAte application.				
2. Navigating ConservAte's interface and features is user-friendly.				
3. Users can quickly locate food products or vendor listings.				

4. Interacting with vendors and placing orders through ConservAte is simple.				
Personal Innovativeness (PI) with ConservAte	**4**	**3**	**2**	**1**
1. Users are excited to try innovative platforms like ConservAte.				
2. Users prefer exploring new technologies that address food waste.				
3. Users are interested in discovering new e-commerce features in ConservAte.				
4. Users proactively try digital solutions that promote sustainability.				
Social Influence (SI) on ConservAte	**4**	**3**	**2**	**1**
1. Users' family or peers encourage using ConservAte.				
2. Community trends or groups influence users to adopt ConservAte.				
3. Seeing others shop or donate on ConservAte inspires users to participate.				
4. Users are more likely to try ConservAte if recommended by trusted people.				
Information Awareness (IA) on ConservAte	**4**	**3**	**2**	**1**
1. Users are aware that ConservAte promotes food waste reduction.				
2. Users know how ConservAte helps communities access food affordably.				
3. Users are informed about ConservAte's unique features and tools.				
4. Users are aware of ConservAte's ongoing efforts and impact.				
Perceived Cost (PC) of ConservAte	**4**	**3**	**2**	**1**
1. ConservAte offers reasonable pricing for the value it delivers.				
2. Users find ConservAte's marketplace to be cost-effective.				
3. The benefits of using ConservAte outweigh any associated fees.				
4. Users believe that ConservAte provides fair pricing for surplus items.				
Willingness to Adopt (WTA) ConservAte	**4**	**3**	**2**	**1**
1. Users are willing to use ConservAte for buying and redistributing food.				
2. Users plan to continue using ConservAte regularly.				
3. Users are willing to recommend ConservAte to others.				
4. Users would encourage others to use ConservAte for sustainable food practices.				

USER MANUAL

All Users

1. Landing Page - The Landing Page serves as the system's main entry point, giving an overview of its features and purpose. Users can hover the "(A) Account" button and click the "(B) Login" or "(C) Sign Up" buttons to access or create an account, with buttons highlighting on hover for easy navigation and a smooth start to using the system.

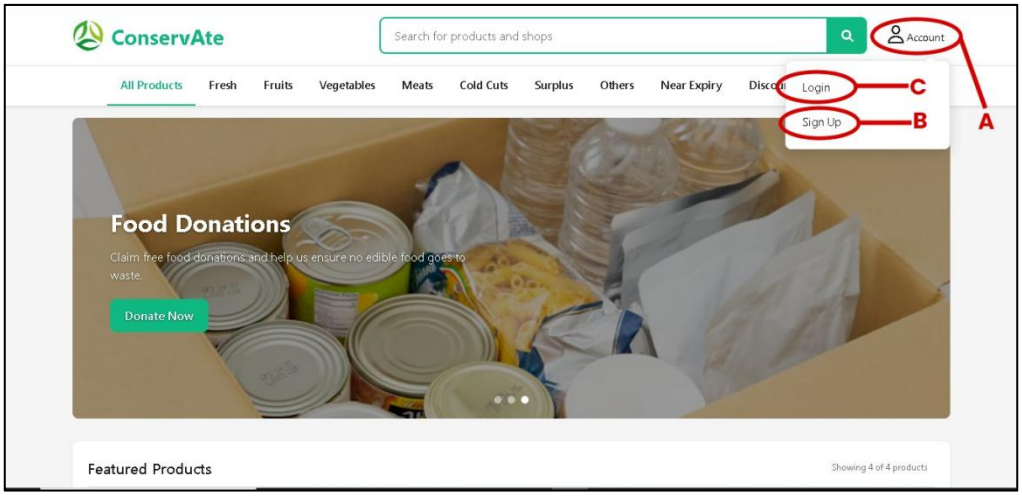


Figure 1. ConservAte's Landing Page

2. Login Page - The Login Page allows users to securely access their accounts. It includes a username field (A) and a password field (B) where users enter their credentials, followed by a login button (C) to submit them. Below these, the "Forgot Password" link (D) helps users recover their accounts if they forget their password, while the "Create Account" link (E) directs new users to the registration page.

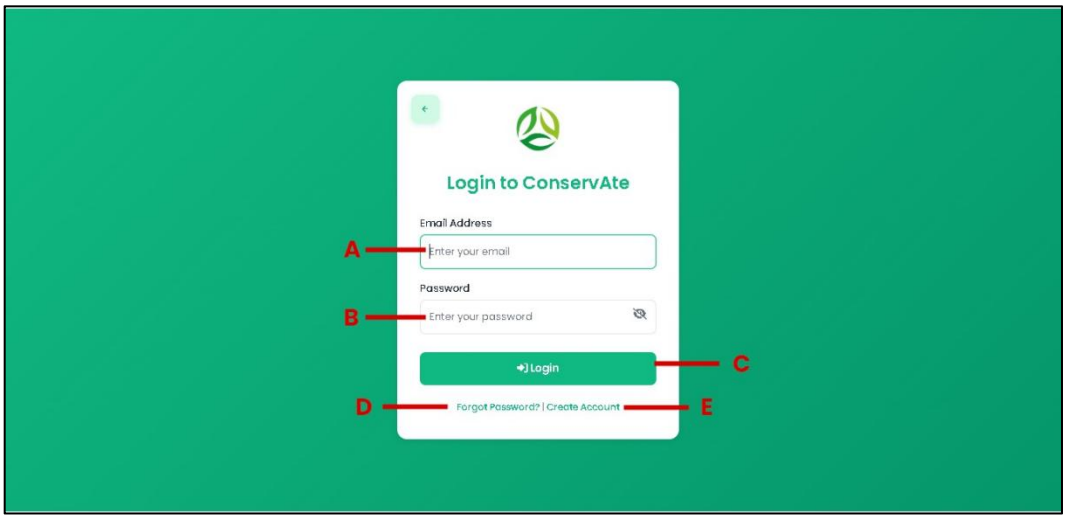


Figure 2. ConservAte's Login Page

User Manual for Customer

1. Signup Page - The Signup Page allows users to create a new account by filling out the required text fields. If applicable, they may also enter a referral code (A) provided by a friend. For the agreement and terms (B), users can click the links to

view the details before proceeding. After completing the form, they can click the signup button (C) to register. If they already have an account, they can click the “Login here” link (D), and if they wish to register as a seller instead, they can click the “Signup as Seller” option (E).

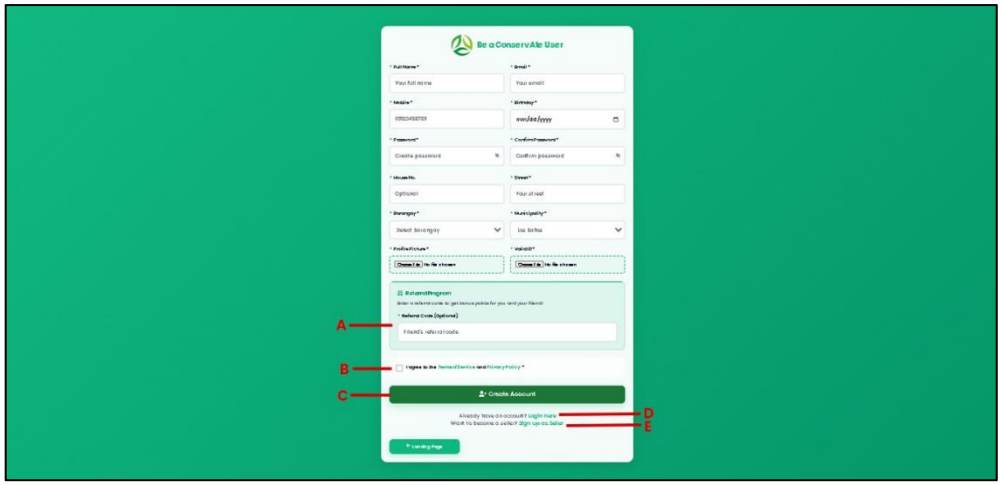


Figure 3. Customer’s Signup Page

2. Customer Dashboard - The Customer Dashboard includes a notification button (A) where users can view all alert notifications related to their account and activities. It also displays all available products in ConservAte, each shown in a product card (B) that includes the product’s image, status, price, and stock quantity. Each card has a message button (C) for contacting the seller, an add to cart button (D) for saving the item, and a buy now button (E) for direct purchase, with displayed customer reviews.

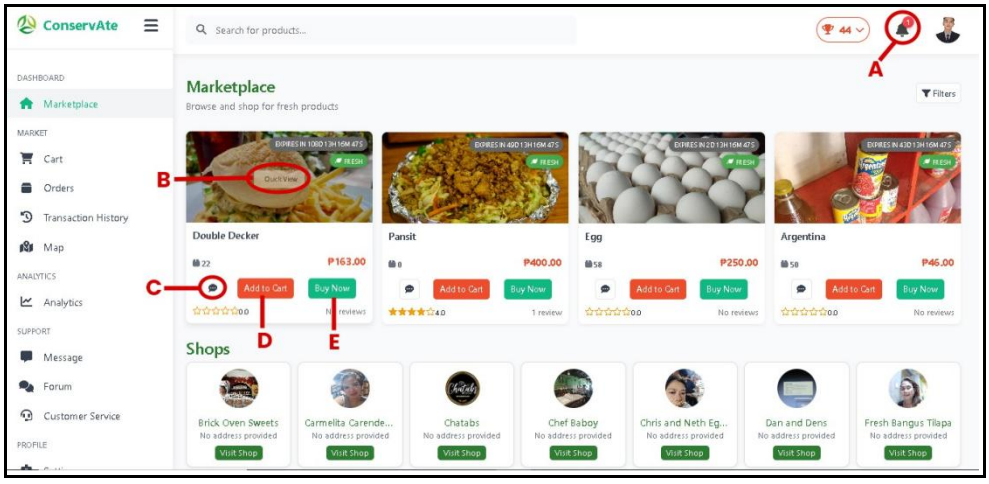
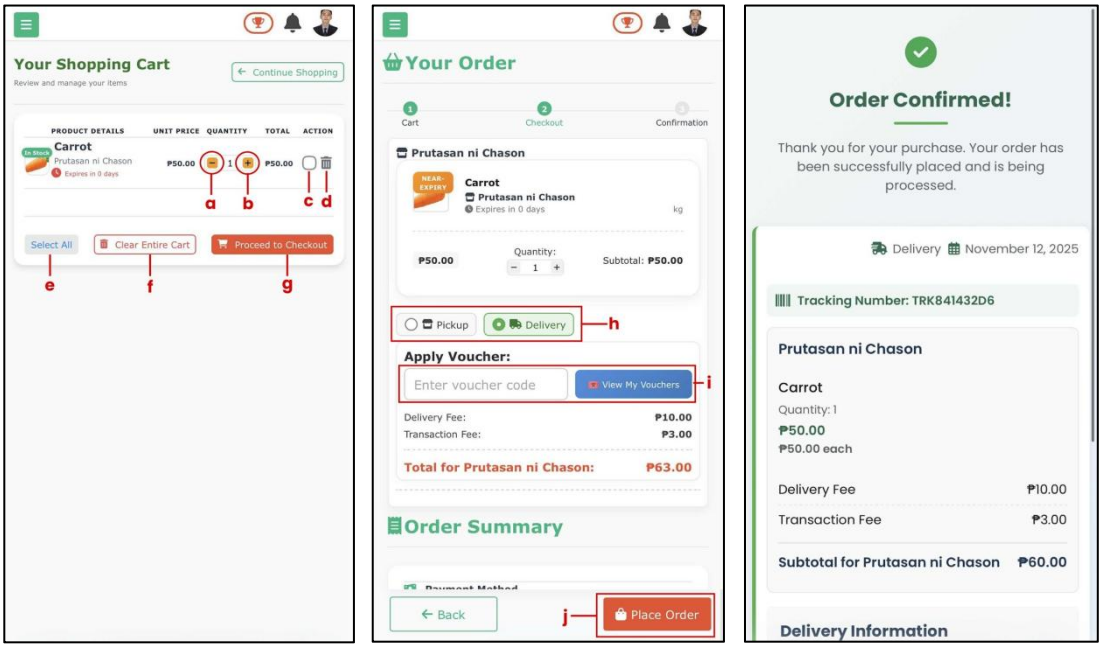


Figure 4. Customer Dashboard Page

3. Cart Page - The Cart section (A) displays a table showing all products the customer has added to their cart, including product details and quantity. Users can click the “-” button (a) to reduce the desired quantity and the “+” button (b) to increase it. Beside each item, the total of the order is shown along with actions, including a checkbox (c) to select the item and a delete button (d) to remove it. Below the table, there are buttons to select all items (e), clear the entire cart (f), and proceed to checkout (g). In the Checkout section (B), users can view the products they have proceeded with from the cart. They can choose their preferred mode of delivery: Pickup or Delivery (h). Below that, there is an option to apply a voucher (i) and review the detailed price breakdown. By clicking the Place Order button (j), users will be directed to the Order Details section (C), where they can see the complete

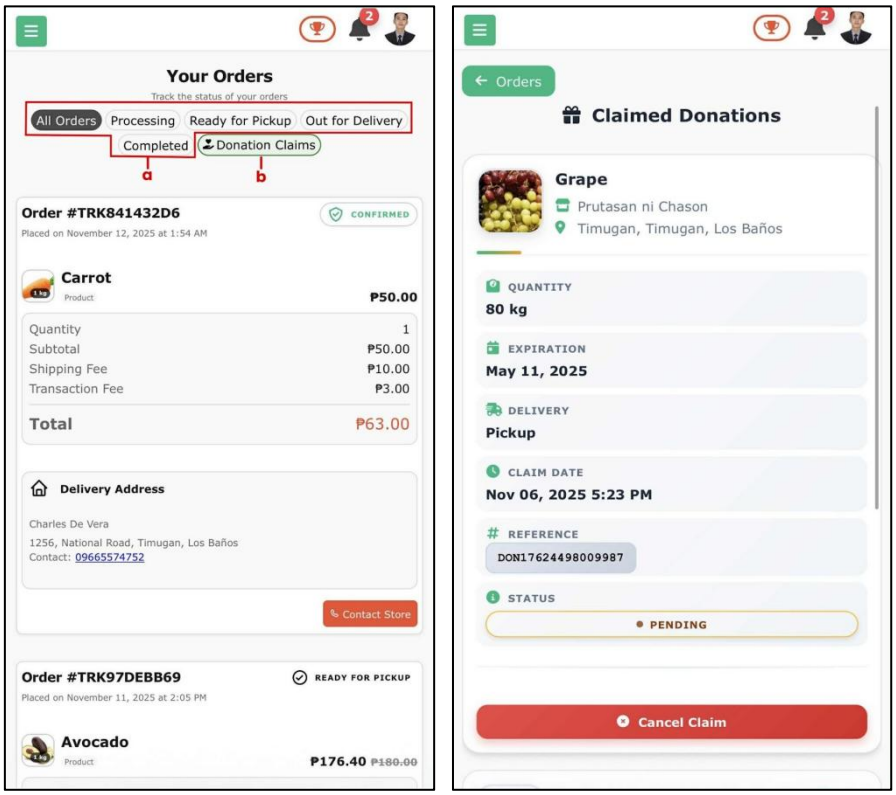
summary of their order, including products, quantities, total cost, delivery information, and order status.



(A) (B) (C)

Figure 5. Customer Cart Page

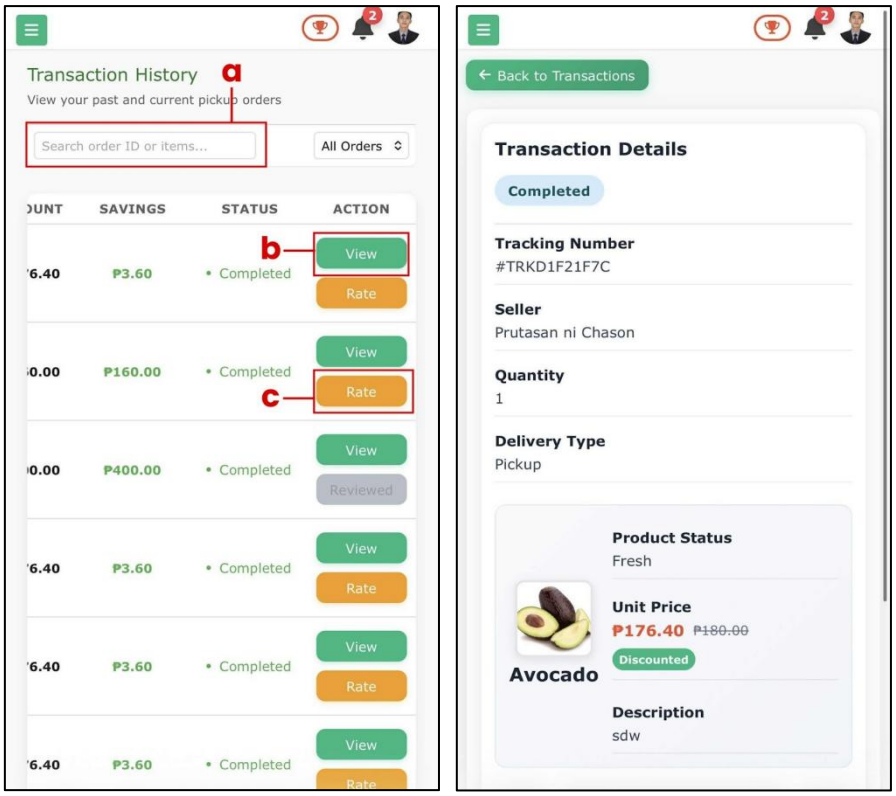
4. Orders Page - In the Orders section, users can view the status of their placed orders and filter them according to their status (a). Additionally, there is a Donations section (b) where users can see the status of their claimed donations (B), allowing them to track both purchases and donations in one place.



(A) (B)

Figure 6. Customer Order Page

5. Transaction Page - In the Transaction page (A), users can search (a) and filter their completed orders by status, whether successful or cancelled. They can also View (B) the details of each order and Rate (c) each order or product based on their experience.



(A) (B)

Figure 7. Customer Transaction Page

6. Map Page - In the Map Section, customers can see nearby stores by clicking on the sellers registered in the system (a), allowing them to locate and explore available sellers in their area.

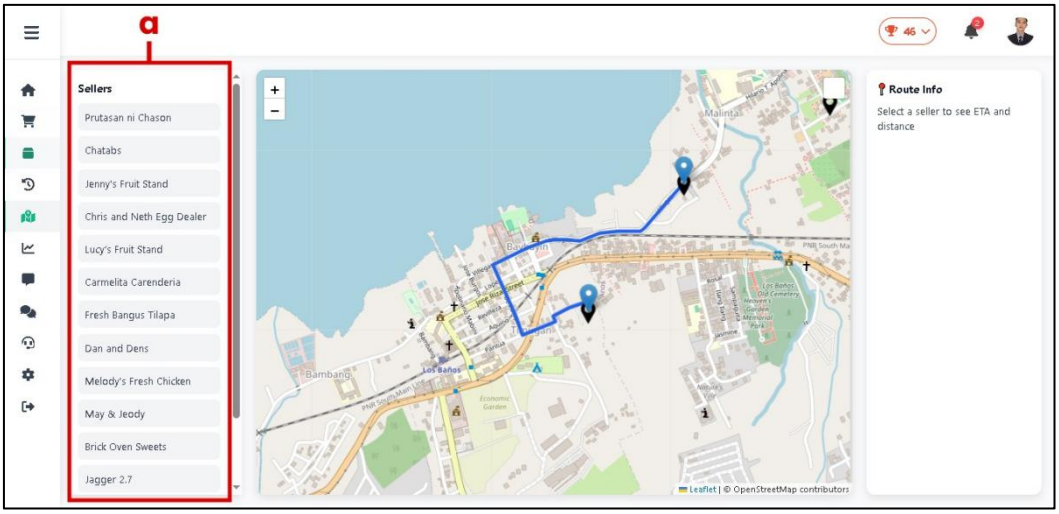


Figure 8. Customer Map Page

7. Analytics Page - The Customer Analytics displays data analytics that provide insights into the customer's activities and transactions. Users can change the data range (a) to view analytics by daily, weekly, or monthly intervals. If they wish to save or download the data, they can click the export button (b) to export the report for their reference.

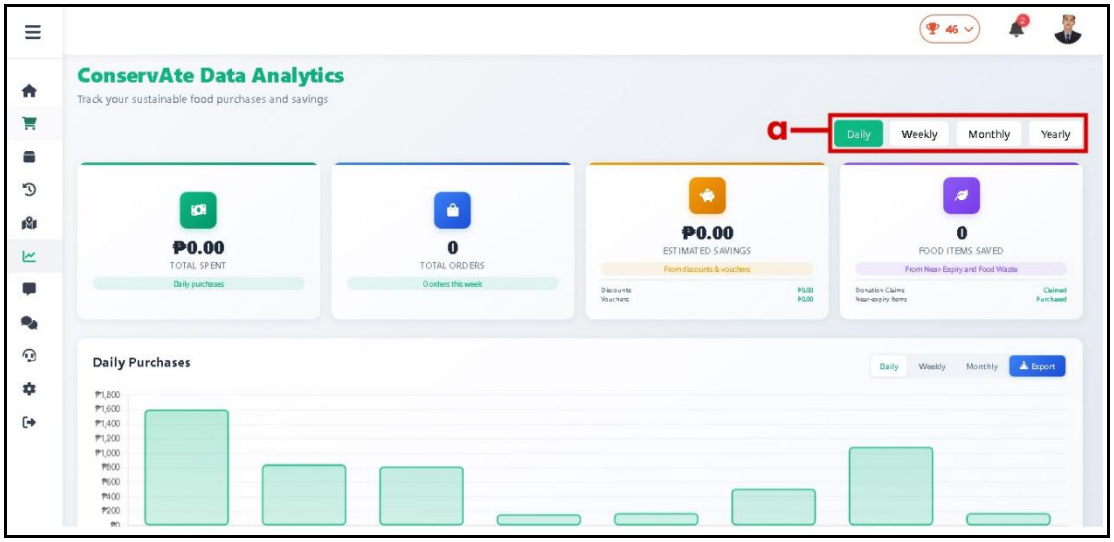


Figure 9. Customer Analytics Page

8. Message Page - In the Message section, users can select a recipient and send a message by clicking the send button (a), allowing them to communicate directly with sellers or support. The message will only display when the customer clicked the conversation head of the seller or when redirected from the message feature from the product cards.

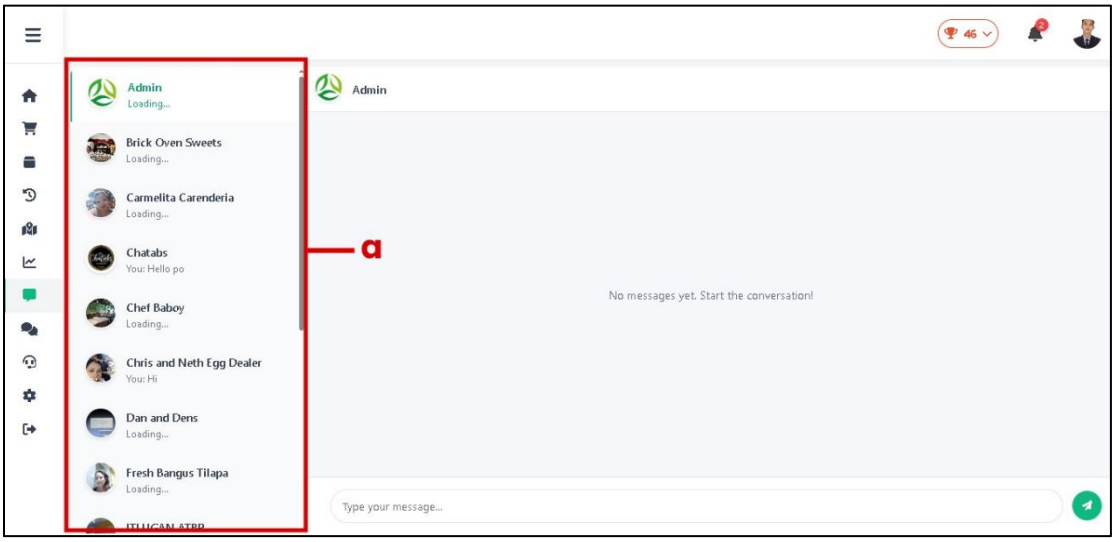
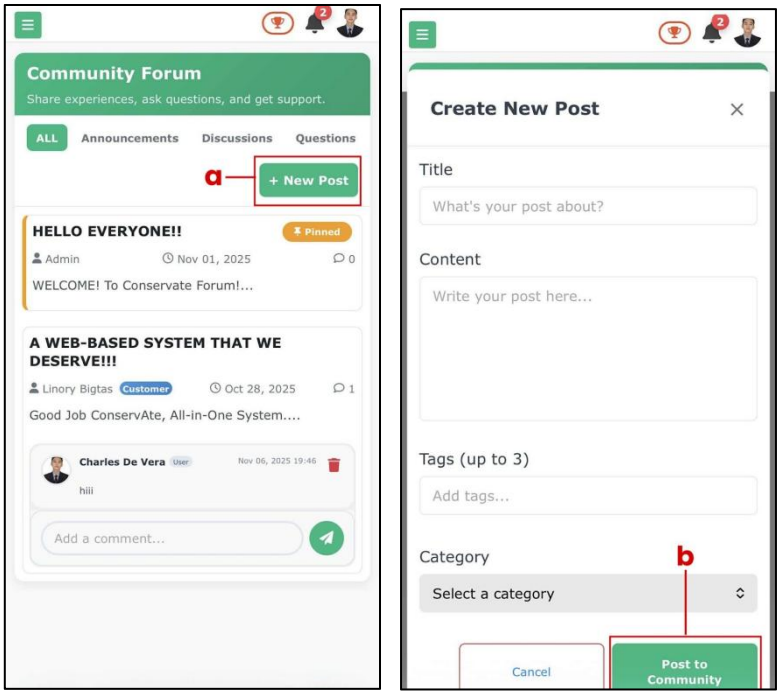


Figure 10. Customer Message Page

9. Forum Page - In the Forum section (A), customers can see ongoing discussions within ConservAte or create their own posts by clicking the New Post button (a). In the Create Post modal (B), they can fill out the required text fields and click the Post button (b) to publish their post. They can also edit and delete the posted content in the forum.



(A) (B)

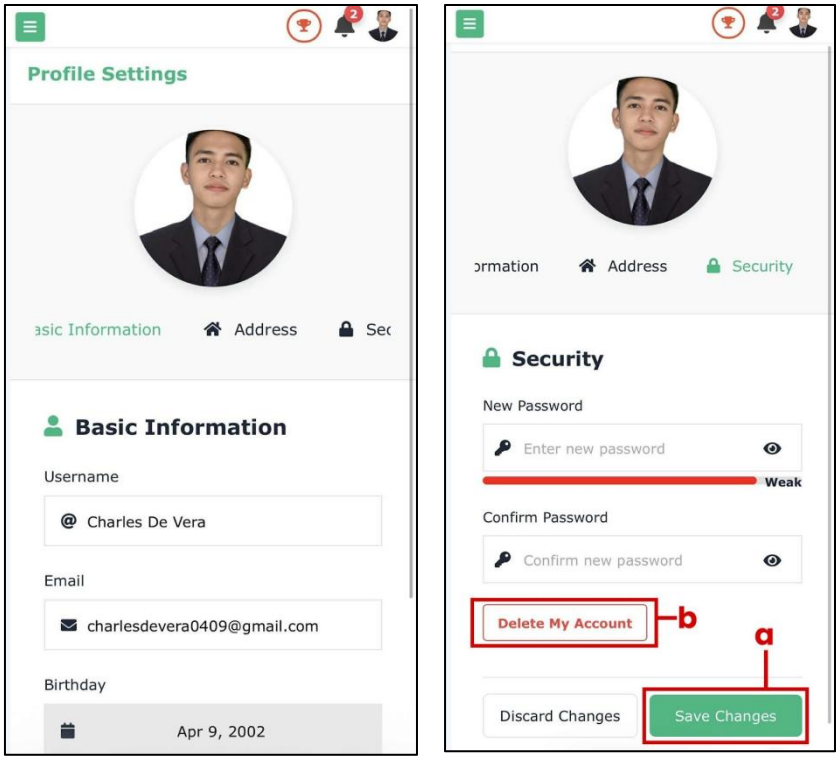
Figure 11. Customer Forum Page

10. Customer Service Page - In the Customer Service page, users can fill out a form to submit their reports or inquiries and click the Submit button (a) to send their message to the support team or admin of the system for them to able to respond quickly in the user concerns and reports.



Figure 12. Customer Service Page

11. Settings Page - In the Setting page, users can view their profile details (A) and update their security key (B) by clicking the Confirm Change button (a). They also have the option to delete their account (b) if needed. With these features, the modifications of personal information of the customers will be easier to do and check their data for flawless and smooth transactions.



(A) (B)

Figure 13. Customer Settings Page

12. Reward Page - In the Rewards page, users can click the Reward button (a) to view redeemable vouchers (A) and redeem them by clicking the Redeem button (b). They can also copy (c) their referral codes to share with friends for future use.

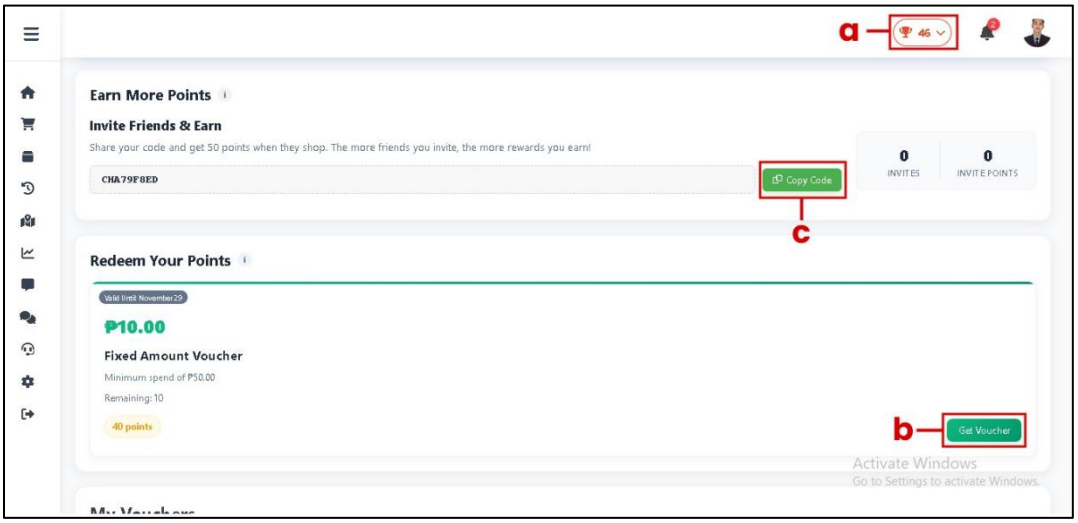


Figure 13. Customer Reward Page

User Manual For Seller

1. Seller Dashboard - The Seller Analytics displays data analytics that provide insights into the seller's activities and transactions. Users can change the data range (a) to view analytics by daily, weekly, or monthly intervals. If they wish to save or download the data, they can click the export button (b) to export the report for their reference.

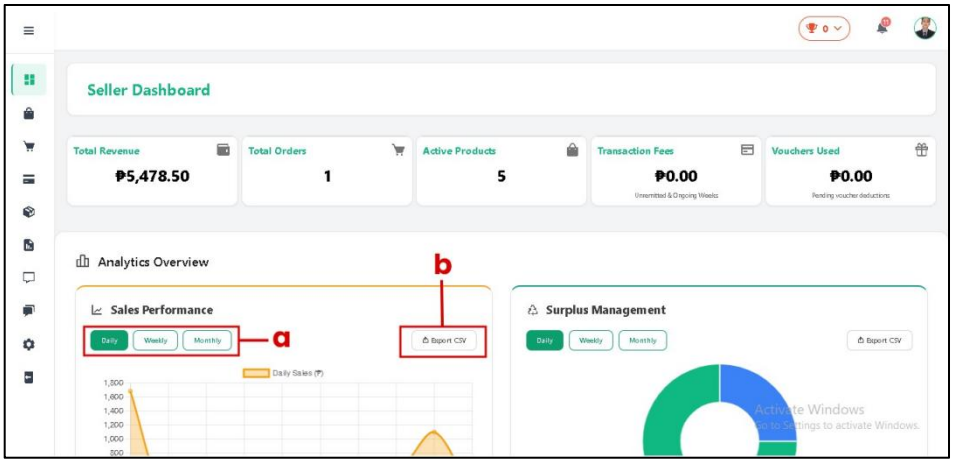


Figure 14. Seller Dashboard Page

2. Manage Products - In the Seller Manage Products section, sellers can add products by filling out the textboxes (a) and clicking the Add button (b). The added products will then appear in the My Products section, where they can view and manage their uploaded products (c).

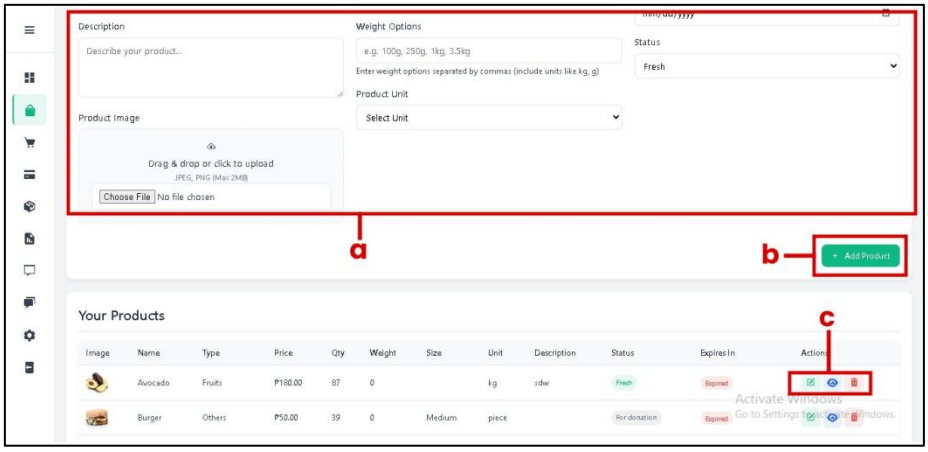


Figure 15. Seller Manage Products Page

3. Seller Manage Orders Page - In the Manage Orders section, sellers can manage orders placed by customers in their shop by updating the status of each order (a) to Confirmed, Out for Delivery/Ready for Pickup, or Complete. They can also print (b) the order details and cancel (c) orders if necessary, same as for Donation(d).

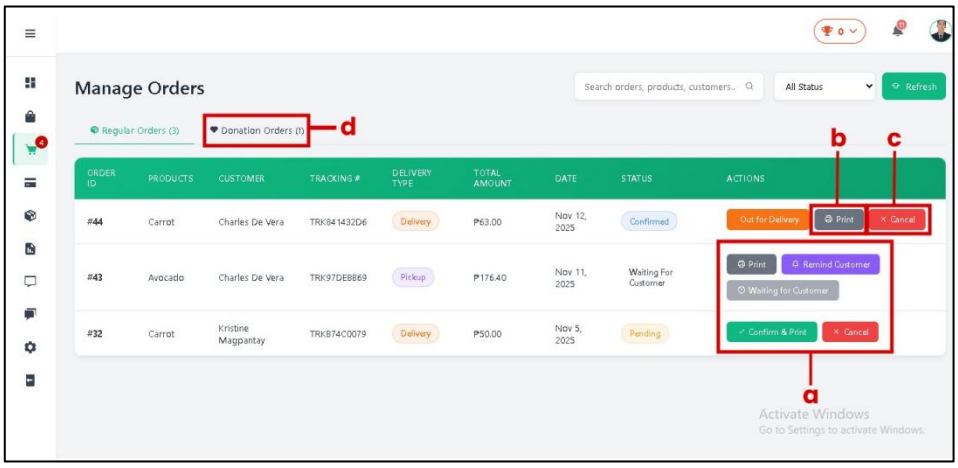


Figure 16. Seller Manage Orders Page

4. Seller Transaction Page - In the Transaction page, users can search (a) and filter their completed orders by status, whether successful or cancelled. They can also print (b) the details of each order.

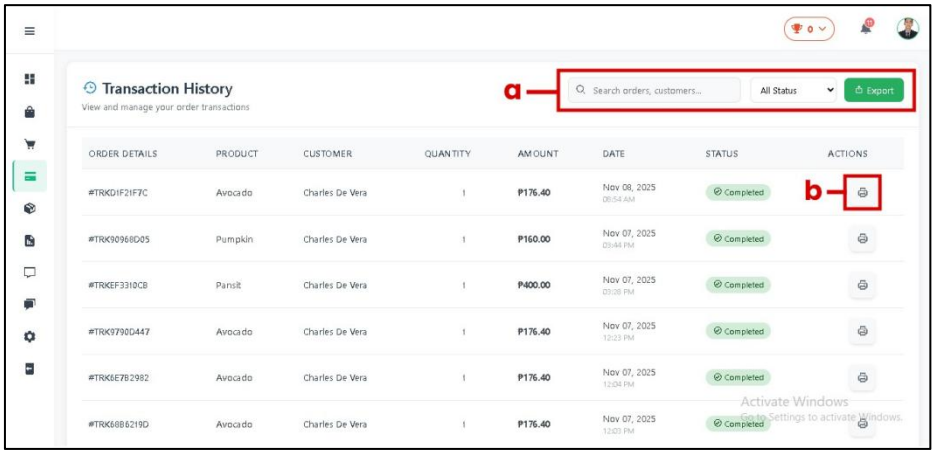


Figure 17. Seller Transaction Page

5. Surplus Management Page - In the Surplus Management section, sellers can monitor the status of their products and choose to donate or apply a discount by clicking the Action button (a).

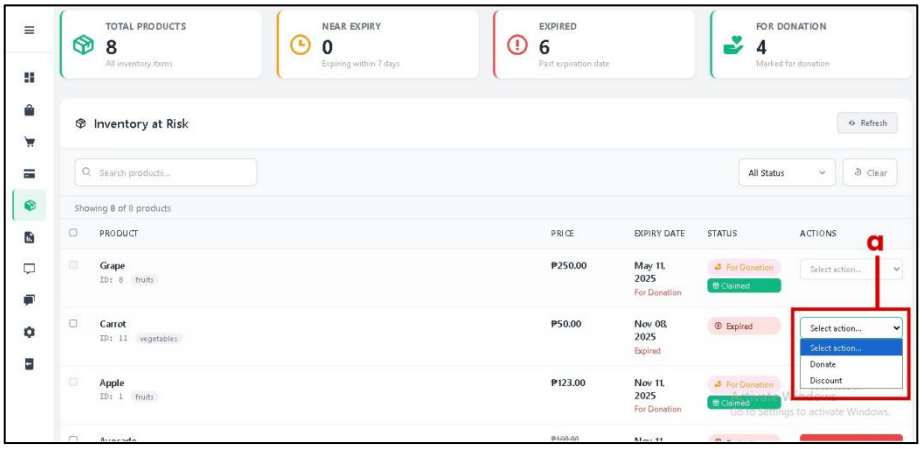


Figure 18. Surplus Management Page

6. Seller Report Page - In the Seller Report page, users can fill out a form to submit their reports or inquiries and click the Submit button (a) to send their message.



Figure 19. Seller Report Page

7. Seller Message Page - In the Message section, users can select a recipient and send a message by clicking the send button (a), allowing them to communicate directly with sellers or support.

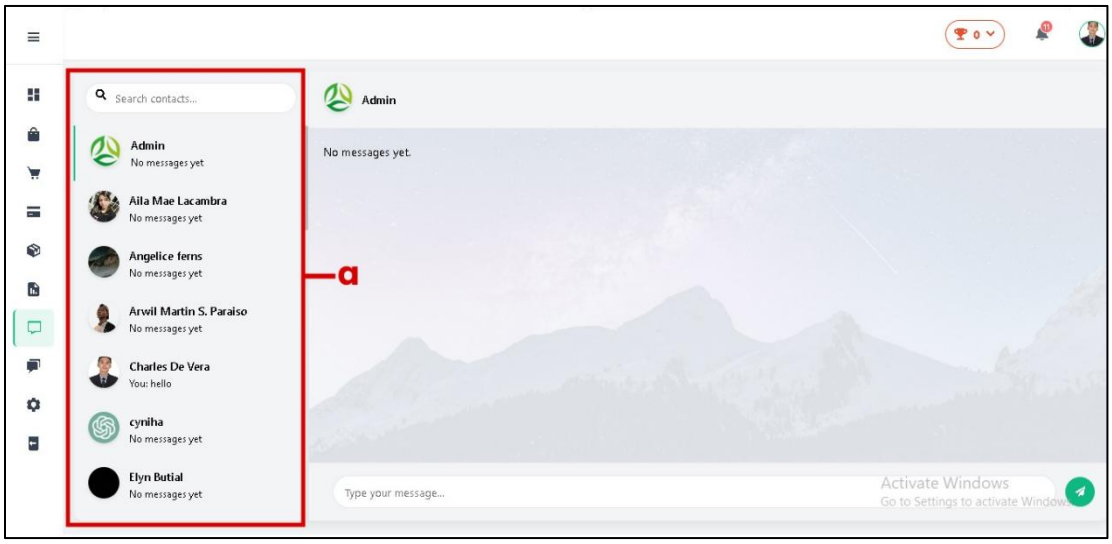
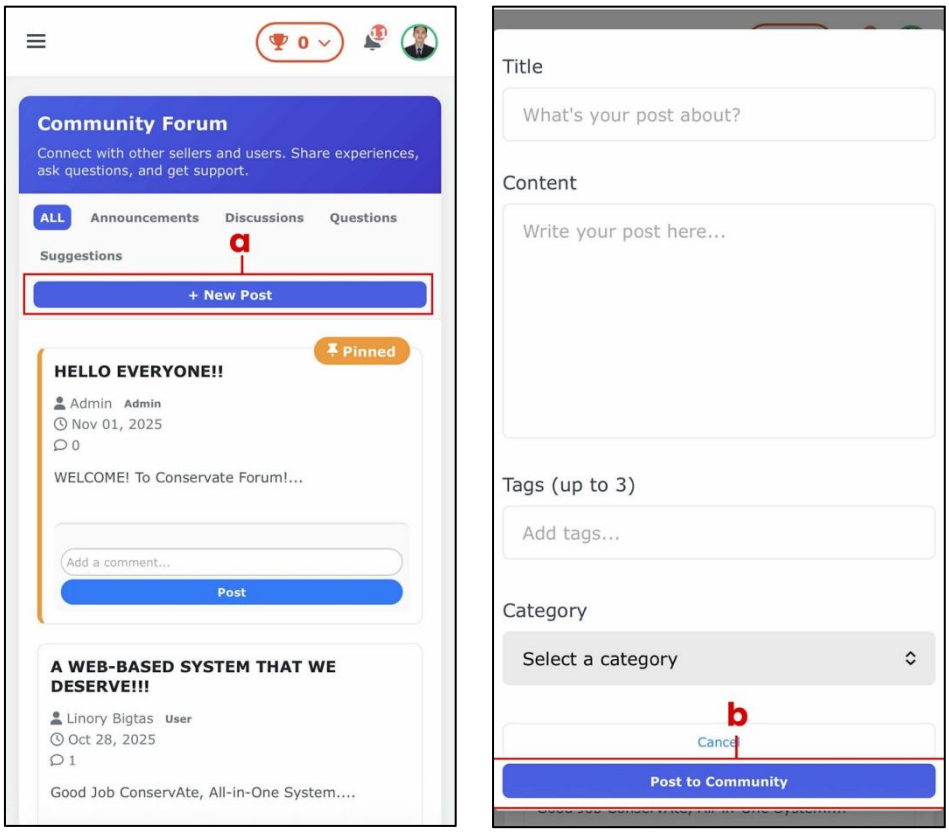


Figure 20. Seller Message Page

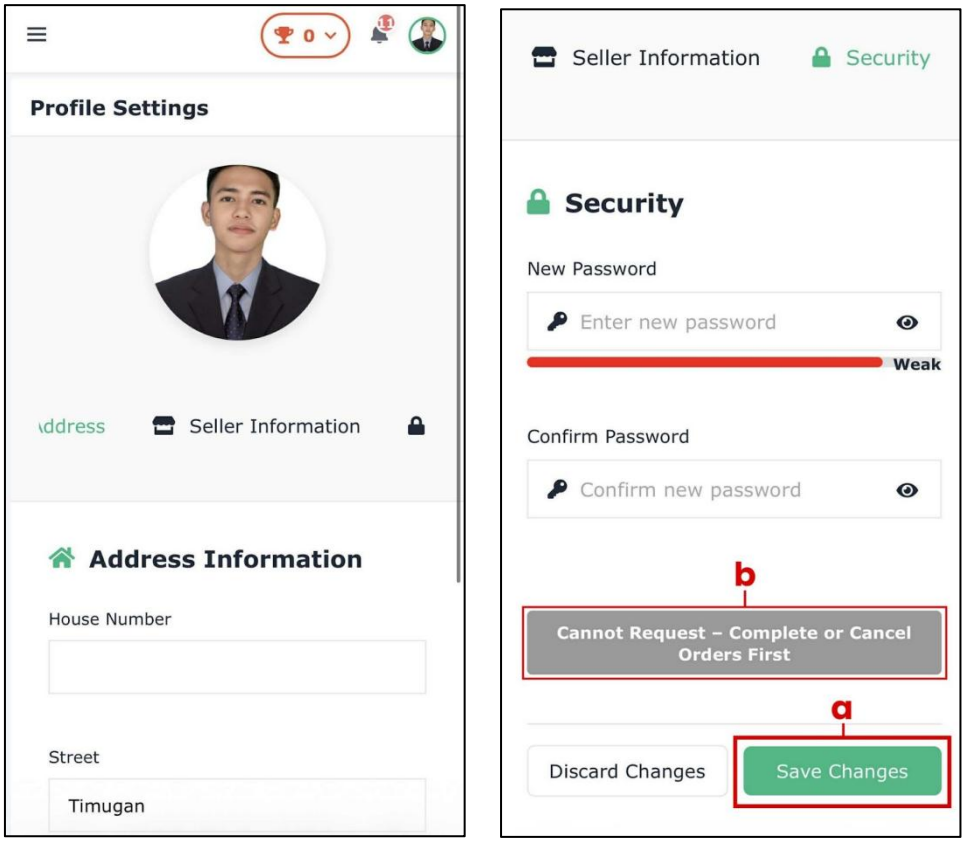
8. Seller Forum Page - In the Forum section (A), customers can see ongoing discussions within ConservAte or create their own posts by clicking the New Post button (a). In the Create Post modal (B), they can fill out the required text fields and click the Post button (b) to publish their post.



(A) (B)

Figure 21. Seller Forum Page

9. Seller Settings Page - In the Setting page, users can view their profile details (A) and update their security key (B) by clicking the Confirm Change button (a). They also have the option to delete their account (b) if needed.



(A) (B)

Figure 22. Seller Settings Page

10. Seller Rewards Page - In the Rewards page, users can click the Reward button (a) to view redeemable vouchers (A) and redeem them by clicking the Redeem button (b). They can also copy (c) their referral codes to share with friends for future use.

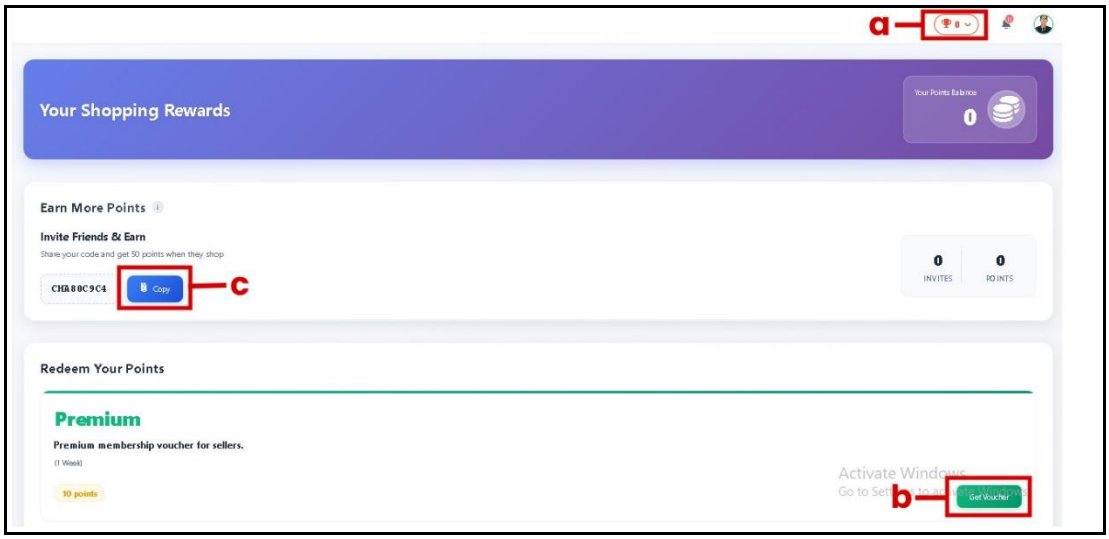


Figure 23. Seller Rewards Page

User Manual for Super Admin

1. Super Admin Dashboard Page - The Super Admin Analytics displays data analytics that provide insights into system activities and transactions. Admin can change the data range (a) to view analytics by daily, weekly, or monthly intervals. If they wish to save or download the data, they can click the export button (b) to export the report for their reference.

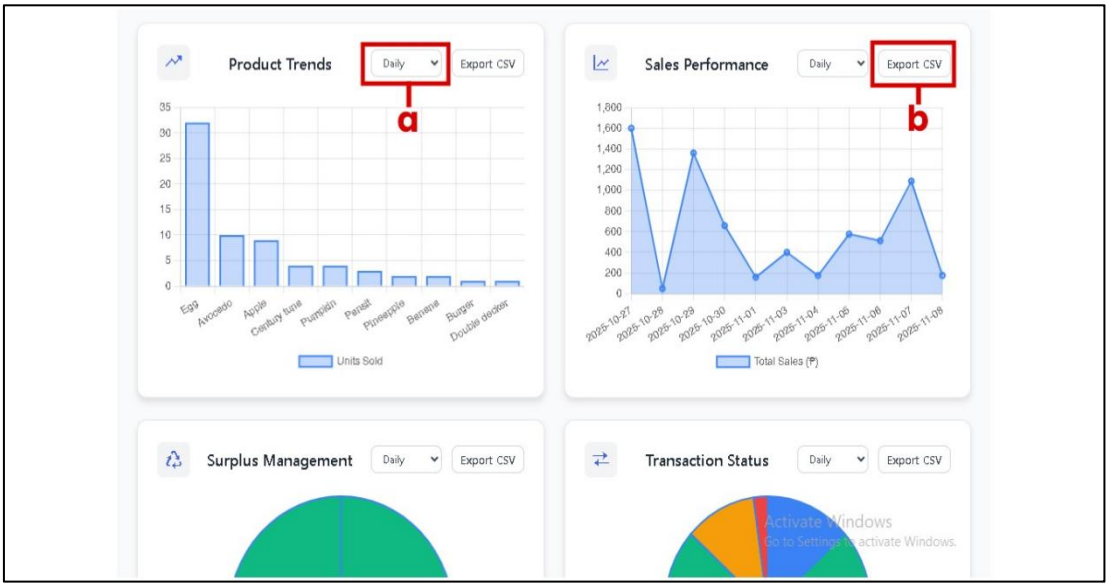


Figure 24. Super Admin Dashboard Page

2. Manage Customers Page - In the Manage Users section, the Super Admin can view all customers signed up in the system. They can view user credentials (a), verify or unverify accounts (b), suspend accounts (c), and deactivate accounts (d) as needed.

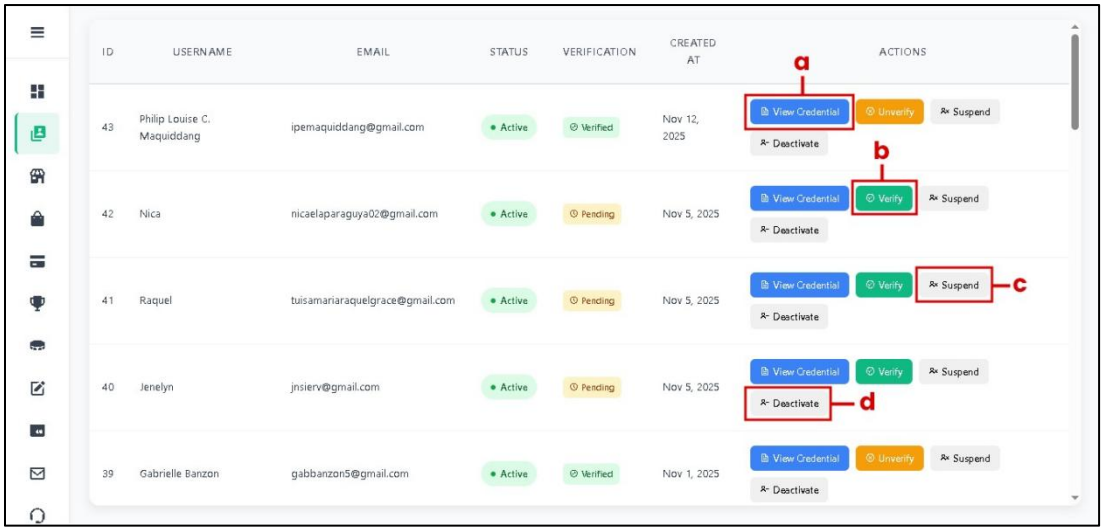


Figure 25. Manage Customers Page

3. Manage Sellers Page - In the Manage Seller section, the Super Admin can view all customers signed up in the system. They can view user credentials (a), verify or unverify accounts (b), suspend accounts (c), and deactivate accounts (d) as needed especially when there are reports from other users due to their policy violations.

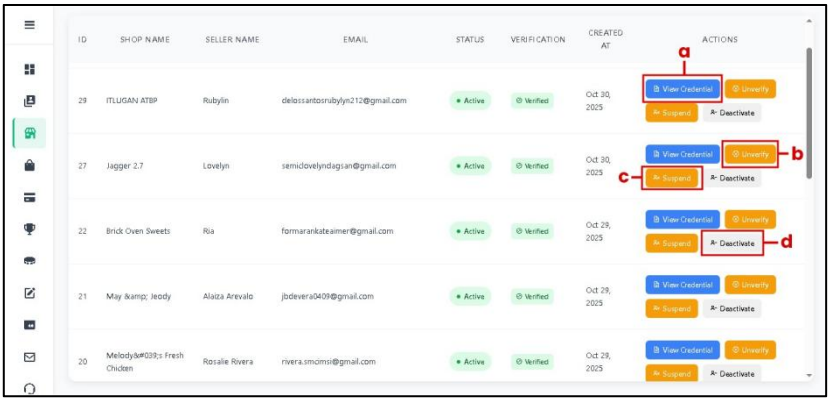


Figure 26. Manage Sellers Page

4. Super Admin Product Management Page - In the Super Admin section, they can view all products from sellers and take desired actions, such as a rejection, or view product details. All sellers can monitor the super admin, the status of their products and choose to donate or apply a discount by clicking the Action button (a).

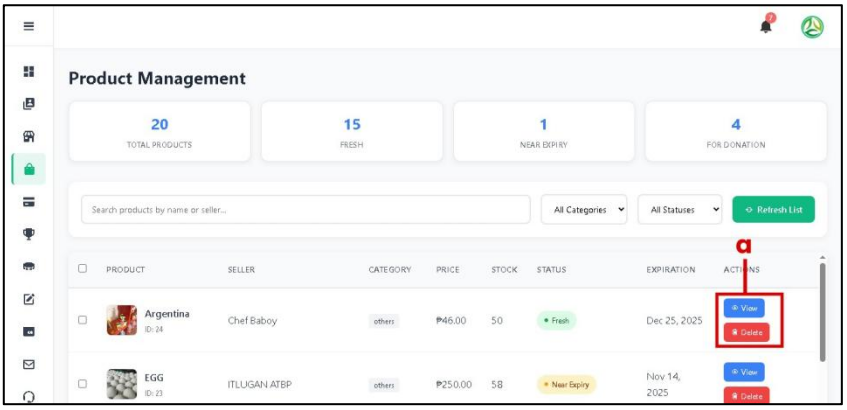
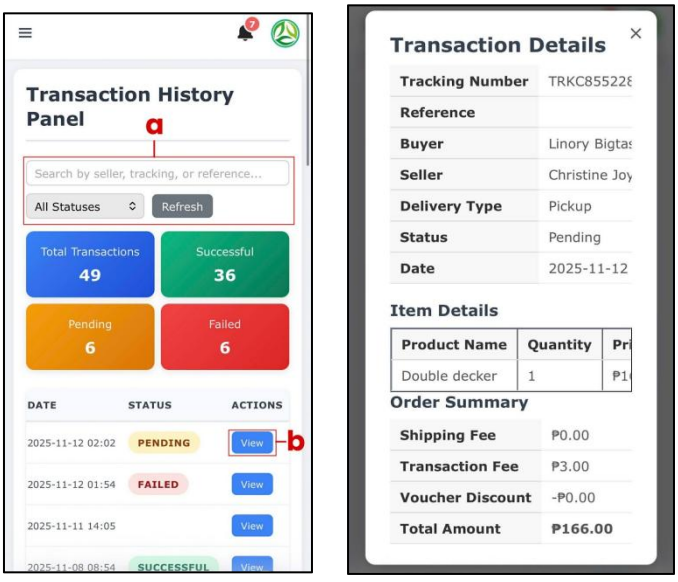


Figure 27. Super Admin Product Management Page

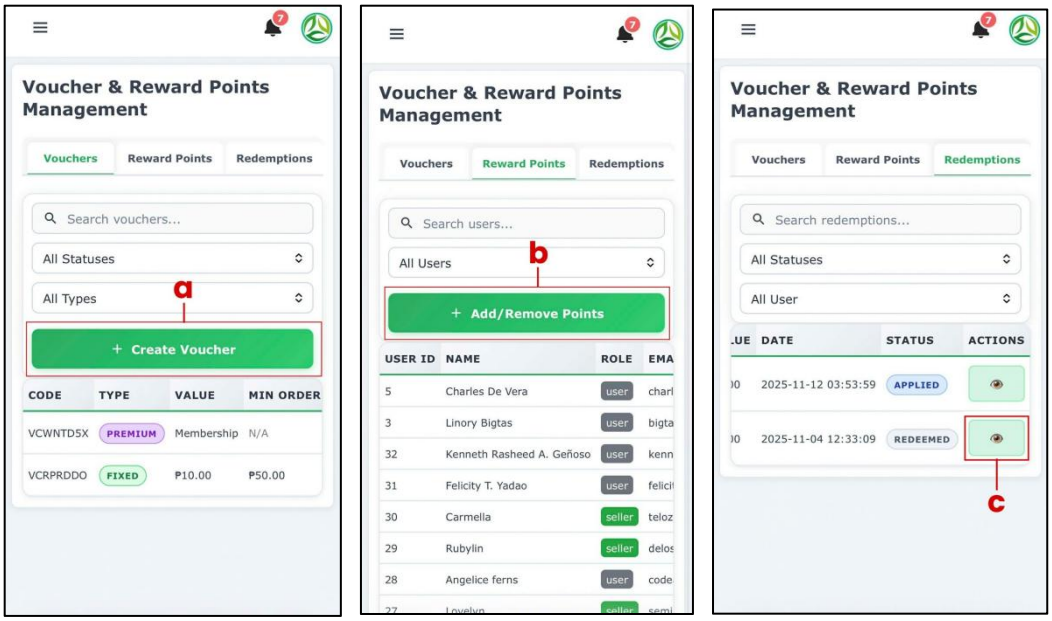
5. Super Admin Transaction Page - In the Transaction page (A), users can search (a) and filter the completed orders by status, whether successful or cancelled. They can also View (B) the details of each order.



(A) (B)

Figure 28. Super Admin Transaction Page

6. Reward Management Page - In the Reward Management section (A), the Super Admin can create vouchers by clicking the Create Voucher button (a). In the Reward Points section (B), they can add or remove points by filling out the given fields and clicking the Confirm button (b). In the Redemptions section (C), they can see which accounts have used the vouchers and view details by clicking the View button (c).



(A) (B) (C)

Figure 29 .Reward Management Page

7. Remit Management Page - In the Remit Management section, the Super Admin can view (a) the remittance status of each seller.

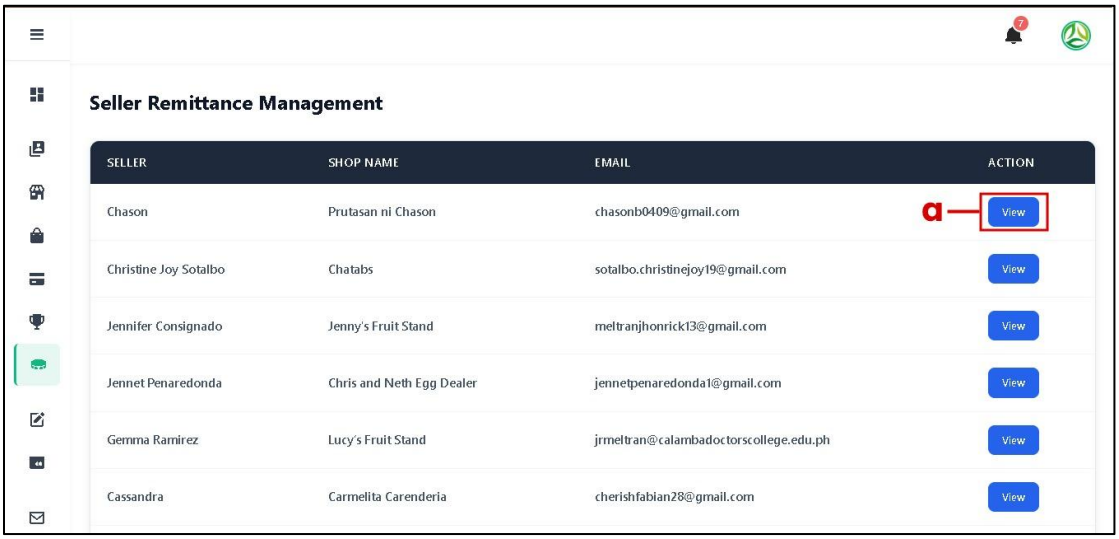


Figure 30. Remit Management Page

8. Forum Content Management Page - In the Content Management section, the Super Admin can view all posts created by users and manage them by issuing a warning (a), deleting (b), or unpinning (c) posts as needed For adding an announcement, it is by clicking the “Announcement” button (d).



Figure 31 .Forum Content Management Page

9. Subscription Management Page - In the Subscriptions section, the Super Admin can view all subscribed premium sellers and take actions by revoking (a) or messaging (b) the accounts.

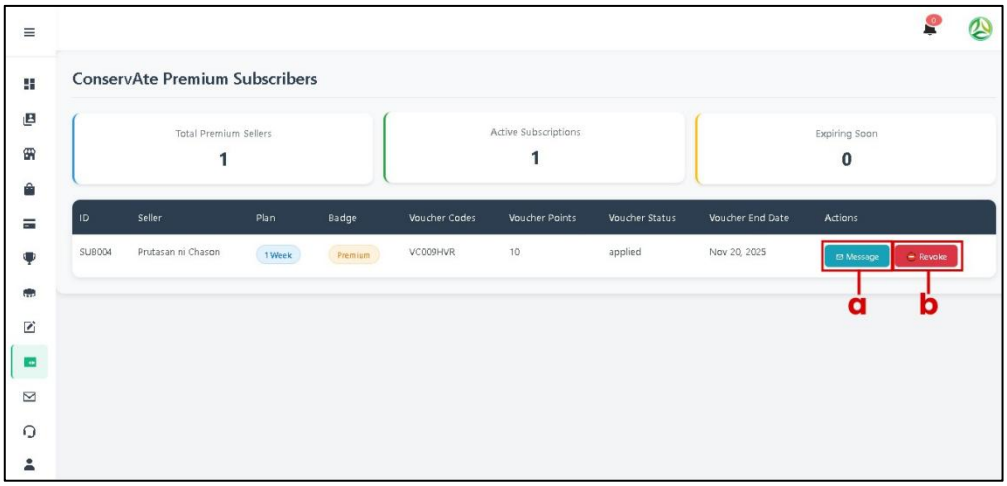


Figure 32. Subscription Management Page

10. Super Admin Message Page - In the Message section, Admin can select a recipient and send a message by clicking the send button (a), allowing them to communicate directly with sellers or support.

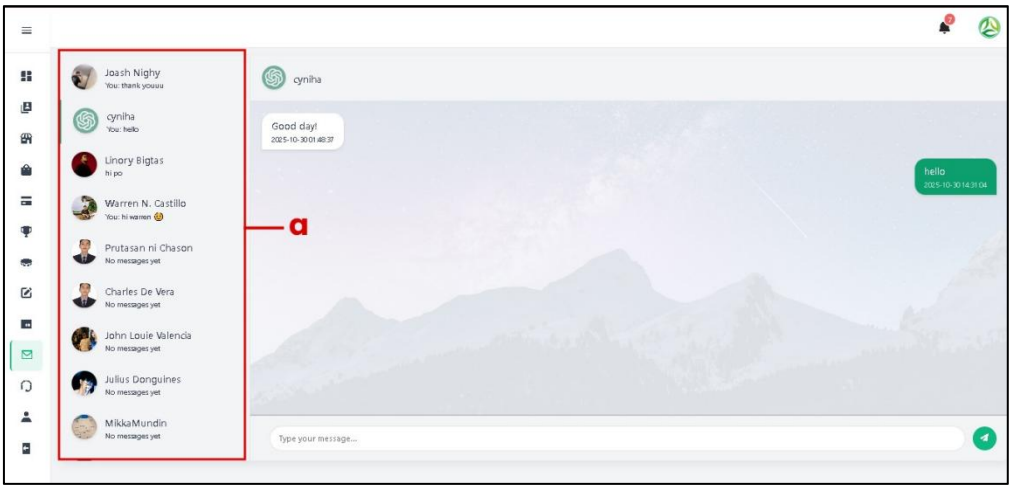


Figure 33. Super Admin Message Page

11. Customer Support Page - In the Customer Support section, the Admin can view all ticket reports from users and sellers and manage them by clicking Resolve (a), Open (b), or Close (c) and view the attachments of images for report tickets (d).

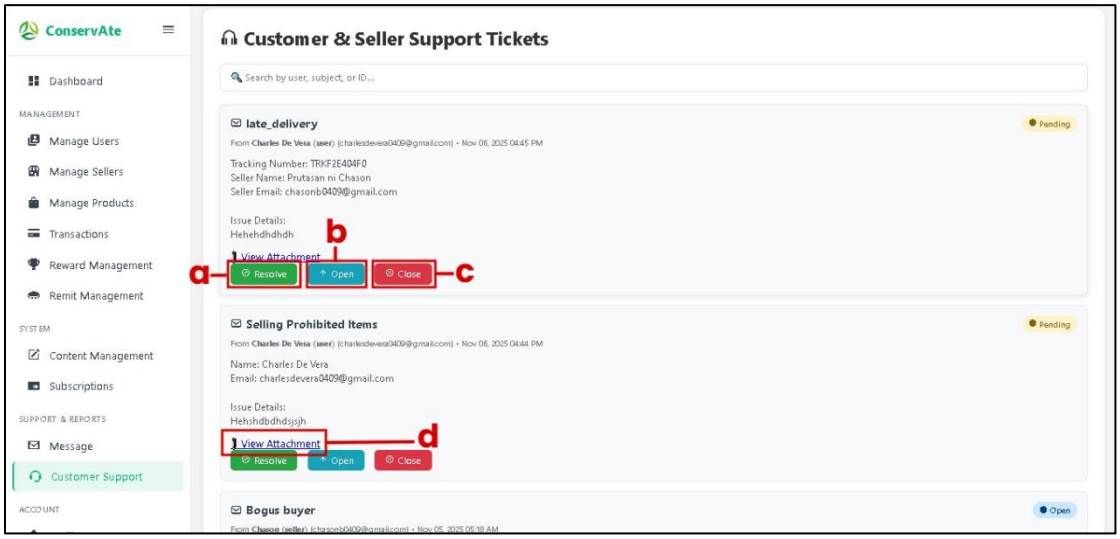


Figure 34. Customer Support Page

12. Super Admin Settings Page - In the Setting page, users can view their profile details and update their security key by clicking the Confirm Save Changes button (a).

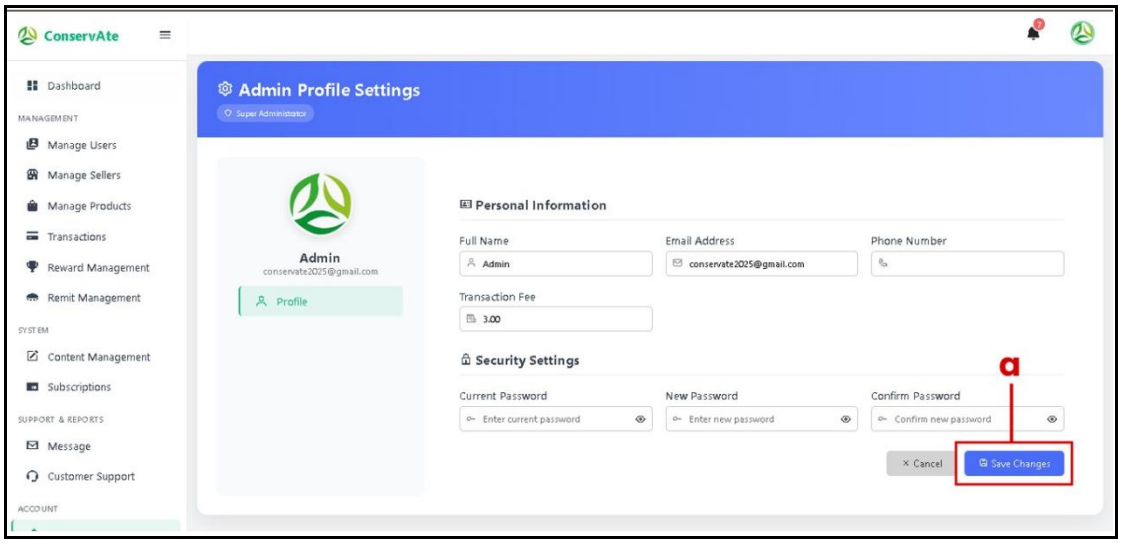


Figure 3. Super Admin Settings Page

APPENDIX

B

Source Code

FLOWCHARTS

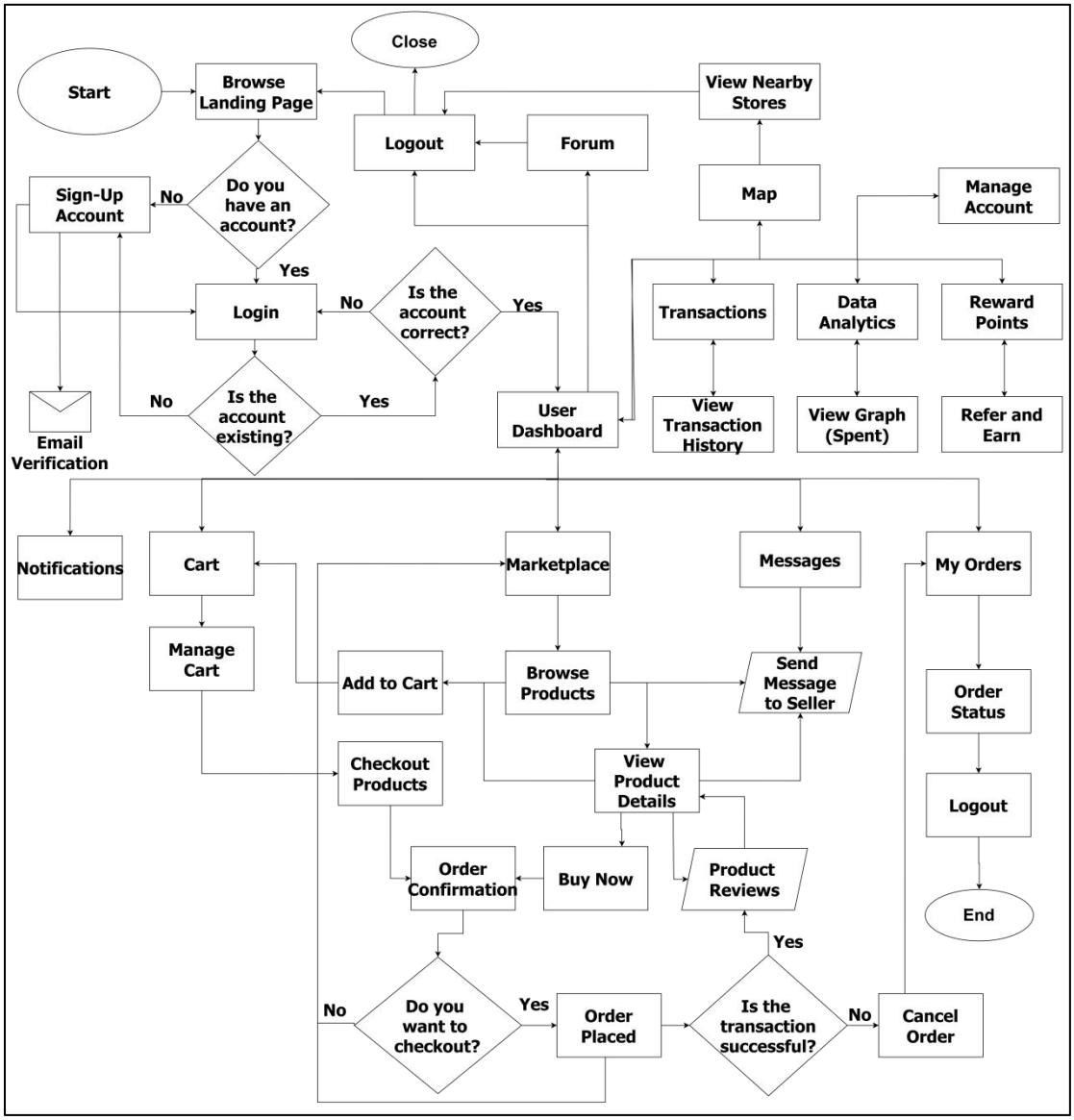


Figure _. Customer Flowchart of Conservate

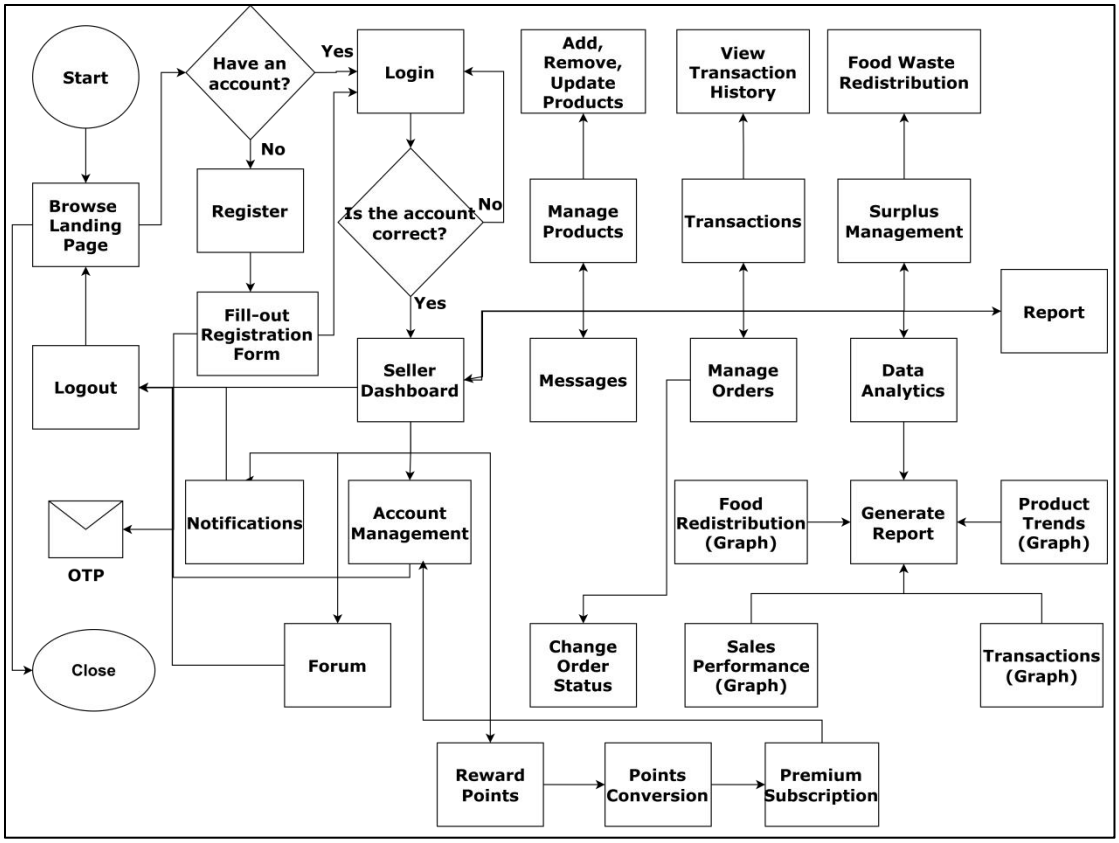


Figure _. Seller Flowchart of Conservate

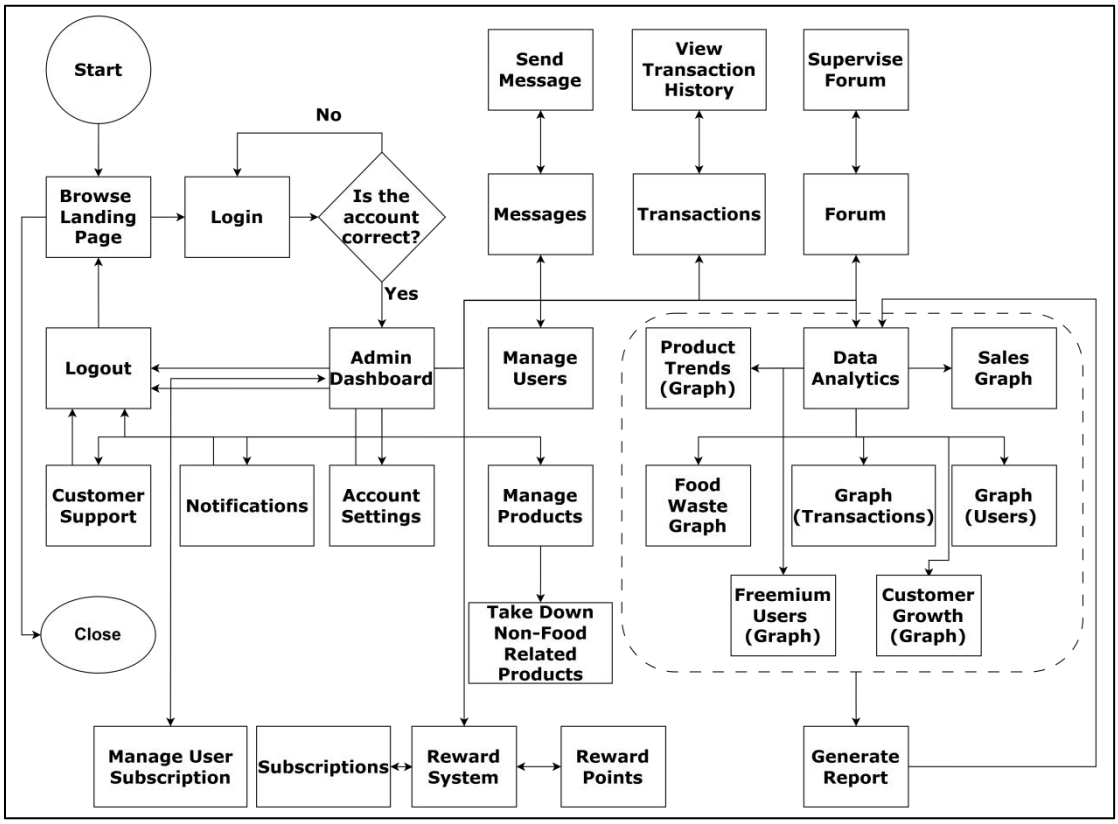
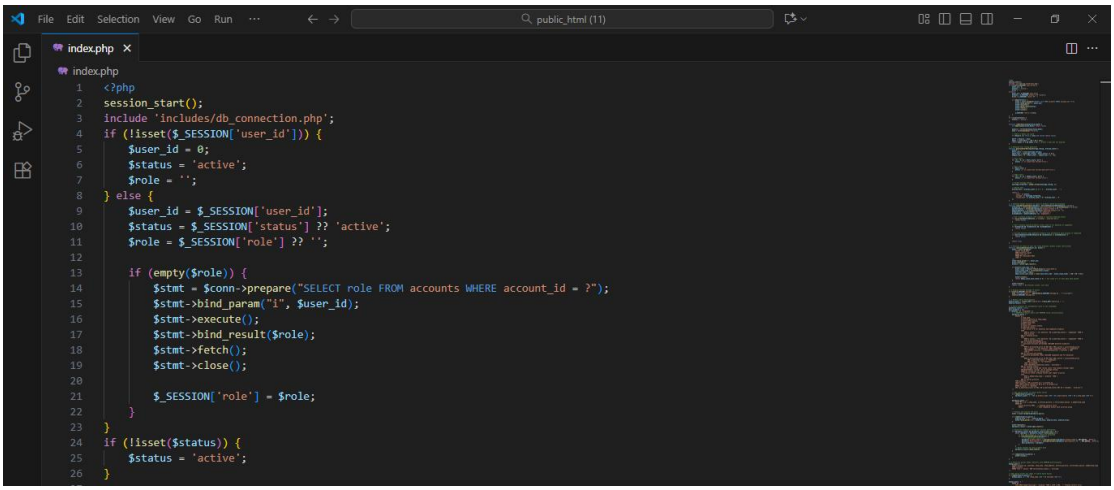


Figure _. Admin Flowchart of Conservate

SOURCE CODE

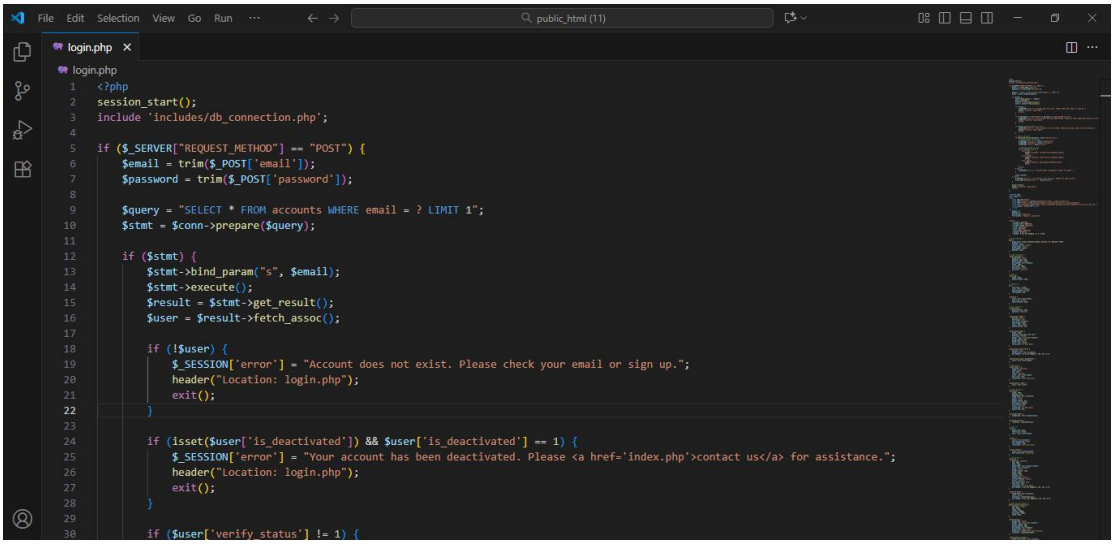


```php
<?php
session_start();
include 'includes/db_connection.php';
if (!isset($_SESSION['user_id'])) {
    $user_id = 0;
    $status = 'active';
    $role = '';
} else {
    $user_id = $_SESSION['user_id'];
    $status = $_SESSION['status'] ?? 'active';
    $role = $_SESSION['role'] ?? '';

    if (empty($role)) {
        $stmt = $conn->prepare("SELECT role FROM accounts WHERE account_id = ?");
        $stmt->bind_param("i", $user_id);
        $stmt->execute();
        $stmt->bind_result($role);
        $stmt->fetch();
        $stmt->close();

        $_SESSION['role'] = $role;
    }
}
if (!isset($status)) {
    $status = 'active';
}
```

Figure _. Landing Page Source Code



```php
<?php
session_start();
include 'includes/db_connection.php';

if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $email = trim($_POST['email']);
    $password = trim($_POST['password']);

    $query = "SELECT * FROM accounts WHERE email = ? LIMIT 1";
    $stmt = $conn->prepare($query);

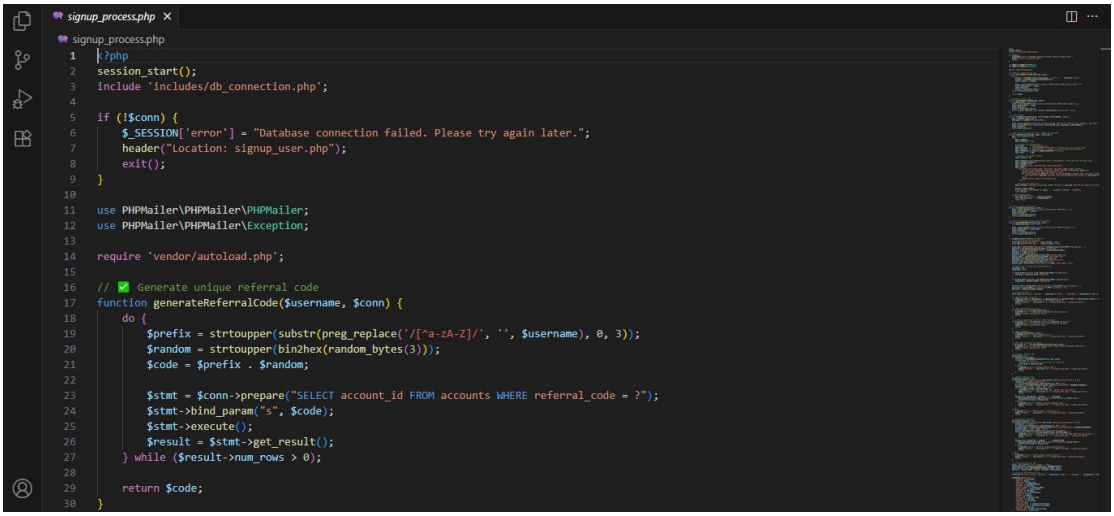
    if ($stmt) {
        $stmt->bind_param("s", $email);
        $stmt->execute();
        $result = $stmt->get_result();
        $user = $result->fetch_assoc();

        if (!$user) {
            $_SESSION['error'] = "Account does not exist. Please check your email or sign up.";
            header("Location: login.php");
            exit();
        }

        if (isset($user['is_deactivated']) && $user['is_deactivated'] == 1) {
            $_SESSION['error'] = "Your account has been deactivated. Please <a href='index.php'>contact us</a> for assistance.";
            header("Location: login.php");
            exit();
        }

        if ($user['verify_status'] != 1) {
```

Figure _. Login Source Code



```php
<?php
session_start();
include 'includes/db_connection.php';

if (!$conn) {
    $_SESSION['error'] = "Database connection failed. Please try again later.";
    header("Location: signup_user.php");
    exit();
}

use PHPMailer\PHPMailer\PHPMailer;
use PHPMailer\PHPMailer\Exception;

require 'vendor/autoload.php';

// ✅ Generate unique referral code
function generateReferralCode($username, $conn) {
    do {
        $prefix = strtoupper(substr(preg_replace('/[^a-zA-Z]/', '', $username), 0, 3));
        $random = strtoupper(bin2hex(random_bytes(3)));
        $code = $prefix . $random;

        $stmt = $conn->prepare("SELECT account_id FROM accounts WHERE referral_code = ?");
        $stmt->bind_param("s", $code);
        $stmt->execute();
        $result = $stmt->get_result();
    } while ($result->num_rows > 0);

    return $code;
}
```

Figure _. Signup Process Source Code

APPENDIX

C

Communication Letters and Approvals

APPROVAL SHEET

The Capstone Project entitled **"ConservAte: A WEB-BASED PLATFORM FOR FOOD SALES AND REDISTRIBUTION OF SURPLUS AND NEAR-EXPIRY FOOD PRODUCTS"** prepared and submitted by **Linory L. Bigtas, Charles I. De Vera, and Jhon-Rick L. Meltran** in partial fulfillment of the requirements for the degree Bachelor of Science in Information Technology specialized in Service Management Program, is hereby recommended for approval and acceptance.

JOCELYN O. PADALLAN
Adviser

Approved and Accepted by the Committee on Oral Examination with a grade of_____.

MIKE PHILIP T. RAMOS
Member

SHERWIN B. SAPIN
Member

MYRA V. PAICAGLINO
Member

JEFFERSON L. LERIOS
Member

Accepted in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology specialized in Service Management Program.

LOYD S. ECHALAR
Associate Dean, College of Computer Studies

OFELIA B. MANINGAS, EdD.
Chairperson, Research and Development Office

_____
Date Signed

RESEARCH CONTRIBUTION NO.	

Republic of the Philippines
Laguna State Polytechnic University
College of Computer Studies
Province of Laguna

March 10, 2025

HON. ANTHONY "TON" GENUINO
Municipal Mayor
Los Baños, Laguna

Subject: Data Request for Capstone Project entitled "ConservAte: A Web-Based System for Sustainable Food Waste Management and Redistribution to Enhance Food Security"

Dear Hon. Genuino,

I hope this message finds you well.

We are currently pursuing a Bachelor's Degree in Information Technology and are working on a project focused on food waste and redistribution in Los Baños, Laguna. As part of this, we kindly request access to raw data on food waste management within the municipality, excluding personal details to maintain anonymity.
Specifically, we are seeking data on:

- Food waste statistics (volume, sources, seasonal trends)
- Food donation programs and registered charities
- Existing waste management policies and regulations
- Environmental impact reports on food waste
- Business participation in food waste reduction efforts

This data will aid our research and may offer valuable insights for improving food waste management in your organization and the community. We assure you that all information will be kept confidential and used solely for academic purposes in compliance with the Data Privacy Act of 2012.

We would appreciate your approval and are open to any terms or conditions for data access. Upon completion of our study, we will share our findings with your office as a token of gratitude.

Thank you for your consideration. We look forward to your response and are happy to provide further information if needed.

Warm regards,
BIGTAS, LINORY L.
Group Representative

Noted by:

JOCELYN O. PADALLAN
Capstone / Thesis Adviser,
College of Computer Studies Faculty

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

THESIS ADVISER NOMINATION

Students/s Name and Signature : DE VERA , CHARLES

: BIGTAS , LINORY

: MELTRAN , JHON-RICK

:

Degree Sought : BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

Major Field : SERVICE MANAGEMENT PROGRAM

I/we, hereby nominate as Thesis Adviser

Name of Faculty : Jocelyn Padallan

College :

Field of Specialization :

I, hereby indicate my willingness to serve as the Thesis Adviser.

Signature of Faculty : _____ 3/07/25

Date Signed

Endorsed by:

_____
Research Coordinator

Recommending Approval:

_____
Dean/Associate Dean

APPROVED/DISAPPROVED

_____
R&D Director/Chairperson

APPENDIX

D

Publishable Paper

ConservAte: A Web-Based Platform for Food Sales and Redistribution of Surplus and Near-Expiry Food Products

Linory L. Bigtas, Charles I. De Vera, Jhonrick L. Meltran
College of Computer Studies (CCS), Laguna State Polytechnic University, Los Baños, Laguna, Philippines

***Abstract* —** Food waste continues to be a problem across the world for food security, environmental sustainability, and economic stability. Despite these, various initiatives have not fully translated to communities having systems that can support accessible surplus or near-expired food management. Therefore, this research addresses the problem of increased food waste by developing ConservAte, a web-based platform meant to handle the sale and redistribution of food. Though previous studies illustrated that technology incorporation offers the support needed for sustainable consumption, most have failed to incorporate location-based and user-centered approaches. Hence, the present study makes such an incorporation by using GIS technology to connect consumers and vendors efficiently. Employing the Agile Scrum methodology for development, the system was evaluated using ISO 25010 and the Technology Acceptance Model. The results obtained from testing indicated that the system met all the quality standards, hence proving efficient, usable, and reliable. This study provides evidence that ConservAte is an innovative yet practical way to advance food sustainability and community food security.

Keywords — GIS, TAM, Redistribution, Food Security

I. INTRODUCTION

Food waste is a global crisis that requires careful planning and strategies for effective management [1]. Each year, approximately 88 million tons of food are wasted, amounting to 174 kilograms per person and resulting in an economic loss of around 143 billion euros [2]. Alarmingly, food waste contributes 8–10% of global greenhouse gas emissions, worsening the climate crisis [3]. Despite this surplus, millions of people still suffer from hunger and malnutrition, highlighting a significant imbalance in the global food system.

Consumer behavior significantly contributes to food waste through overbuying, improper food storage, and misreading expiration dates, while inefficiencies across the food supply chain—from manufacturing to retail—further exacerbate the issue [4]. Technological innovations offer potential solutions for better food waste management. Tools such as smart packaging, tracking systems, and food-sharing applications can extend food shelf life, simplify inventory management, and enable efficient redistribution of near-expiry products [5].

This capstone project aims to develop a web-based system for managing food waste sustainably and redistributing surplus food to enhance food security. The project specifically aims to; Design and develop a web-based system to facilitate sustainable food waste management and efficient redistribution of surplus food, To Implement a tracking and analytics system to monitor food waste trends, optimize redistribution, and ensure secure data management. To evaluate the web-based system for ISO 25010 and Technology Acceptance Model (TAM) for Software Quality Standards

This project supports several United Nations Sustainable Development Goals (SDGs) focused on reducing food waste and promoting sustainability. It aligns primarily with SDG 12: Responsible Consumption and Production (Target 12.3) by encouraging efficient food production, distribution, and consumption to minimize waste. It also contributes to SDG 2: Zero Hunger (Target 2.4) through sustainable food systems, and SDG 13: Climate Action (Target 13.2) by reducing greenhouse gas emissions from food waste. Additionally, it supports SDG 11: Sustainable Cities and Communities (Target 11.6) through urban waste reduction and SDG 1: No Poverty (Target 1.2) by redistributing surplus food to low-income families, improving food access and affordability.

II. RELATED WORKS

Food security ensures that everyone has consistent physical and financial access to sufficient, safe, and nutritious food for a healthy life [6]. However, factors like population growth, climate change, rising food prices, and environmental challenges hinder this goal [7]. Despite some progress in certain countries, the world is still far from achieving Sustainable Development Goals 2.1 (ending hunger) and 2.2 (eliminating malnutrition). As a result, billions of people continue to lack access to safe and adequate food [8].

Food waste is a global problem involving the loss of edible food across the entire supply chain, from farms to households [9]. It often occurs during production, harvesting, or processing, particularly in developing countries with poor infrastructure and storage systems. Nearly one-third of all food produced globally is wasted before reaching consumers [10]. Moreover, studies show that while many people intend to reduce food waste, their actual behaviors—such as poor planning and purchasing habits—often fail to reflect these intentions [11].

The study emphasizes that all stakeholders in the food supply chain—from farmers to consumers—are responsible for reducing food waste. Redistributing surplus food is an effective way to cut waste and combat poverty, though some view it as shifting government responsibility, with Italian research stressing the need for long-term supportive policies [12]. Contributing factors to food waste include poor expiry date management, lack of plans for leftovers, and pricing systems that encourage overbuying. Nevertheless, environmentally conscious families and those buying discounted products tend to waste less food [13].

Digital technologies for food waste reduction are still in early development, with limited experimentation, prototyping, and application deployment [14]. Moreover, there is a lack of data on the social, environmental, and economic benefits of these innovations [15]. In the Philippines, eCommerce experienced significant growth over the past five years due to the COVID-19 pandemic, which accelerated digital transactions. The sector is projected to grow at a substantial 17% Compound Annual Growth Rate (CAGR) between 2021 and 2025, potentially reaching 2.5 times its current size by 2025 [16][17].

The Geographic Information System (GIS) enables users to visualize, analyze, and interpret spatial patterns by mapping data such as streets, landmarks, and establishments on a single platform. It captures, stores, and displays information about Earth's features using computer-based mapping technology [18]. When combined with e-commerce, GIS helps businesses understand customer demographics and behavior based on location. This allows vendors to create targeted marketing strategies, personalized promotions, and region-specific campaigns to enhance user engagement [19].

III. METHODOLOGY

This section covers the project design and development, project materials, business model, data collection instrument, statistical treatment, and project testing needed for this study and system development.

Project Design

This section provides the design of the project. To understand clearly, the researchers made a visual of the system's flow.

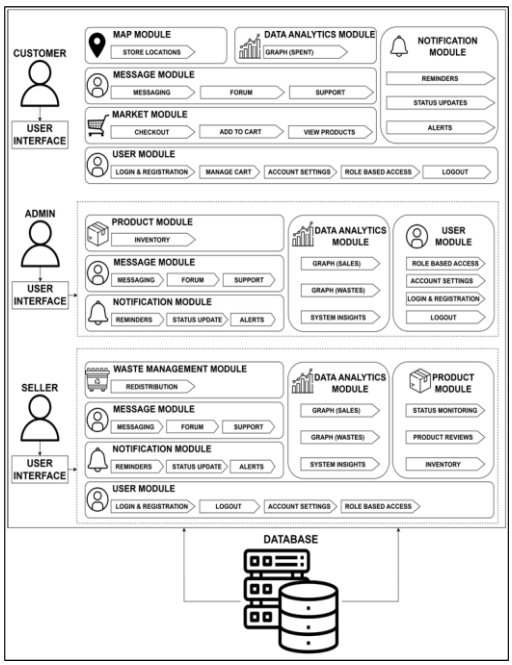


Figure 1. ConservAte's System Architecture

Figure 1 illustrates the architecture of the ConservAte web-based application, which integrates modules such as user, market, product, order, notification, messaging, waste management, and data analytics. Each module works together to ensure efficient system performance and a smooth user experience. The system allows users to manage accounts, monitor products, send messages, and receive

real-time updates on orders and notifications. It also includes waste management and data analytics features that track food waste, display visual reports, and support data-driven decision-making.

Project Development

This section indicates what type of methodology is used by the developers.

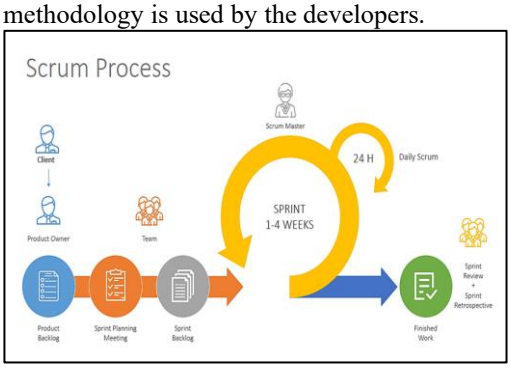


Figure 2. Agile Scrum Methodology

Figure 2 illustrates the implementation of the Agile Scrum model in developing the ConservAte web-based system. Agile Scrum is a flexible, collaborative methodology that emphasizes continuous feedback and improvement. The process begins with a product backlog outlining all system functions, followed by sprint planning, where high-priority tasks are chosen for completion in each cycle. Its iterative framework with short design phases allows the team to quickly adapt to changing requirements and evolving system needs [20].

Project Flowcharts

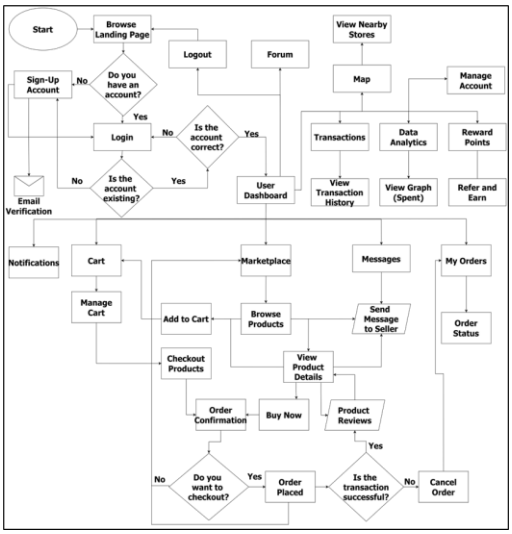


Figure 3. System Flowchart for Customer

Figure 3 shows the customer workflow in the ConservAte web app, starting from browsing products on the landing page to registering and logging in. The dashboard allows customers to place orders, manage carts, message vendors, track transactions, locate stores, redeem rewards, and participate in refer-and-earn programs. The system provides real-time notifications, product reviews, spending analytics, and a user forum. Customers can securely log out, returning to the landing page.

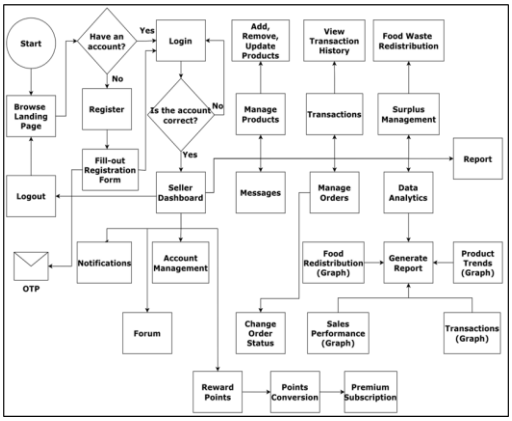


Figure 4. System Flowchart for Seller

Figure 4 illustrates the vendor workflow in ConservAte, beginning with registration and email verification. Approved vendors use the dashboard to manage products, orders, and customer interactions, including handling near-expiry or surplus food through discounts or donations to earn rewards and boost marketplace visibility. The system offers real-time alerts, sales analytics, a user forum, and secure logout to ensure account safety.

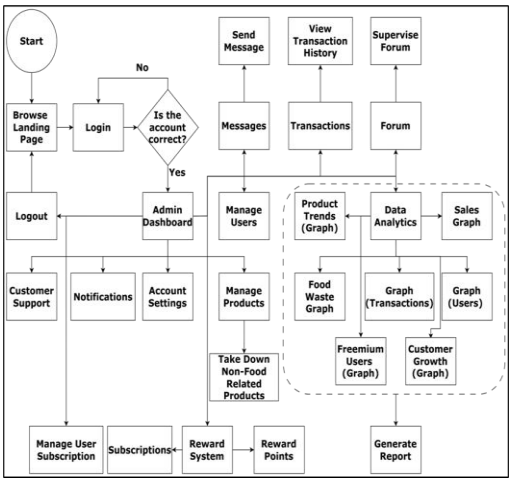


Figure 5. System Flowchart for Super Admin

Figure 5 illustrates the admin functions in ConservAte. Admins access a dashboard to monitor products, manage transactions, resolve disputes, and guide food waste reduction. The system provides sales analytics, trend reports, user activity data, and real-time notifications to oversee all user

actions. Admins also supervise forums and ensure secure logouts, maintaining smooth and efficient platform operations.

Project Business Model

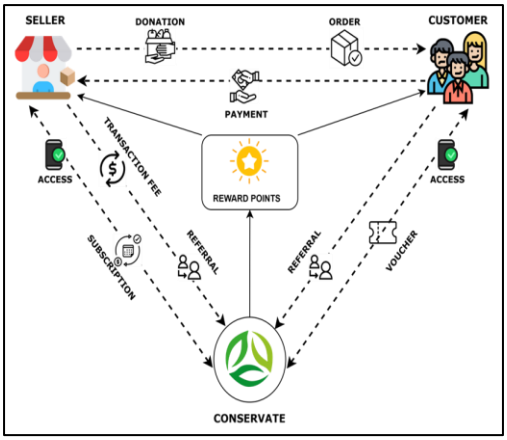


Figure 6. ConservAte's Business Model

Figure 6 illustrates the business model of the ConservAte web-based system, showing benefits for customers, vendors, and admins. Vendors can subscribe to tiered plans and use reward points from referrals and donations to boost their products in the marketplace. Customers can browse products, earn reward points from transactions, redeem them for discount vouchers, and use the referral feature. The platform generates revenue from transaction fees, while promoting user engagement, food accessibility, and food waste reduction for sustainability.

Project Testing and Evaluation

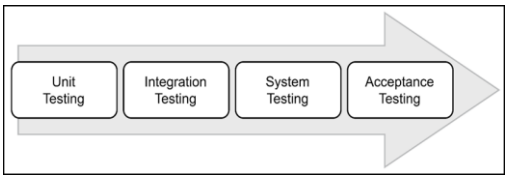


Figure 7. Level of Testing

Figure 7 shows the project testing process, including unit, integration, system, and acceptance testing, to identify errors and ensure system reliability. This thorough evaluation verifies functionality and enhances user experience, allowing the researchers to deliver a high-quality, user-satisfactory system.

IV. RESULT AND DISCUSSIONS

This section presents and discusses the findings of the study, detailing the system's functionalities and features, evaluating its effectiveness, and demonstrating how it fulfills the project's objectives through various tests.

Results by the Objectives

A. Design and develop a web-based platform, integrated with Geographic Information System (GIS), to support the food sales and the redistribution of near-expiry and surplus food products.

Figure 8 presents the key user interfaces of the ConservAte web platform, supporting the project's first objective. These include the (a) landing page, (b) map feature, and (c) surplus management. The landing page is accessible to all users, displaying the system's purpose, available products, featured sellers, and contact information. The GIS-integrated map helps users locate nearby sellers and their products, while the surplus management feature allows vendors to monitor product status, adjust prices for near-expiry items, or initiate donations to reduce food waste.

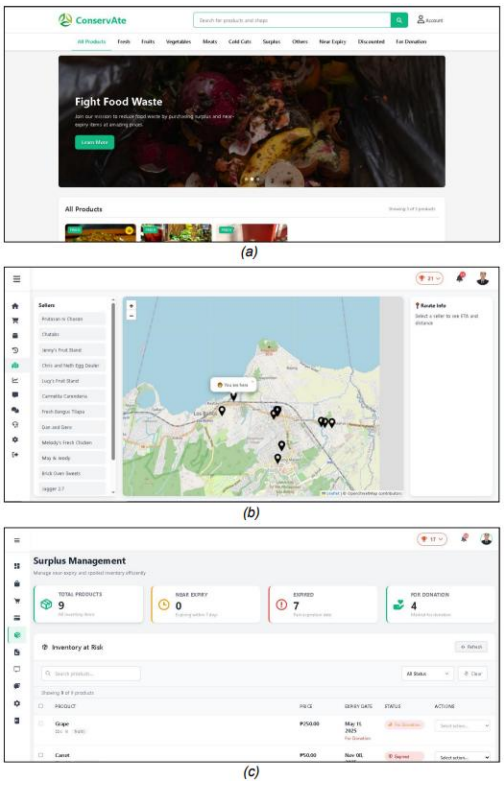
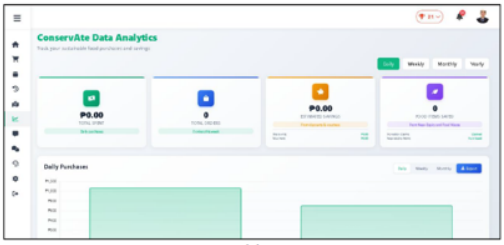


Figure 8 .ConservAte System Interfaces: *(a)* Landing Page, *(b)* Customer Map, *(c)* Seller Surplus Management.

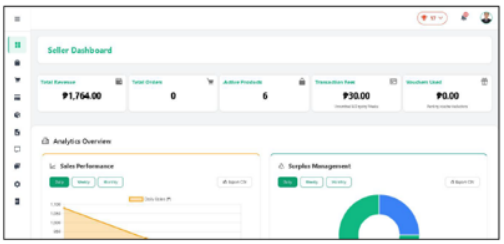
B. To implement a tracking and analytics system to monitor food waste trends, optimize redistribution, and ensure secure data management.

Figure 9 illustrates the analytics features in ConservAte for different user roles. Customers (a) can track total spending, orders,

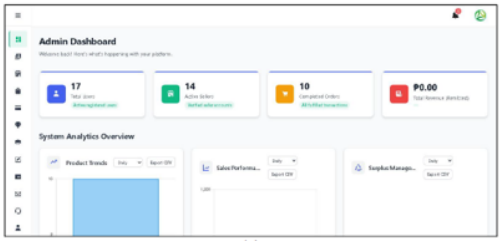
savings, and products saved. Sellers (b) can monitor revenue, active products, ongoing orders, transaction fees, sales performance, product trends, surplus management, and expired versus donated products. The super admin (c) oversees analytics across all sellers to evaluate overall platform performance. Users can view data by daily, weekly, monthly, or yearly ranges to effectively track and assess their performance



(a)



(b)



(c)

Customer Analytics *(a)*, Seller Analytics *(b)*, and Super Admin Analytics *(c)*

C. To evaluate the web-based system's abilities using ISO 25010 and Technology Acceptance Model (TAM) for Software Quality Standards.

Table 1. Technology Acceptance Model Results

Indicators	Mean	SD	Interpretation
Perceived Usefulness	3.58	0.60	Strongly Agree
Perceived Ease of Use	3.52	0.60	Strongly Agree
Personal Innovativeness	3.51	0.59	Strongly Agree
Social Influence	3.59	0.56	Strongly Agree
Information Awareness	3.55	0.59	Strongly Agree
Perceived Cost	3.50	0.66	Strongly Agree
Willingness to Adopt	3.57	0.57	Strongly Agree
Overall Mean	**3.55**	**0.60**	**Strongly Agree**

Legend: 3.25 - 4.00 = Strongly Agree, 2.50 - 3.24 = Agree, 1.75 - 2.49 = Disagree, 1.0 - 1.74 = Strongly Disagree

Table 1 summarizes the constructs of the Technology Acceptance Model (TAM) for ConservAte. Users perceived the platform as useful (mean = 3.58) and user-friendly (mean = 3.52). Personal preference for digital solutions addressing food waste scored a mean of 3.51, while recommendations from trusted sources had the highest influence on usage (mean = 3.59). Users showed awareness of food waste initiatives (mean = 3.55), considered the platform cost-effective (mean = 3.50), and expressed willingness for continued use (mean = 3.57). Overall, the TAM survey yielded a composite mean of 3.55, indicating strong agreement on the platform's usefulness, effectiveness, and efficiency.

Table 2. ISO 25010 Results

Indicators	Mean	SD	Interpretation
Functional Suitability	3.78	0.44	Strongly Agree
Performance Efficiency	3.83	0.41	Strongly Agree
Compatibility	3.75	0.46	Strongly Agree
Usability	3.77	0.35	Strongly Agree
Reliability	3.75	0.44	Strongly Agree
Security	3.78	0.33	Strongly Agree
Maintainability	3.57	0.54	Strongly Agree
Flexibility	3.58	0.50	Strongly Agree
Safety	3.60	0.53	Strongly Agree
Overall Mean	**3.71**	**0.44**	**Strongly Agree**

Legend: 3.25 - 4.00 = Strongly Agree, 2.50 - 3.24 = Agree, 1.75 - 2.49 = Disagree, 1.0 - 1.74 = Strongly Disagree

Table 2 summarizes IT experts' evaluations of the ConservAte web system using ISO 25010, covering functionality, performance, reliability, security, safety, maintainability, flexibility, usability, and compatibility. The composite mean of 3.71 indicates strong agreement that the platform meets its design objectives, functions effectively, and maintains high software quality, making it a reliable, trustworthy, and capable system.

Integration Testing Result

The results indicate that all ConservAte system modules—user, data analytics, product, market, message, notification, waste management, and map—are well-integrated and function cohesively. They effectively share information and perform tasks synchronously, ensuring the platform operates smoothly and efficiently. This level of integration allows real-time data consistency across modules, minimizing errors and delays in system processes. As a result, users experience seamless navigation and reliable system responses, while administrators can efficiently monitor activities, manage

transactions, and generate reports. Overall, the cohesive interaction among the modules supports the system's objectives of improving operational efficiency, enhancing user experience, and ensuring effective management of near-expiration products within the ConservAte platform.

Table 3. Integration Testing Result: Customer, Seller, Super Admin.

Tested Case	Expected Output	Actual Output	Process	Remarks
User Module (Customer, Seller, Super Admin)	User functions should seamlessly integrate with all other modules.	User authentication and profile data successfully shared across all connected modules	3 secs	Passed
Market Module (Customer)	Product browsing, cart, and checkout should integrate with accounts and location.	Customers can browse products, manage carts, and complete purchases using their account and location.	3 secs	Passed
Message Module (Customer, Seller, Super Admin)	The messaging system should connect all users seamlessly.	All users can send and receive messages through their accounts.	3 secs	Passed
Map Module (Customer)	Location services should integrate with market especially store locations	Store locations should display with ETA, linking to product availability and user location.	3 secs	Passed
Notification Module (Customer, Seller, Super Admin)	The alert system should operate across all modules for all users.	Notifications should reach all users for every activity and system update.	2 secs	Passed
Data Analytics Module (Customer, Seller, Super Admin)	Reports should update in real time for all users.	Dashboards should show spending and saved products for all users.	2 secs	Passed
Waste Management Module	Product status should update across inventory and analytics.	Expired and near-expiry products automatically updated in all modules	2 secs	Passed
Product Module (Seller, Super Admin)	Product status should sync with inventory and analytics.	All products appeared correctly across all connected systems	3 secs	Passed

V. CONCLUSION AND FUTURE WORKS

The ConservAte web-based platform effectively brings together sellers, buyers, and beneficiaries to support the sale and redistribution of near-expiry and surplus food products. Through its integration with Geographic Information System (GIS) technology, users can easily locate nearby sellers, markets, and redistribution points, making delivery and pickup more efficient. This location-based feature helps streamline food redistribution, reduce unnecessary waste, and extend support to communities experiencing food insecurity.

ConservAte also includes a tracking and analytics module that provides valuable insights into sales performance, food waste trends, and redistribution activities. These features allow users to make informed, data-driven decisions, while administrators can better plan and implement strategies for sustainable food management. The availability of real-time data improves inventory handling and supports more effective resource allocation across the platform.

The platform was designed with usability and accessibility in mind to ensure a smooth experience for all types of users, including sellers, buyers, and administrators. Its intuitive interface, clear product information, and real-time expiration tracking make it easy for users to navigate the system and complete transactions efficiently. By simplifying the process of listing, purchasing, and redistributing food products, ConservAte encourages consistent user engagement and supports responsible consumption behavior.

Evaluation results based on the ISO 25010 Software Quality Model and the Technology Acceptance Model (TAM) further validated the system's overall performance and acceptance. The findings confirmed that ConservAte is reliable, secure, scalable, and cost-effective, with users expressing strong intentions to adopt and recommend the platform. Overall, ConservAte successfully meets both technical and user-centered requirements, reinforcing its role as a sustainable digital solution that addresses food waste reduction while promoting food accessibility and community support.

Furthermore, ConservAte demonstrates strong potential for future growth and wider implementation. The system's scalable architecture allows for the integration of additional features, such as expanded analytics, mobile application support, and partnerships with local organizations and government units. By continuously improving the platform based on user feedback and evolving needs, ConservAte can further strengthen its impact on reducing food waste and supporting sustainable food distribution practices in the long term.

Recommendations for improvement include adding camera-based image recognition for product freshness, Blockchain for transaction transparency, native iOS and Android apps, dynamic pricing for near-expiry products, and enhanced GIS for real-time delivery tracking. These enhancements would further increase system efficiency, accuracy, and user engagement, while ensuring secure and transparent operations. Implementing such features could also encourage wider adoption of the platform and strengthen its impact on reducing food waste and supporting communities in need.

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APPENDIX

E

Plagiarism Report

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APPENDIX

F

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rare words

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Measures average word length

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Measures average sentence length

24.4

words per sentence

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13	**Delivery**	
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2	Incomplete sentences	
1	Potentially sensitive language	
1	Tone suggestions	
132	**Clarity**	
98	Paragraph can be improved	
18	Passive voice misuse	
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3	Text inconsistencies	
9	**Engagement**	
9	Word choice	

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Measures vocabulary diversity by calculating the percentage of words used only once in your document

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unique words

ConservAte Chapter 3-5

by z

General metrics

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characters	words	sentences	reading time	speaking time

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Writing Issues

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Measures average word length

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Sentence Length

Measures average sentence length

21.4 words per sentence

Writing Issues

4	**Correctness**	
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rare words

APPENDIX

G

Curriculum Vitae



JHON-RICK L. MELTRAN

09948140980 meltranjhonrick13@gmail.com Barandal,Calamba,Laguna

PROFILE

Pursuing a Bachelor of Science in Information Technology specialized in Service Management, a 4-year degree program graduate. A highly motivated and dedicated individual with strong time-management abilities and a proven capacity to perform effectively under pressure. Seeking opportunities in a dynamic organization to further develop expertise and contribute meaningfully to the field.

TECHNICAL SKILLS

- Computer Literate
- Website Development
- Front-End Development
- Microsoft Office
- C#, Java, Phyton, JavaScript

EXPERTISE

- Management Skills
- Digital Marketing
- Negotiation
- Critical Thinking
- Communication Skills

LANGUAGE

- English
- Filipino

EDUCATION

2022 – 2026

Laguna State Polytechnic University - Los Baños Campus

Malinta, Los Baños

- Bachelor of Science in Information Technology, Major in Service Management Program

2020 - 2022

Calamba Doctors College

VIRBOROUGH SUBDIVISION, Calamba, Laguna

- Senior High School

2016 - 2020

Asian Computer College

Doctora St, Calamba, Laguna

- Junior High School

2020 - 2022

Mayapa Elementary School

Maryland Rd, Calamba, Laguna

- Elementary

PROJECTS AND ACHIEVEMENTS

- Tilamseek System
- LSPU-BAO System
- Mic & Lens
- ConservAte System
- 1st Year Dean's Lister
- E-Shop



CHARLES I. DE VERA

09665574751 charlesdevera0409@gmail.com Timugan, Los Baños, Laguna

PROFILE

A diligent 4th-year Bachelor of Science in Information Technology student, majoring in Service Management. He is a driven individual committed to continuously improving his technical, communication, and performance skills. His goal is to contribute to organizational efficiency and deliver enhanced service within a dynamic professional environment.

TECHNICAL SKILLS

- Computer Literate
- Website Development
- Front-End Development
- Microsoft Office
- Knowledgeable in C#, Javascript, Java, HTML, CSS, PHP

EXPERTISE

- Digital Literacy
- Digital Marketing
- Customer Handling
- Critical Thinking
- Communication Skills
- Computer System Servicing

LANGUAGE

- English
- Filipino

EDUCATION

2022 – 2026

Laguna State Polytechnic University - Los Baños Campus

Malinta, Los Baños

- Bachelor of Science in Information Technology, Major in Service Management Program

2018 - 2020

Los Baños Bambang Senior High School

Bambang, Los Baños, Laguna

- Senior High School

2014 - 2018

Los Baños National High School - Poblacion

Timugan, Los Baños, Laguna

- Junior High School

2008 - 2014

Los Baños Central Elementary School

Timugan, Los Baños, Laguna

- Elementary

PROJECTS AND ACHIEVEMENTS

- Tilamseek System
- LSPU-BAO System
- PurrFect System
- ConservAte System
- 1st - 4th Year Dean's Lister
- Little Municipal Assessor (2018)



LINORY L. BIGTAS

0921-744-3138 | bigtaslinory26@gmai.com | Maahas, Los Baños, Laguna

PROFILE

Taking Bachelor of Science in Information Technology Specialized in Service Management Program 4-year degree Graduate and a highly driven and passionate individual with excellent time-management skills and the ability to operate under pressure in any situation. Looking for new chances in a vibrant organization to gain expertise in this subject on the ground.

TECHNICAL SKILLS

- Computer Literate
- Website Development
- Front-End Development
- Microsoft Office
- Proficient in any Programming Language (C#, Java, Python, JavaScript)

EXPERTISE

- Cybersecurity
- Digital Marketing
- Customer Handling
- Critical Thinking
- Communication Skills
- Computer System Servicing

LANGUAGE

- English
- Filipino

EDUCATION

2022 – 2026

Laguna State Polytechnic University - Los Baños Campus

Malinta, Los Baños

- Bachelor of Science in Information Technology, Major in Service Management Program

2019 - 2021

Manacsac High School

Manacsac, Guimba, Nueva Ecija

- Senior High School

2015 - 2019

Manacsac High School

Manacsac, Guimba, Nueva Ecija

- Junior High School

2000 - 2015

Bay Central Elementary School

Bay, Laguna

- Elementary

PROJECTS AND ACHIEVEMENTS

- Tilamseek System
- LSPU-BAO System
- Mic&Lens Studio System
- ConservAte System
- 1st - 4th Year Dean's Lister
- JITU ID Automated Proccessing System